This is CS50

This is CS50

- On today's snack menu, chocolate chip cookies. 🍪
- If you'd like to attend CS50 Lunch this Fri 10/10 at 1:15pm, RSVP at cs50.ly/lunch!
  - We'll be joined by alumni from Meta.
- Do say hi or ask questions during break or after class!
  - What do today's binary bulbs spell?

This is CS50

data structures

abstract data types

queues

FIFO

enqueue

dequeue

```
const int CAPACITY = 50;

typedef struct
{
    person people[CAPACITY];
    int size;
} queue;
```

stacks

LIFO

push

pop

```
const int CAPACITY = 50;

typedef struct
{
    person people[CAPACITY];
    int size;
} stack;
```

dictionaries

word	definition

name	number

key	value

Lists

Contacts

Search

B

Bowser

Bowser Jr.

D

Daisy

Diddy Kong

Donkey Kong

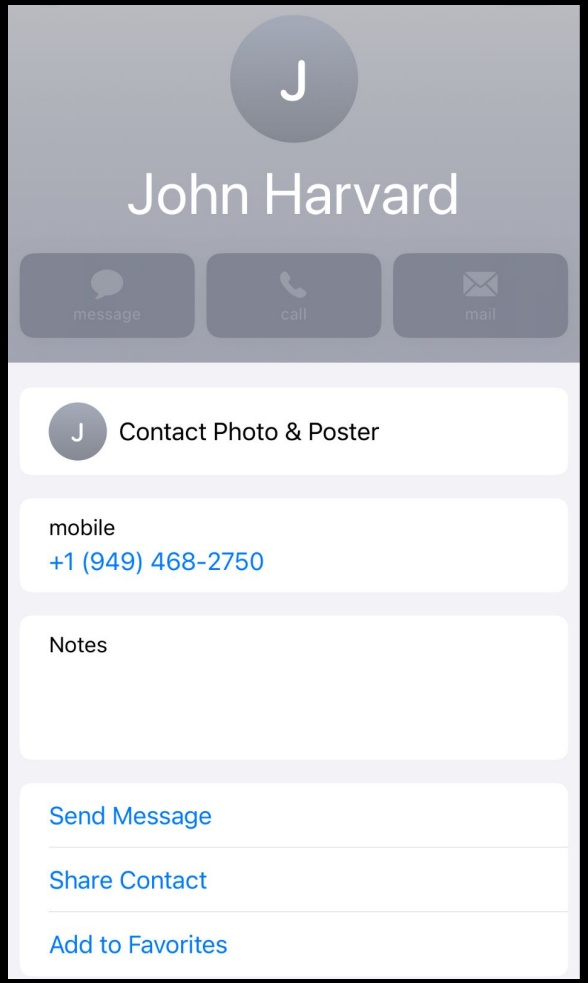
L

Luigi

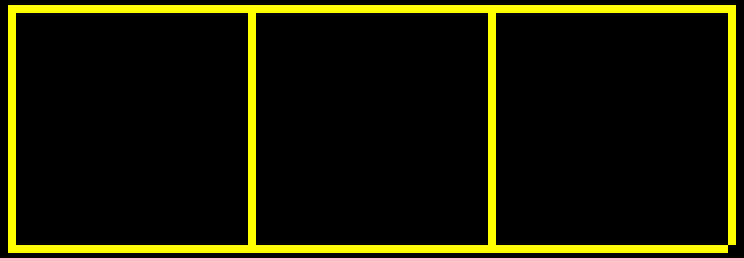
M

Mario

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z #


J
John Harvard
message
call
mail
J
Contact Photo & Poster
mobile
+1 (949) 468-2750
Notes
Send Message
Share Contact
Add to Favorites

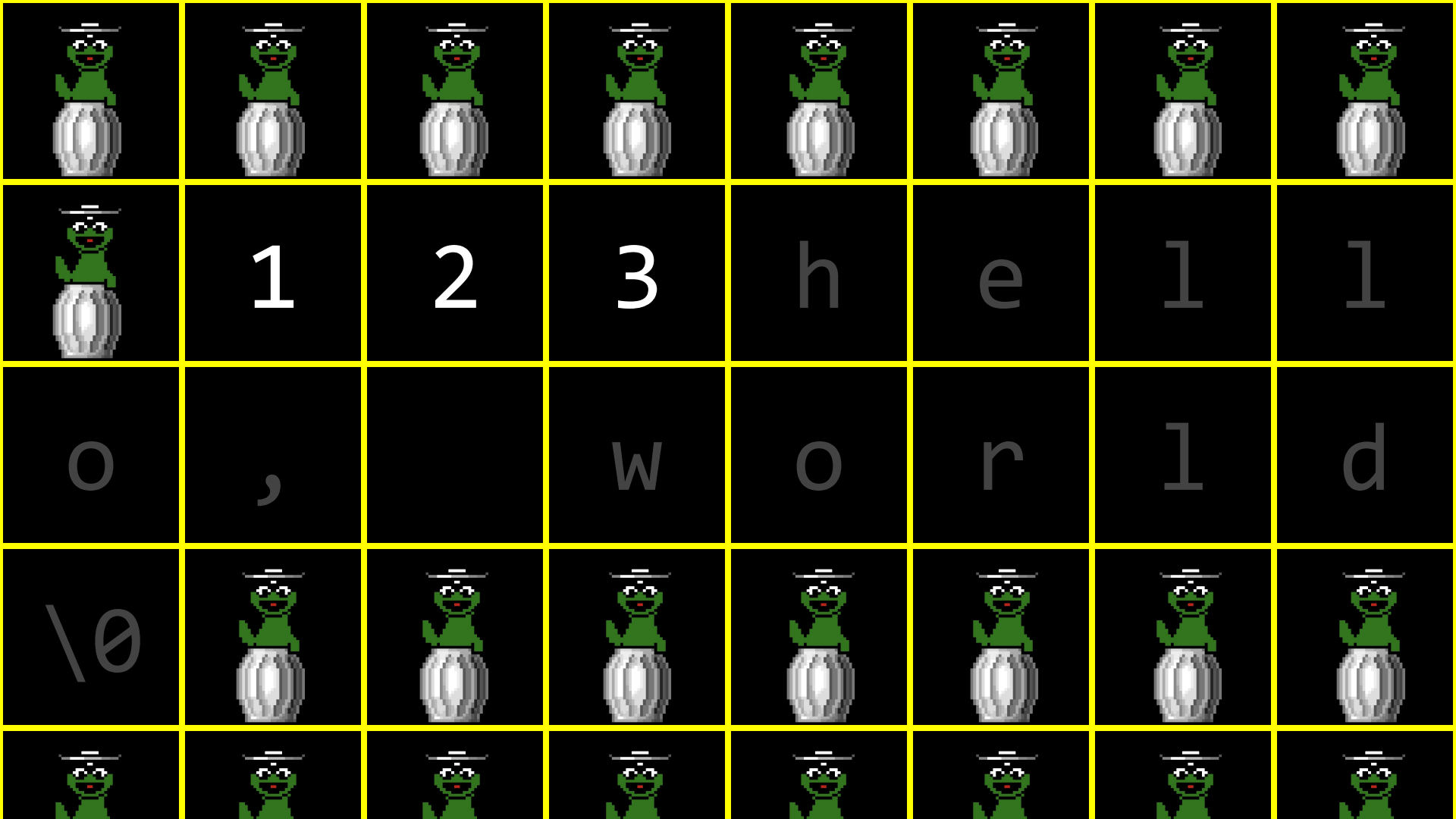
arrays



1	2	3

1	2	3	

1 2 3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o , w o r l d
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1	2	3

1			

1	2	3

1	2		

1	2	3

1	2	3	

1	2	3

1	2	3	4

1	2	3	4

data structures

struct

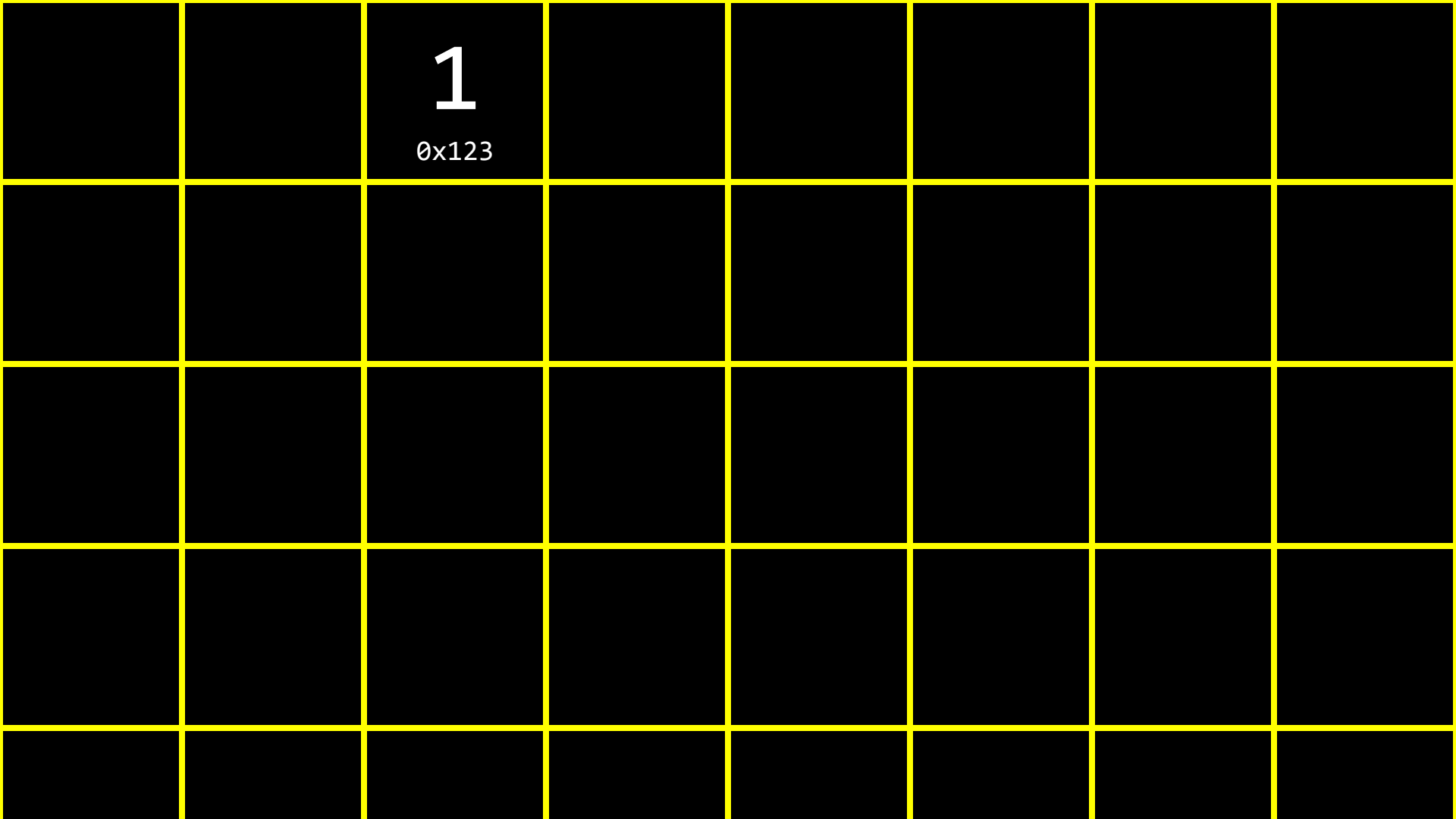
.

*

struct

->

linked lists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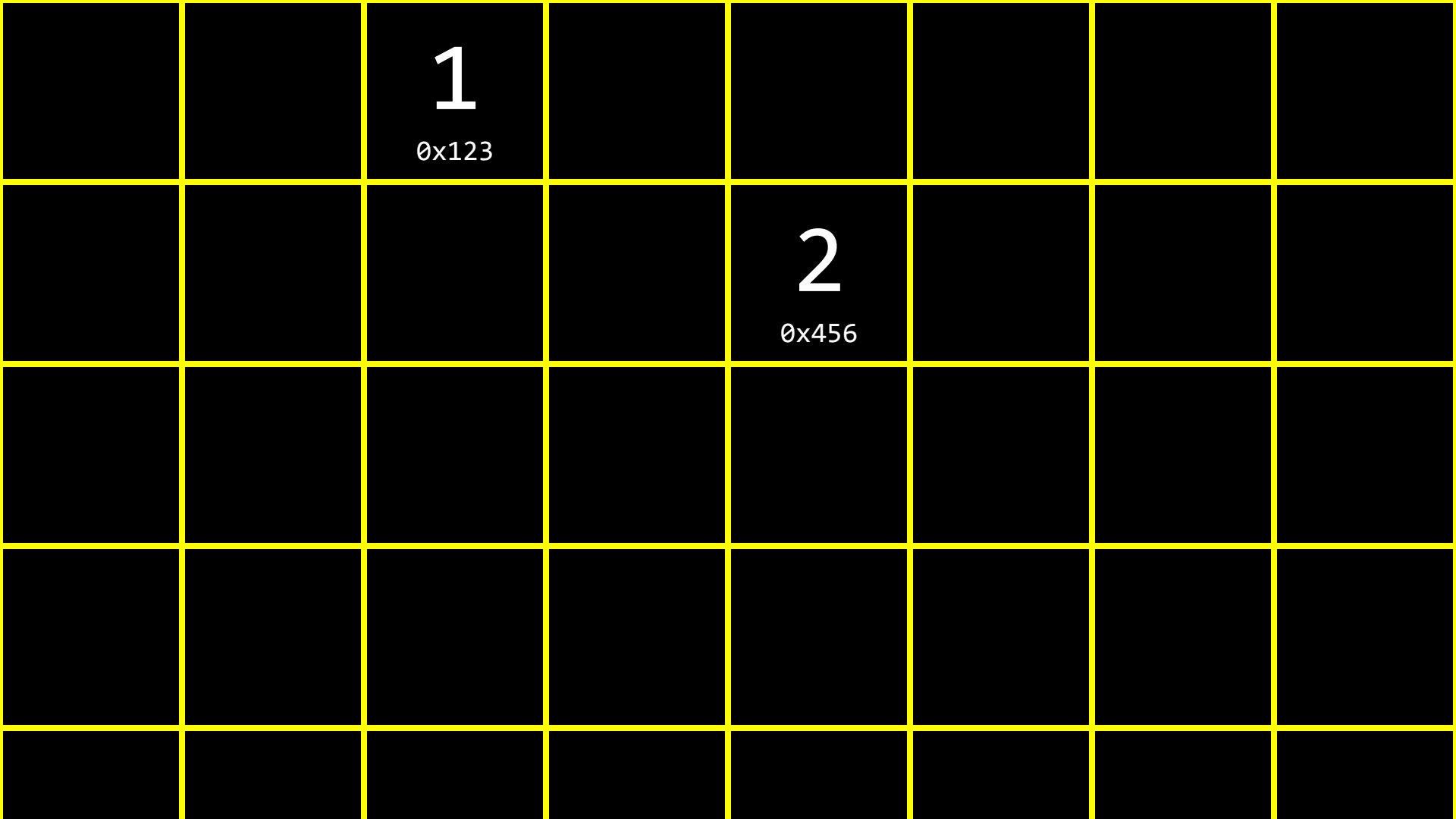
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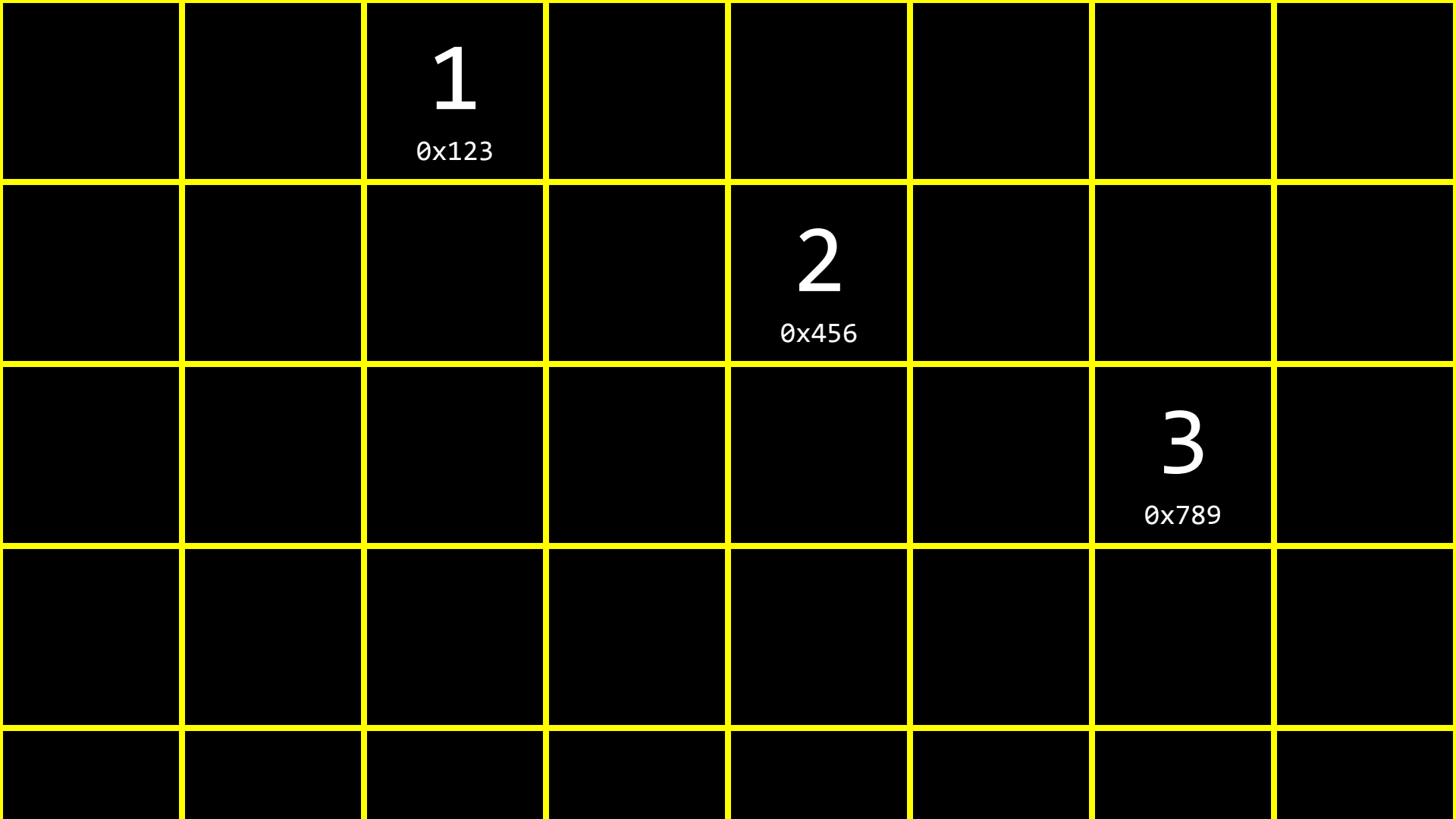
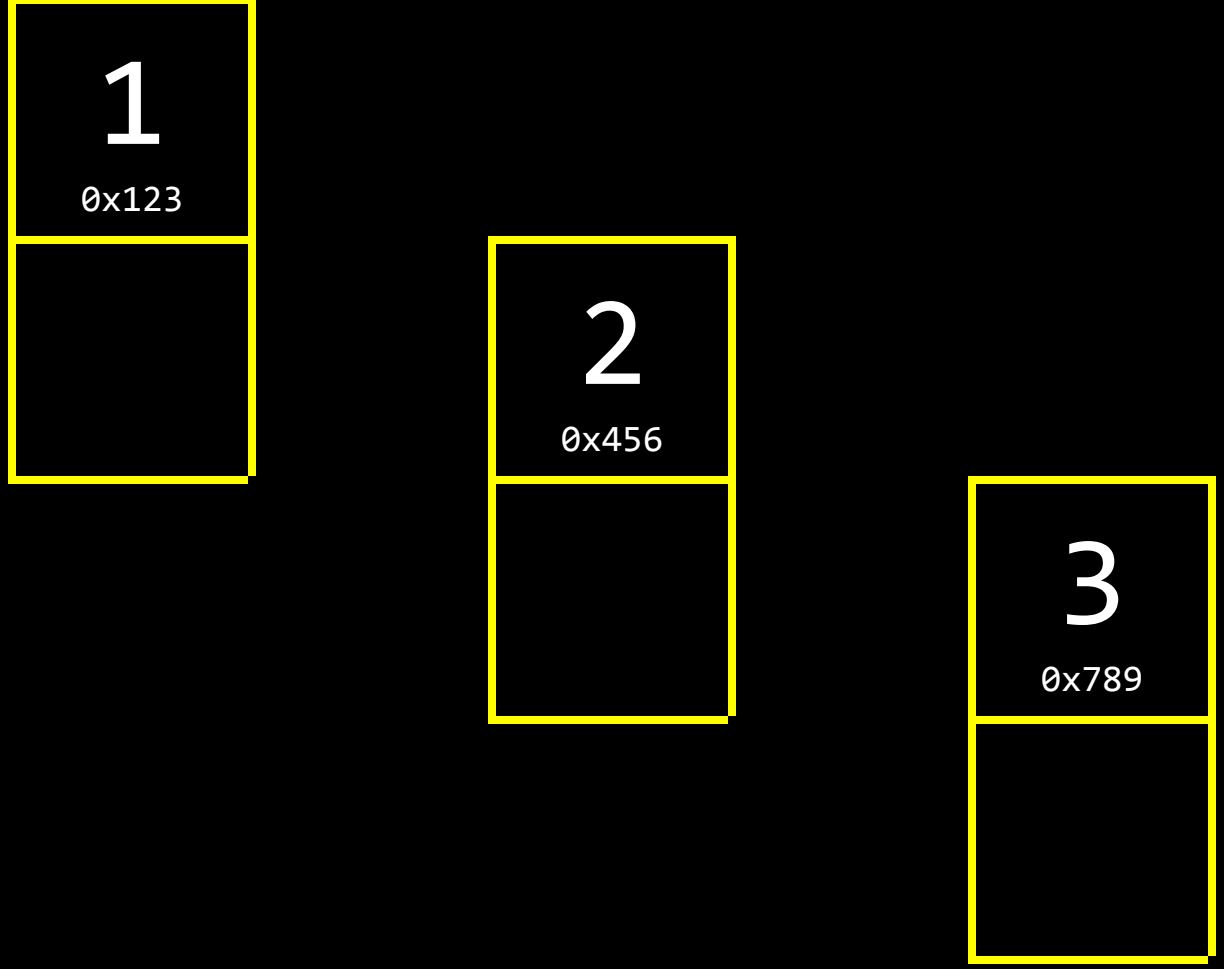
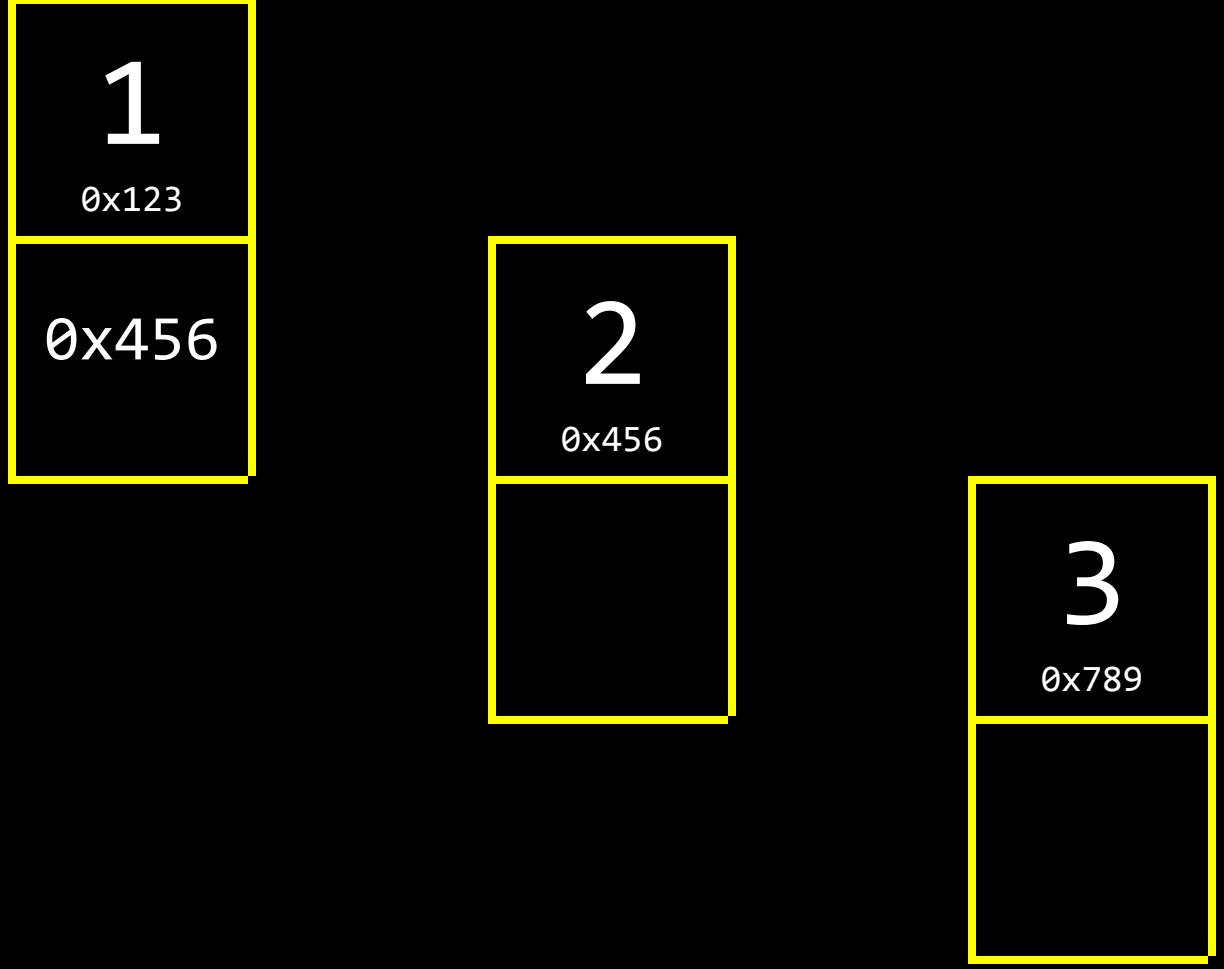
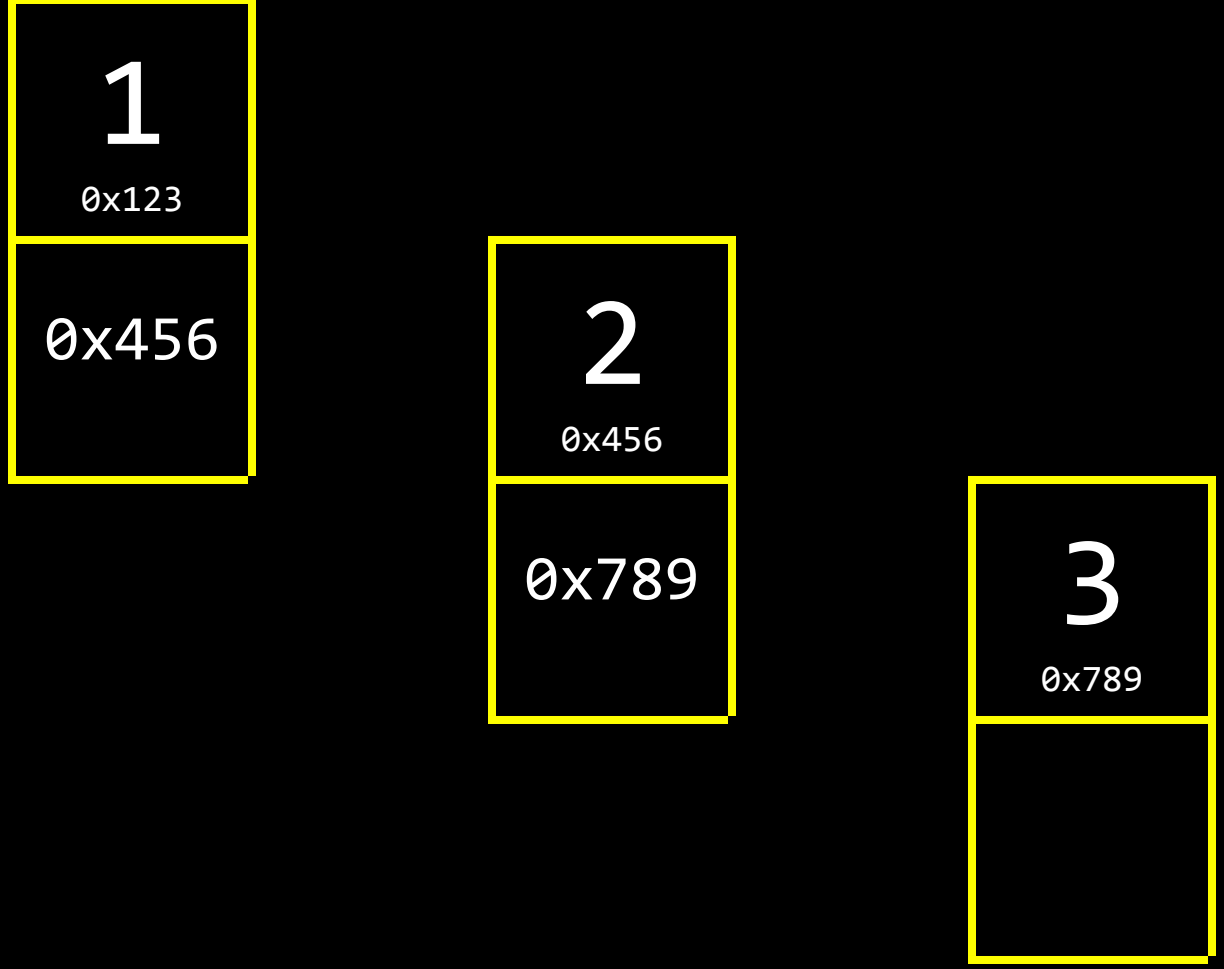
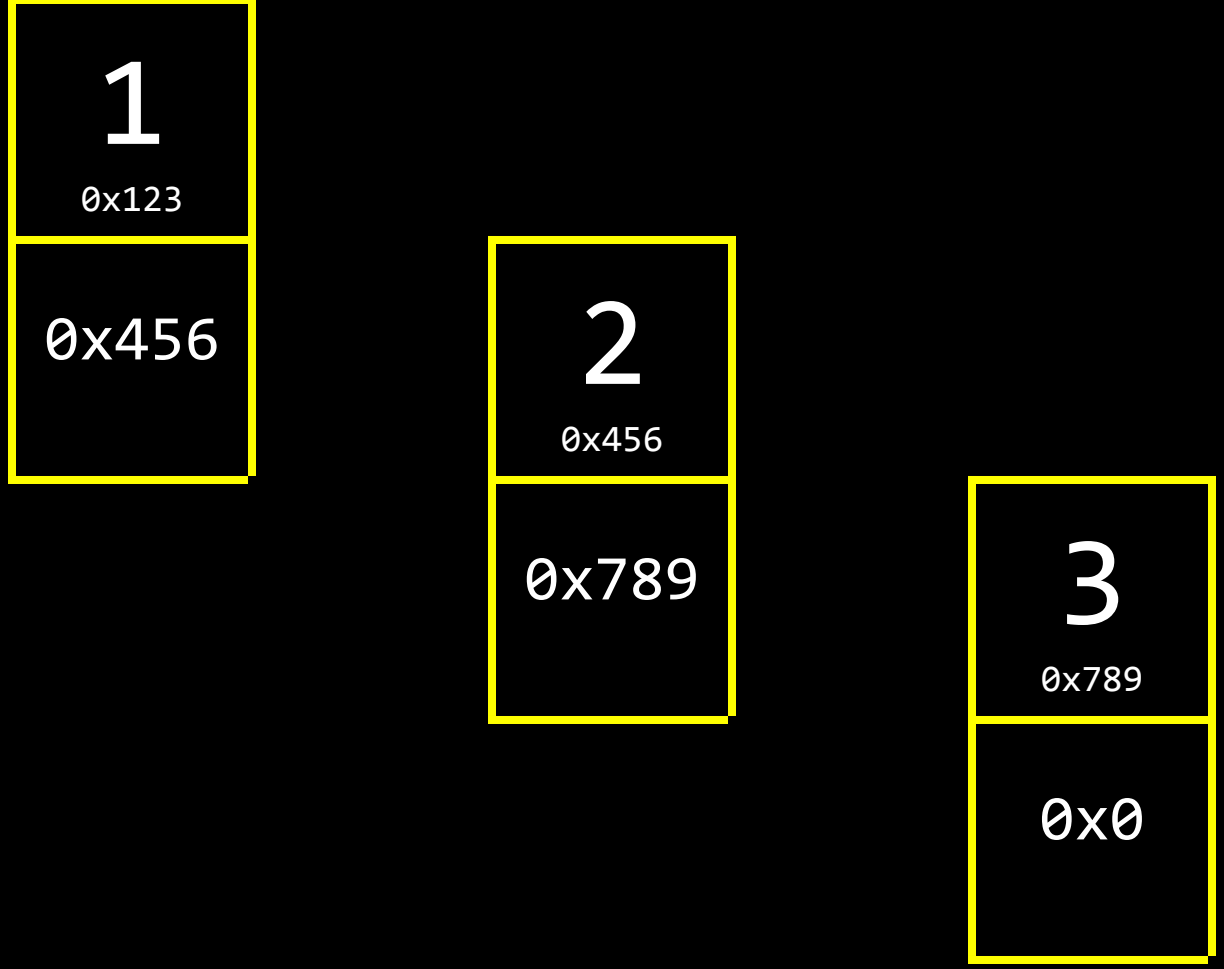
0x123
2
0x456

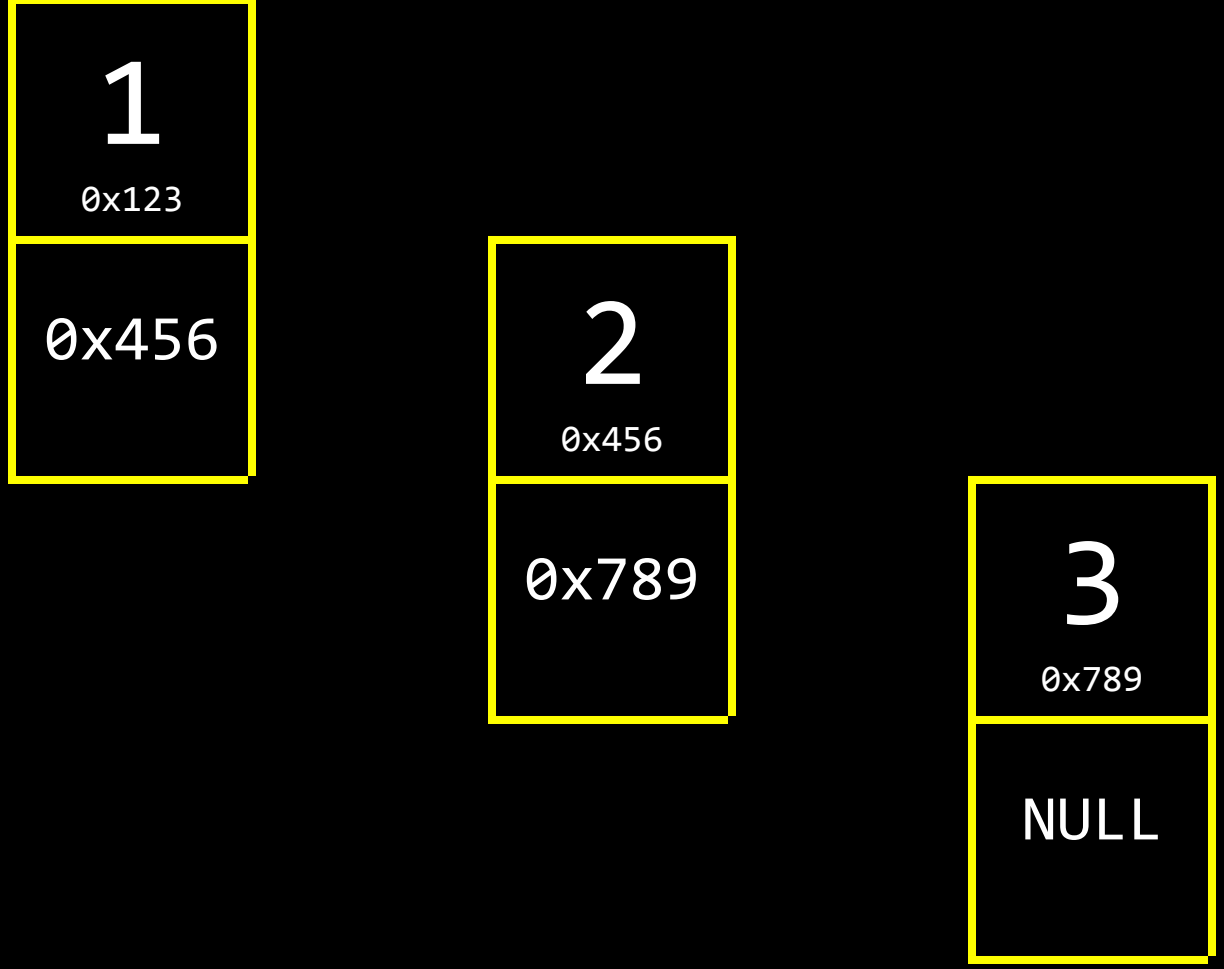

1
0x123
2
0x456
3
0x789

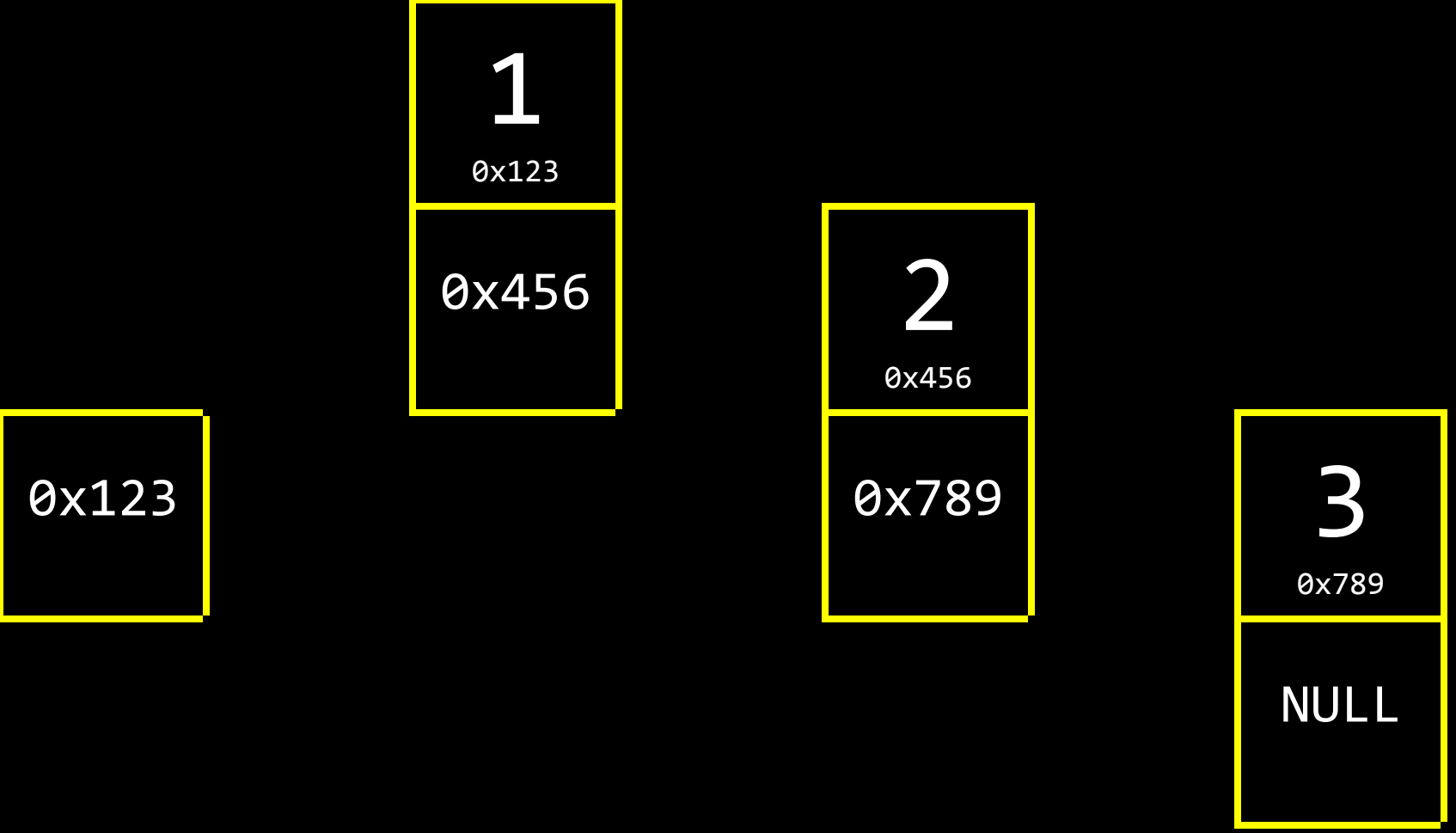

1
0x123
2
0x456
3
0x789

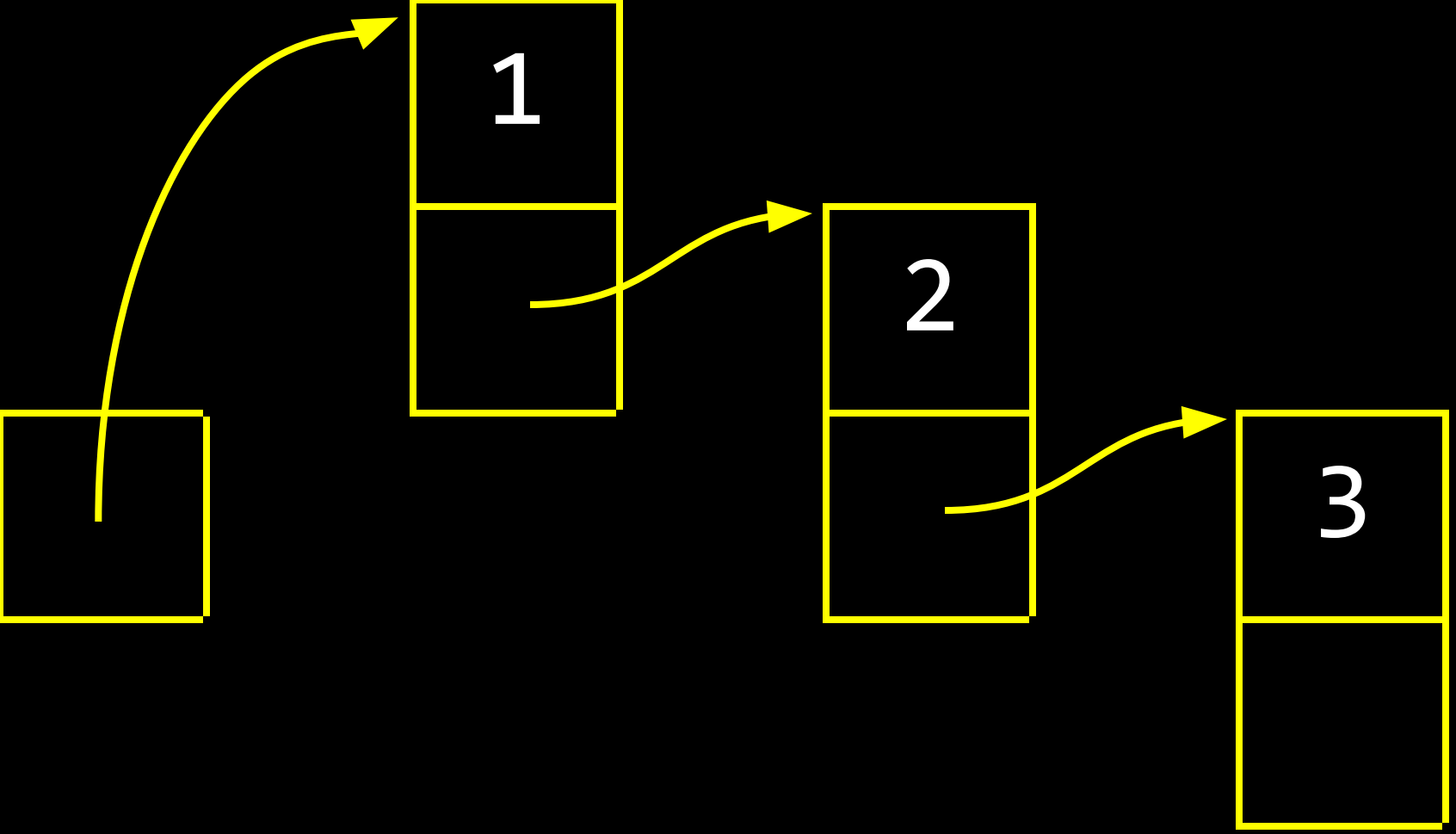

1
0x123
0x456
2
0x456
3
0x789


1
0x123
0x456
2
0x456
0x789
3
0x789


1
0x123
0x456
2
0x456
0x789
3
0x789
0x0


1
0x123
0x456
2
0x456
0x789
3
0x789
NULL


1
0x123
0x456
2
0x456
0x123
0x789
3
0x789
NULL


1
2
3

```
typedef struct
{
    string name;
    string number;
} person;
```

```
typedef struct
{
    char *name;
    char *number;
} person;
```

```
typedef struct
{


} person;
```

```
typedef struct
{


} node;
```

```
typedef struct
{
    int number;

} node;
```

```
typedef struct
{
    int number;
    node *next;
} node;
```

```
typedef struct node
{
    int number;
    node *next;
} node;
```

```
typedef struct node
{
    int number;
    struct node *next;
} node;
```

```
node *list;
```

```
node *list;
```

list

```
node *list = NULL;
```

list

```
node *list = NULL;
```

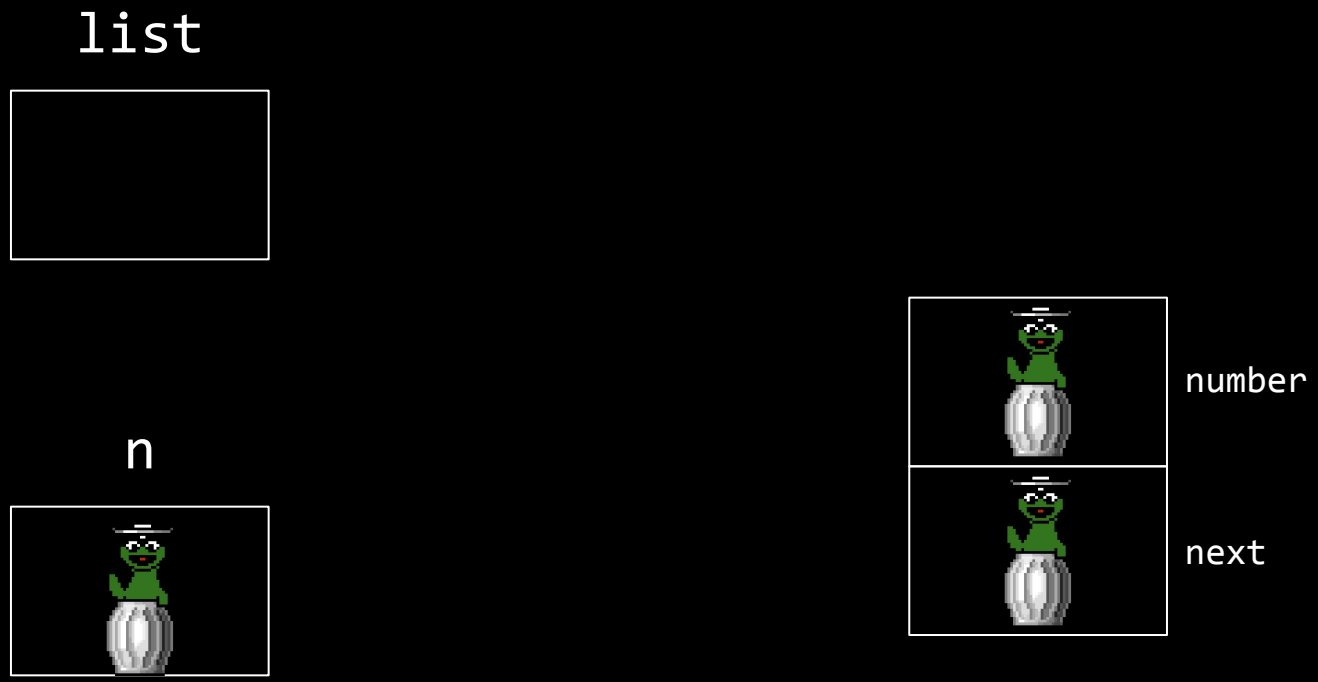
```
node *n = malloc(sizeof(node));
```

```
node *n = malloc(sizeof(node));
```

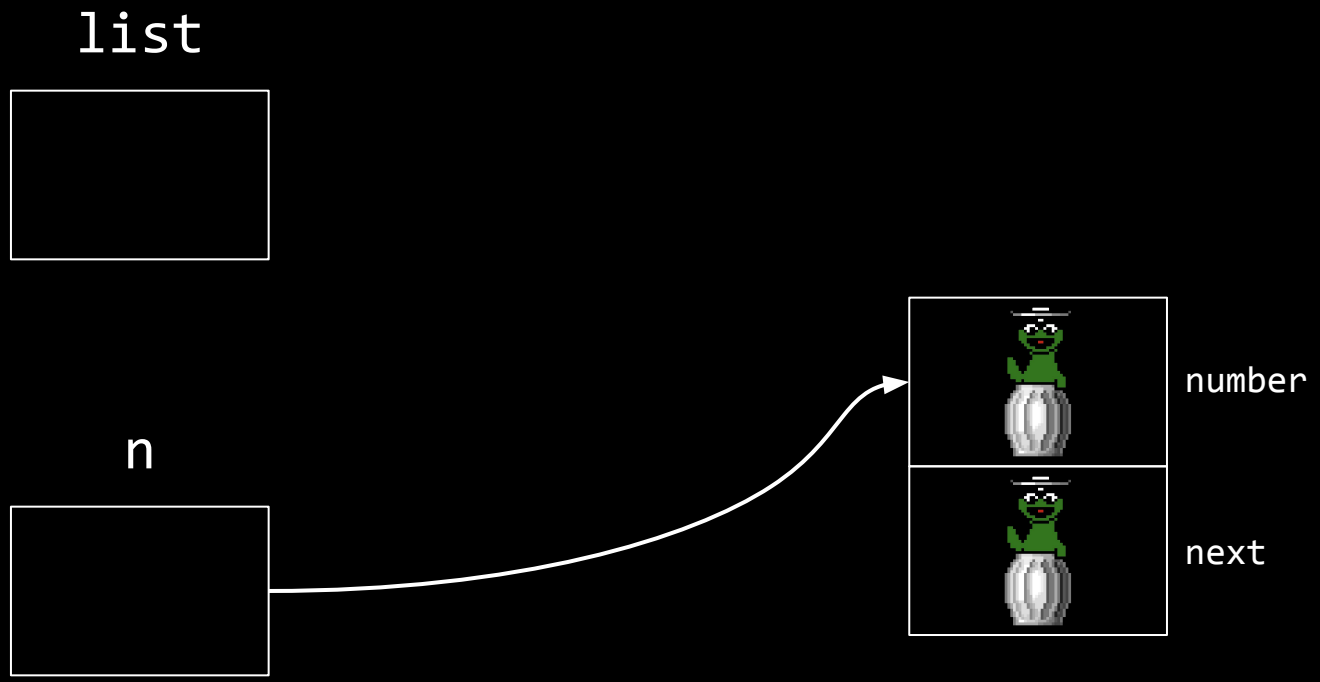
list

n

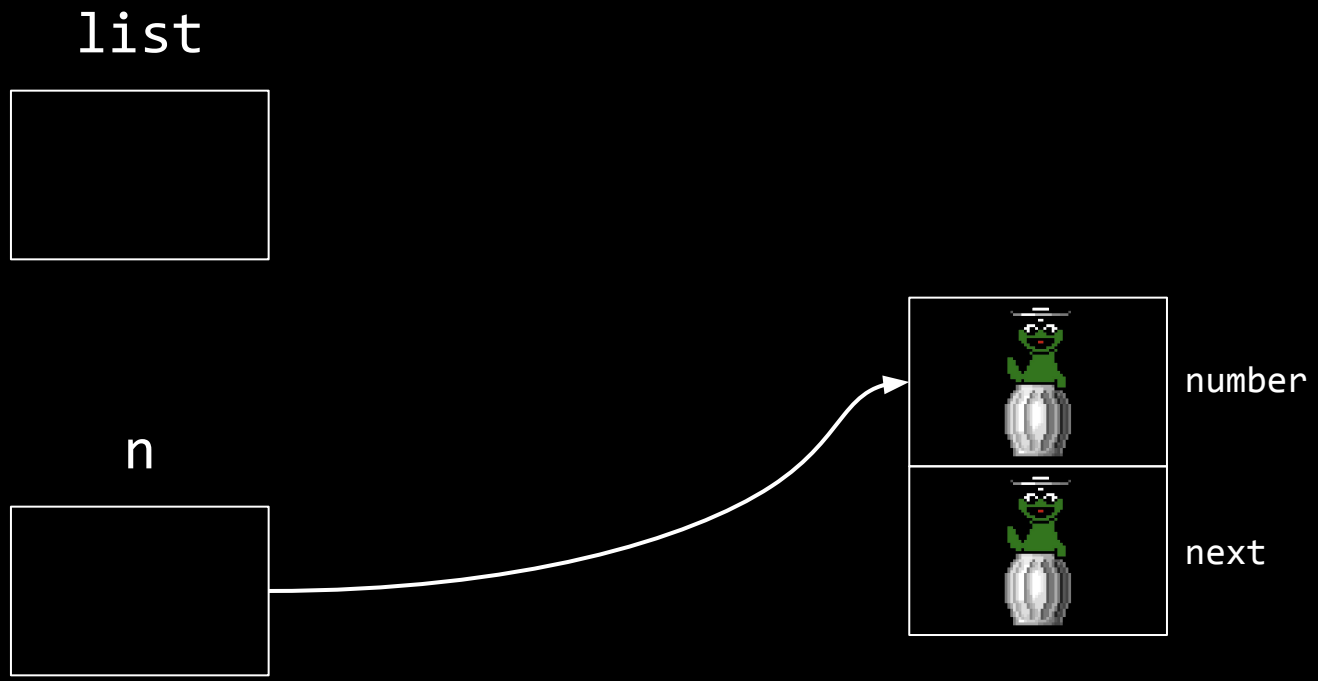
```
node *n = malloc(sizeof(node));
```



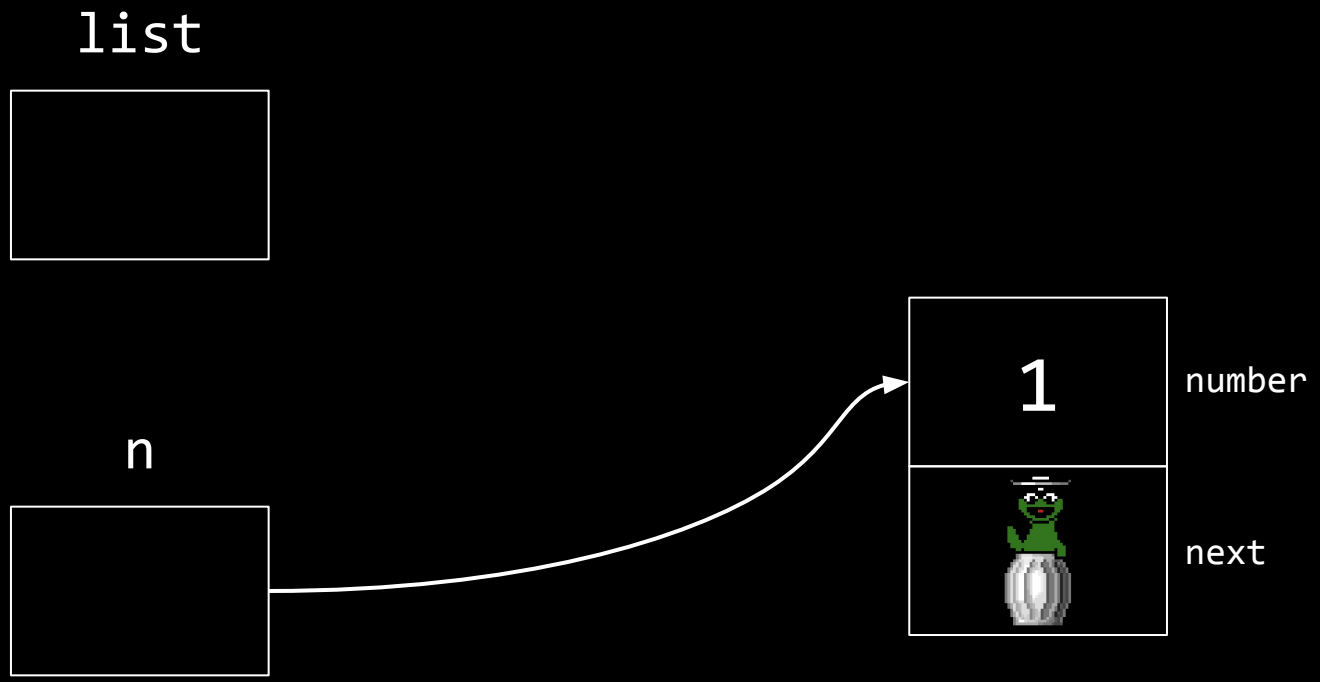
```
node *n = malloc(sizeof(node));
```



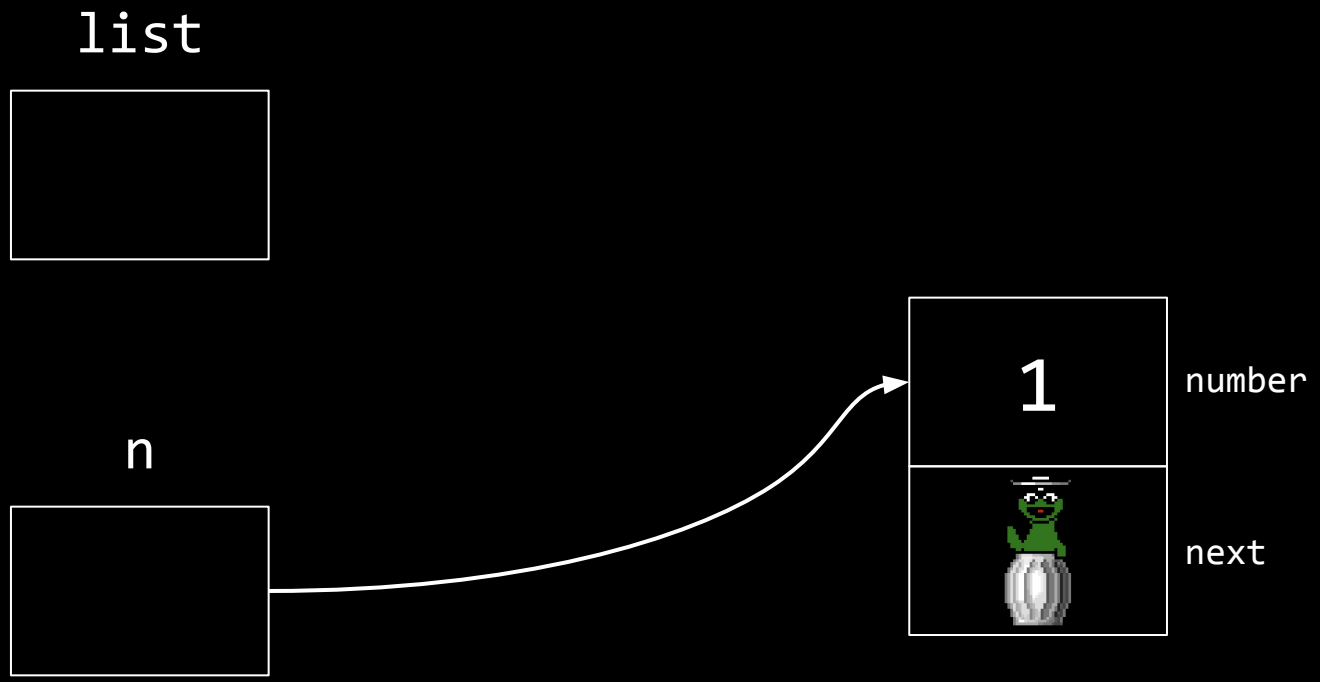
```
(*n).number = 1;
```



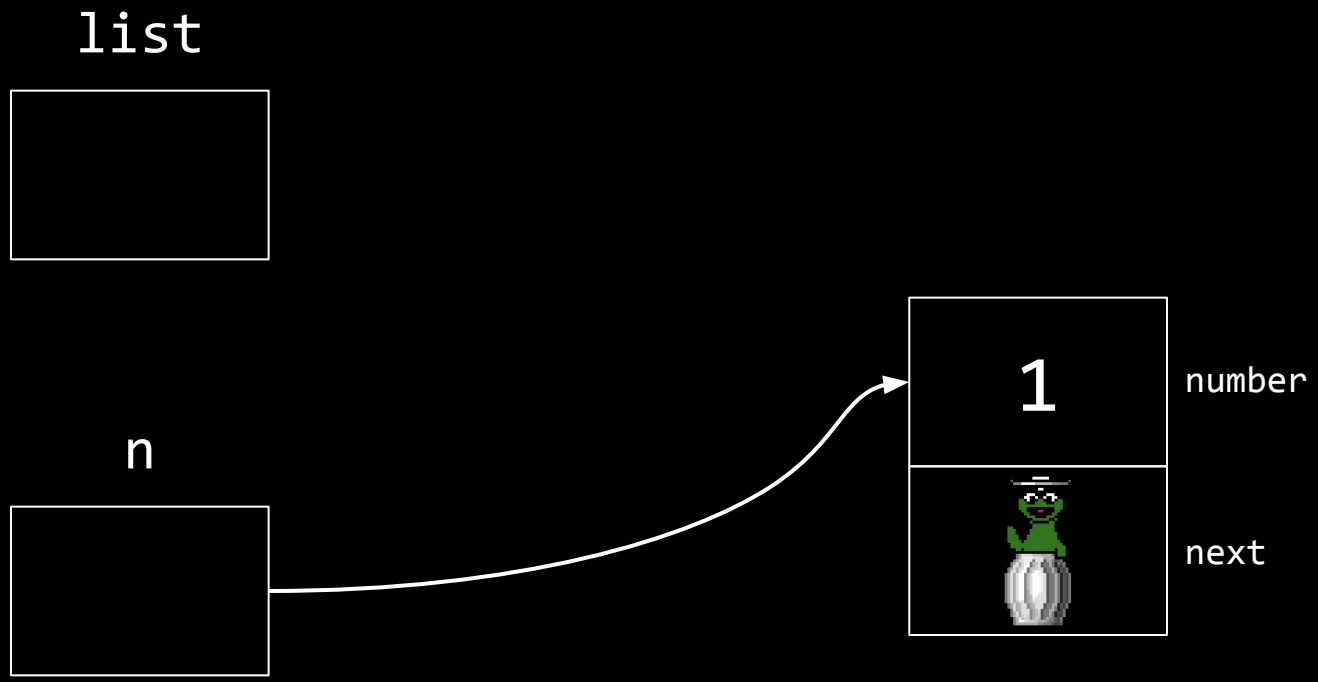
```
(*n).number = 1;
```



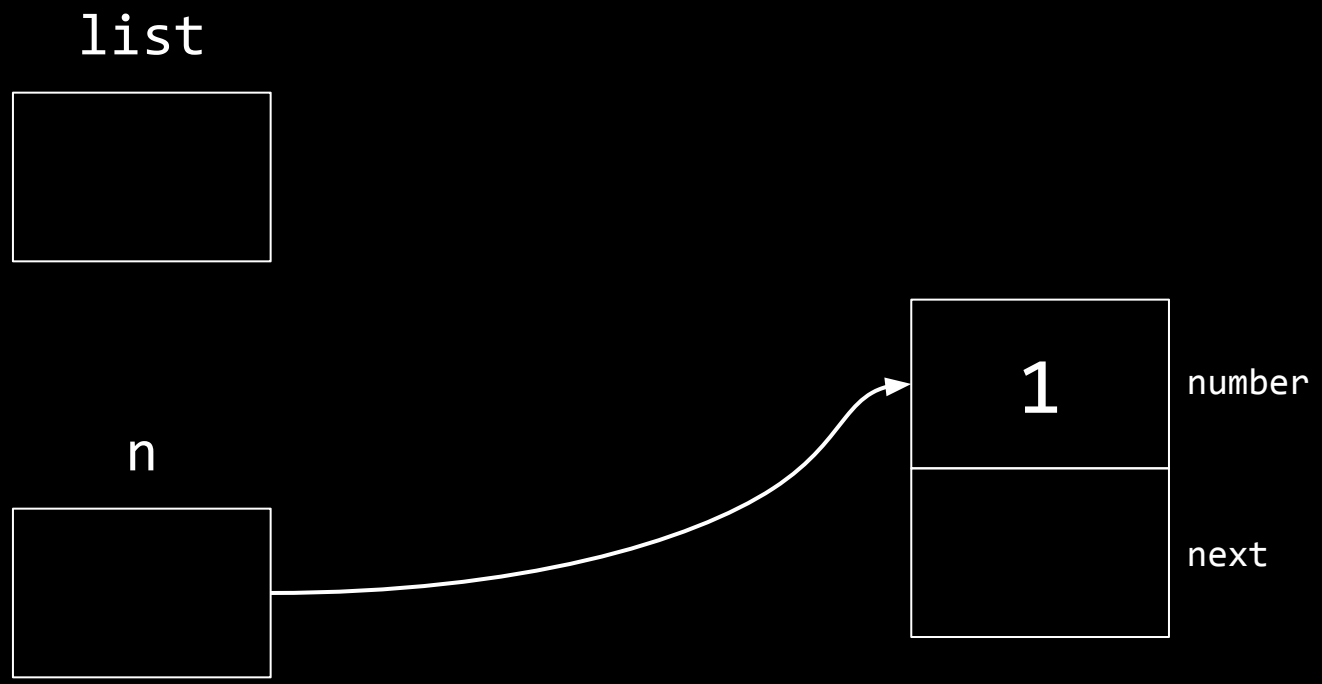
```
n->number = 1;
```



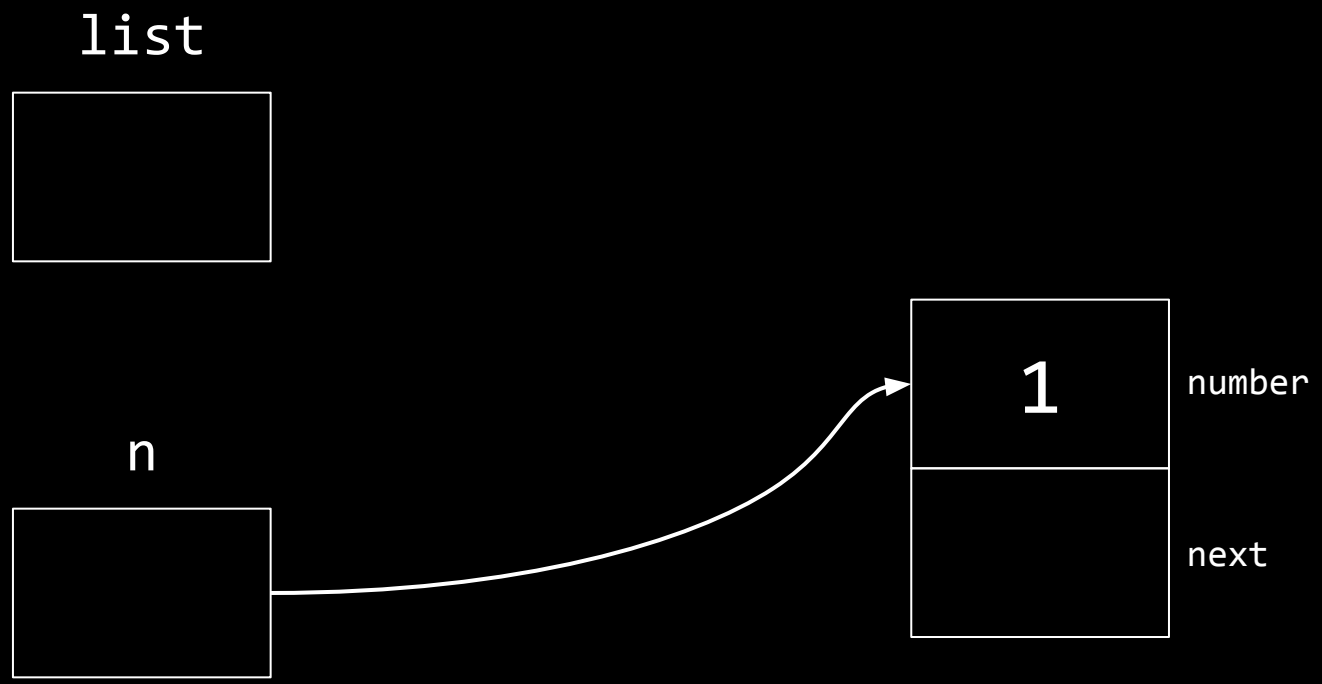
```
n->next = NULL;
```



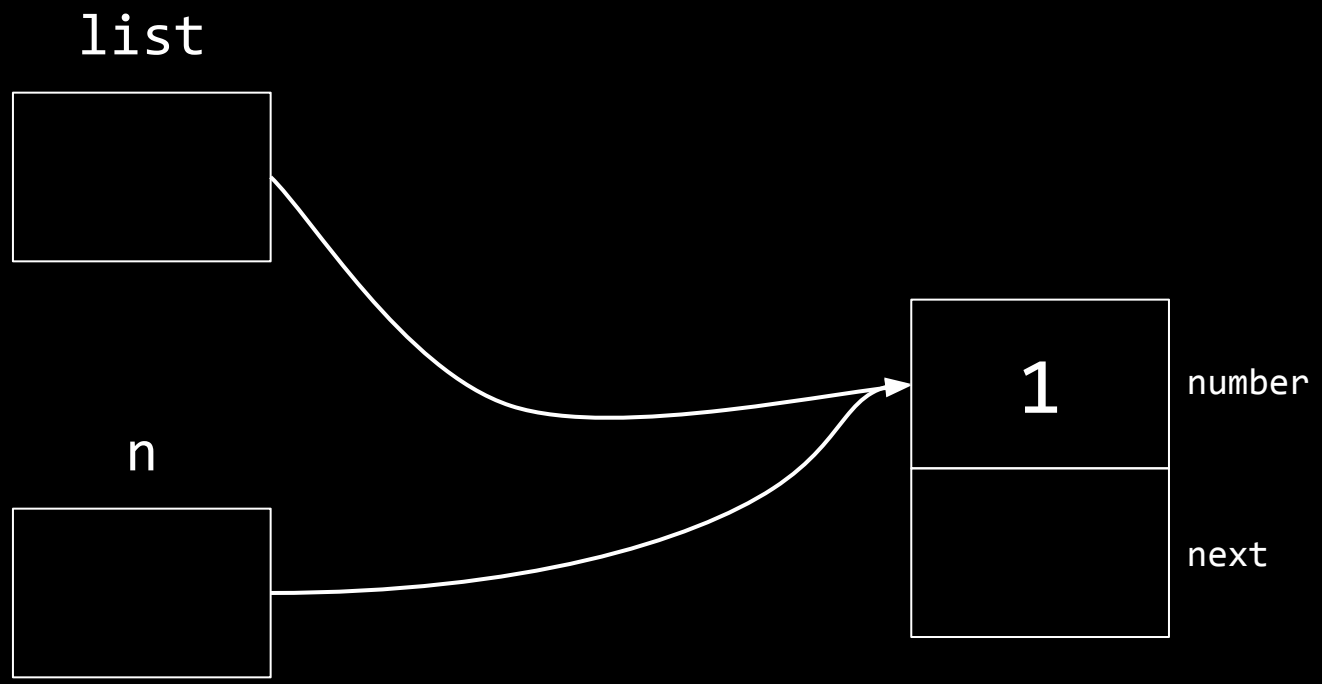
```
n->next = NULL;
```

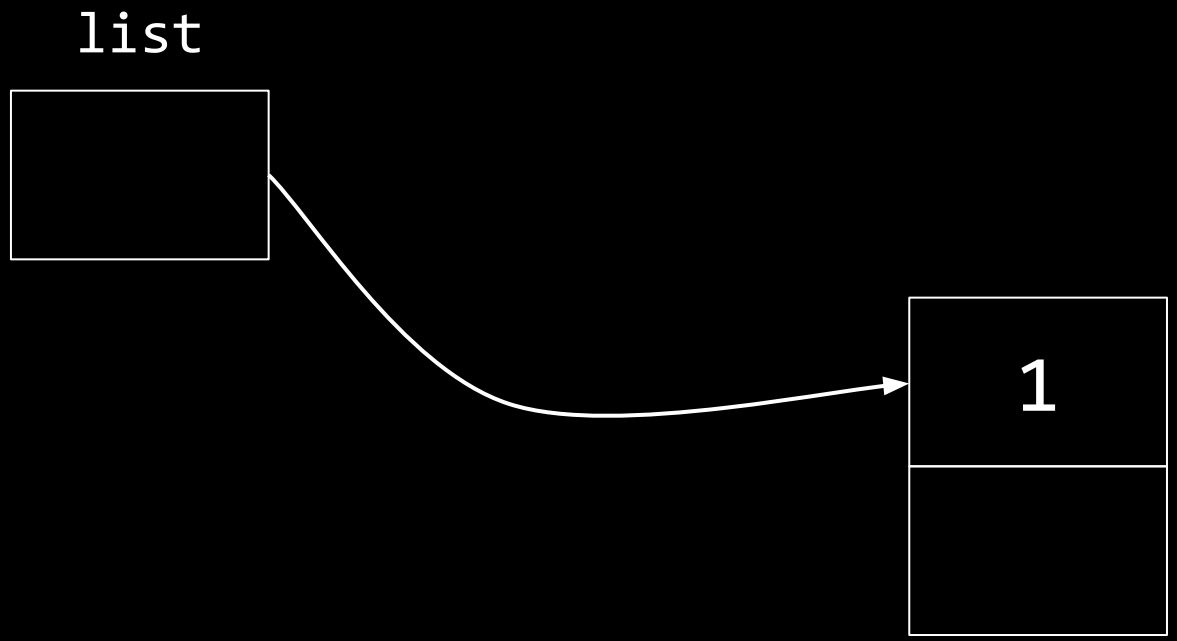


```
list = n;
```

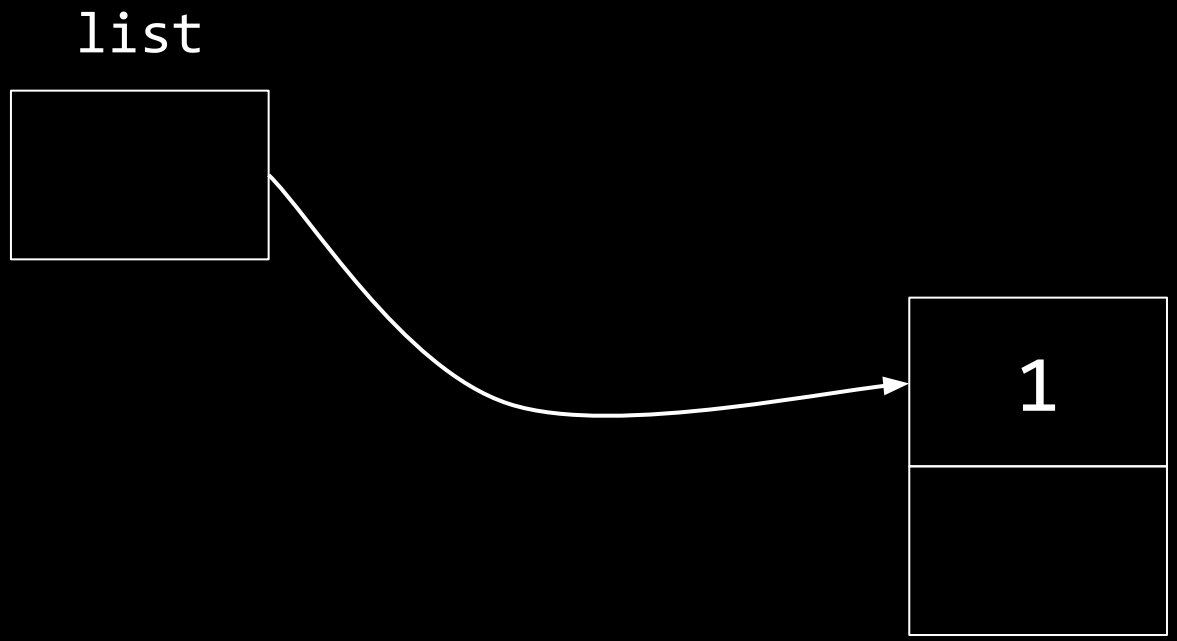


```
list = n;
```

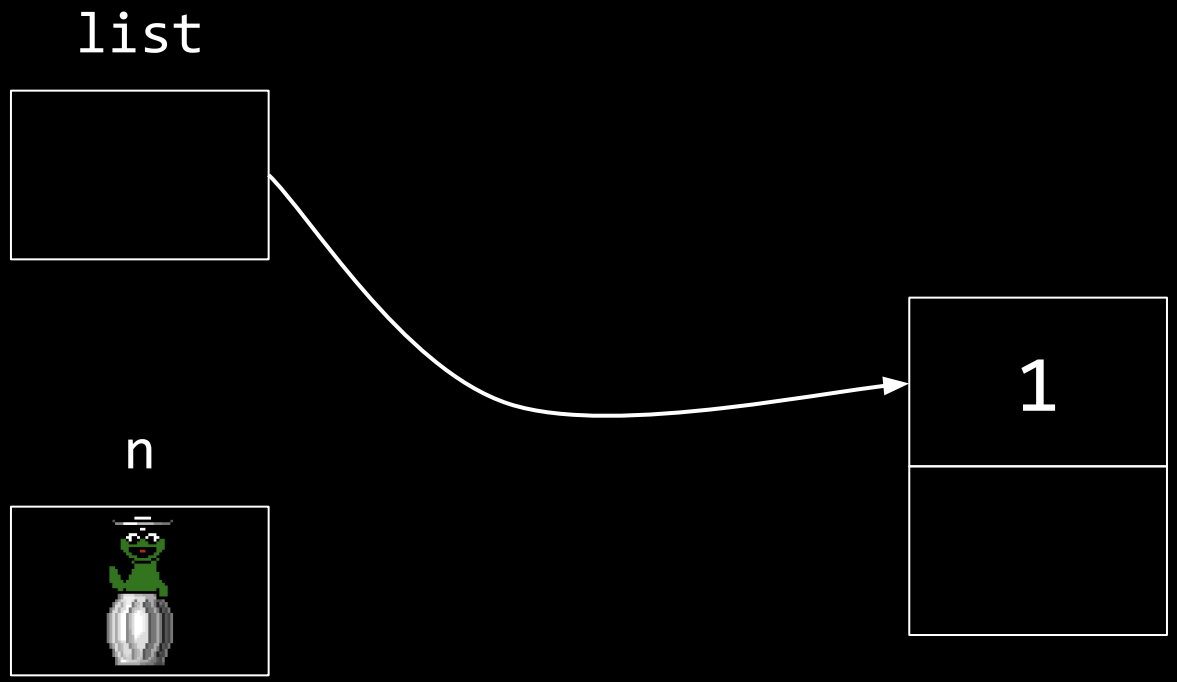



list
1

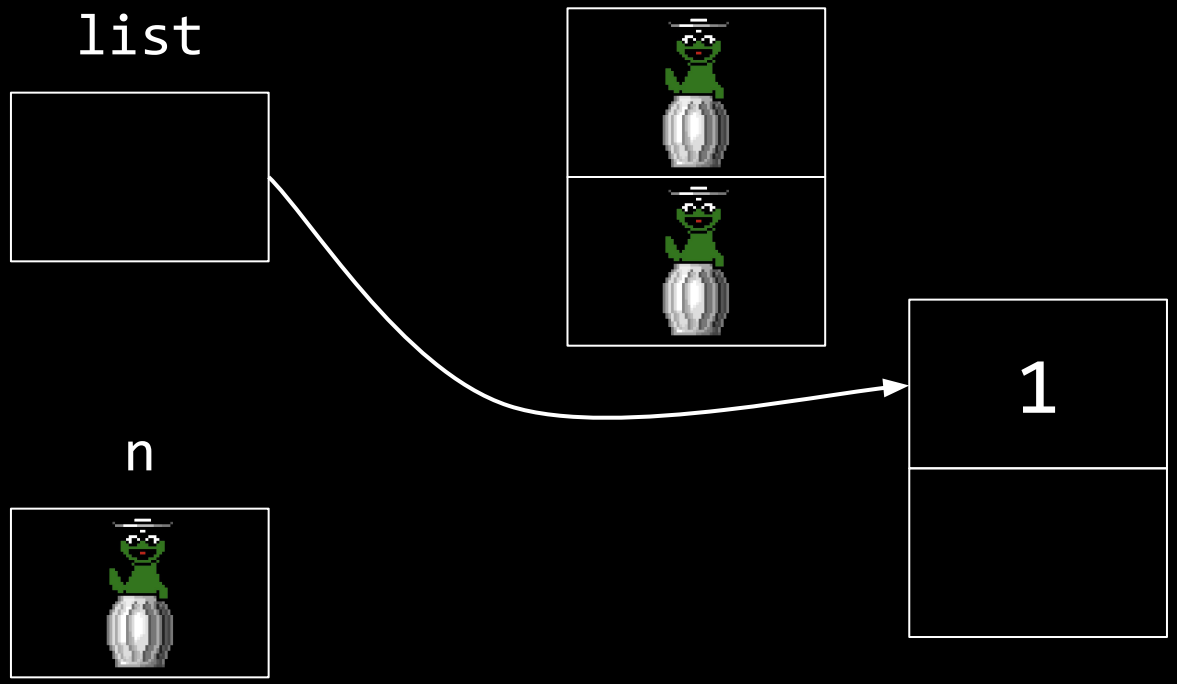
```
node *n = malloc(sizeof(node));
```



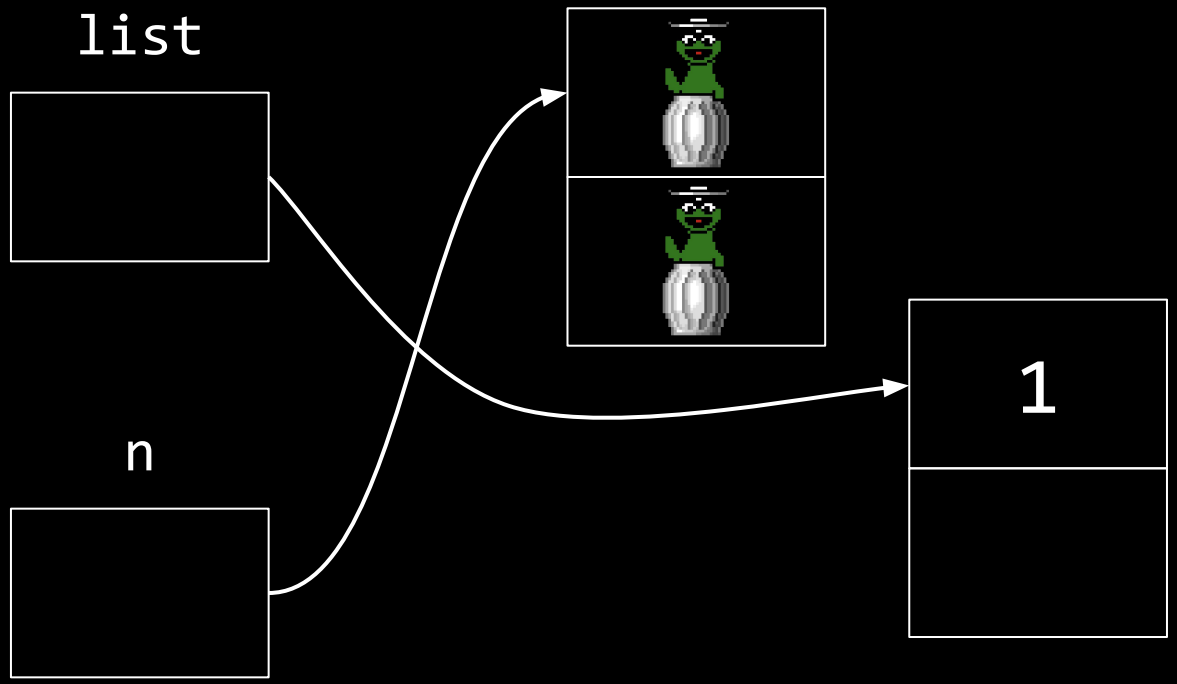
```
node *n = malloc(sizeof(node));
```



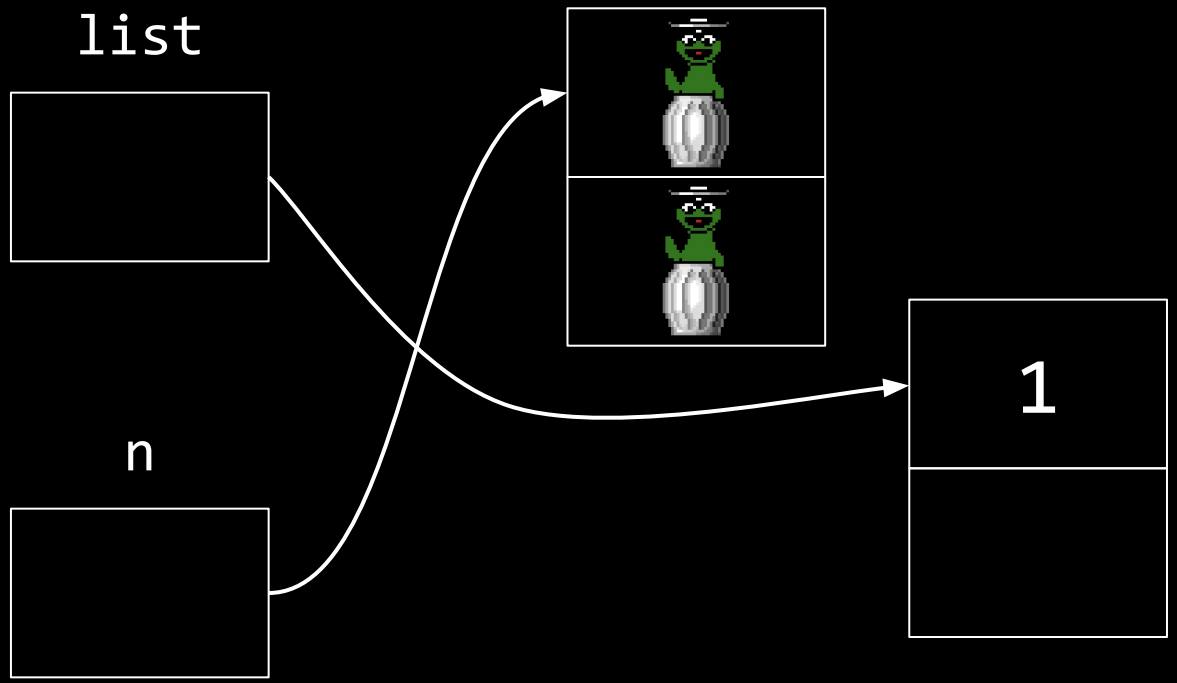
```
node *n = malloc(sizeof(node));
```



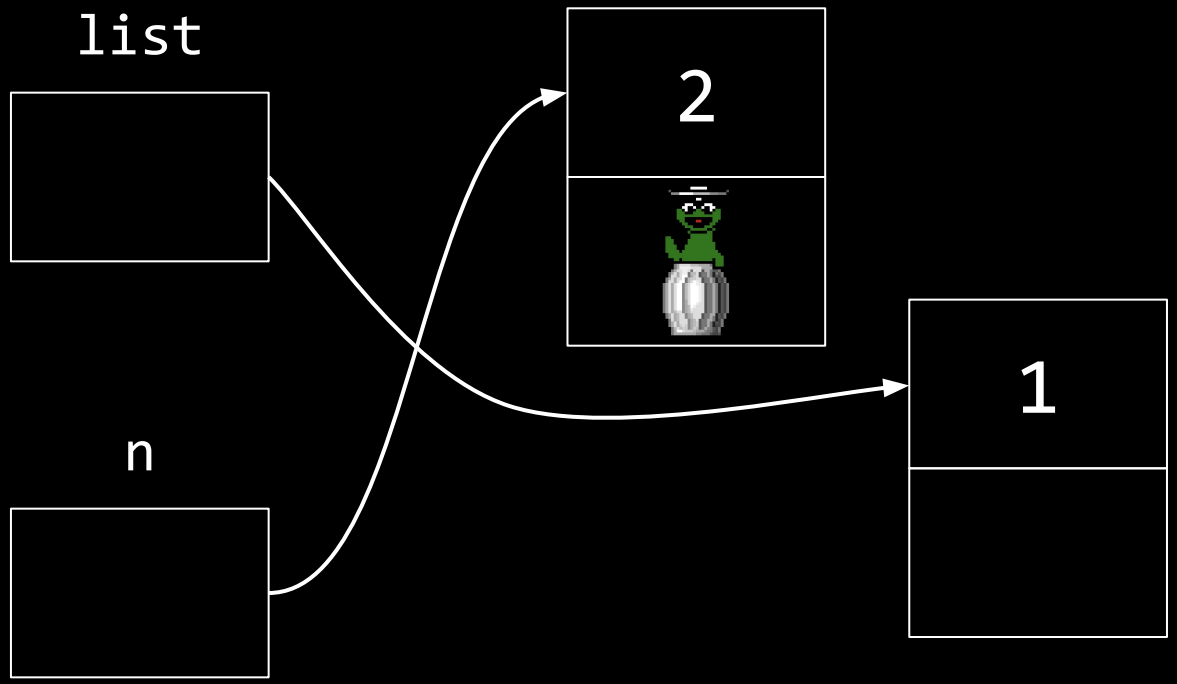
```
node *n = malloc(sizeof(node));
```



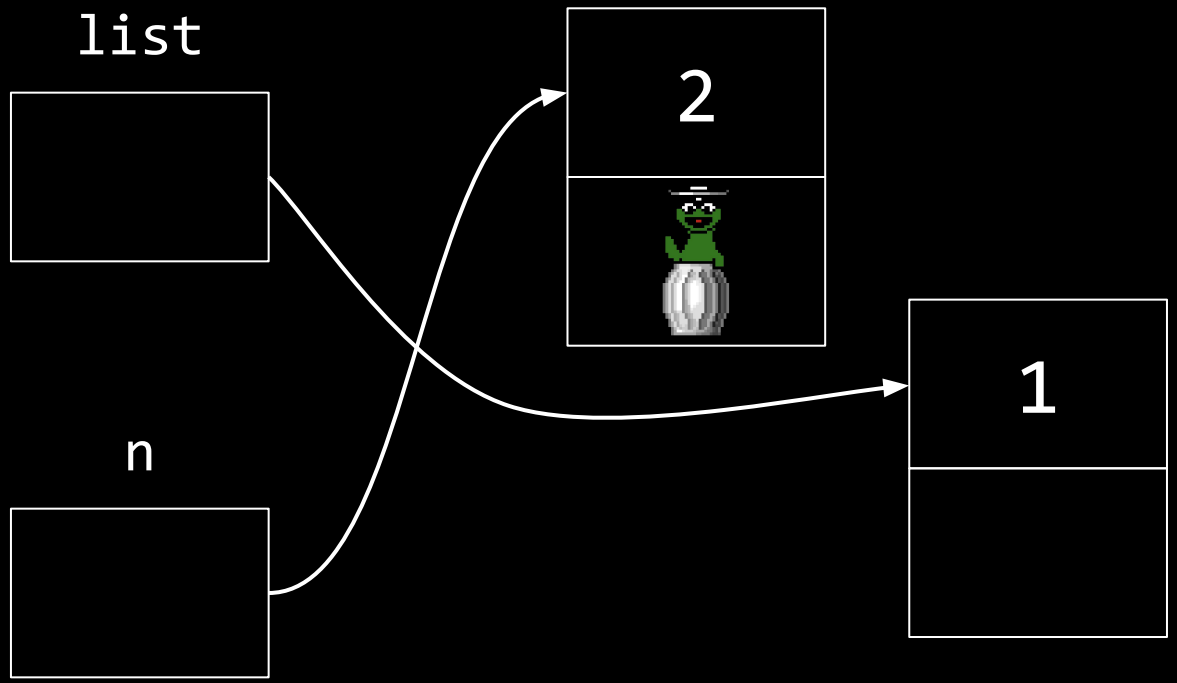
```
n->number = 2;
```



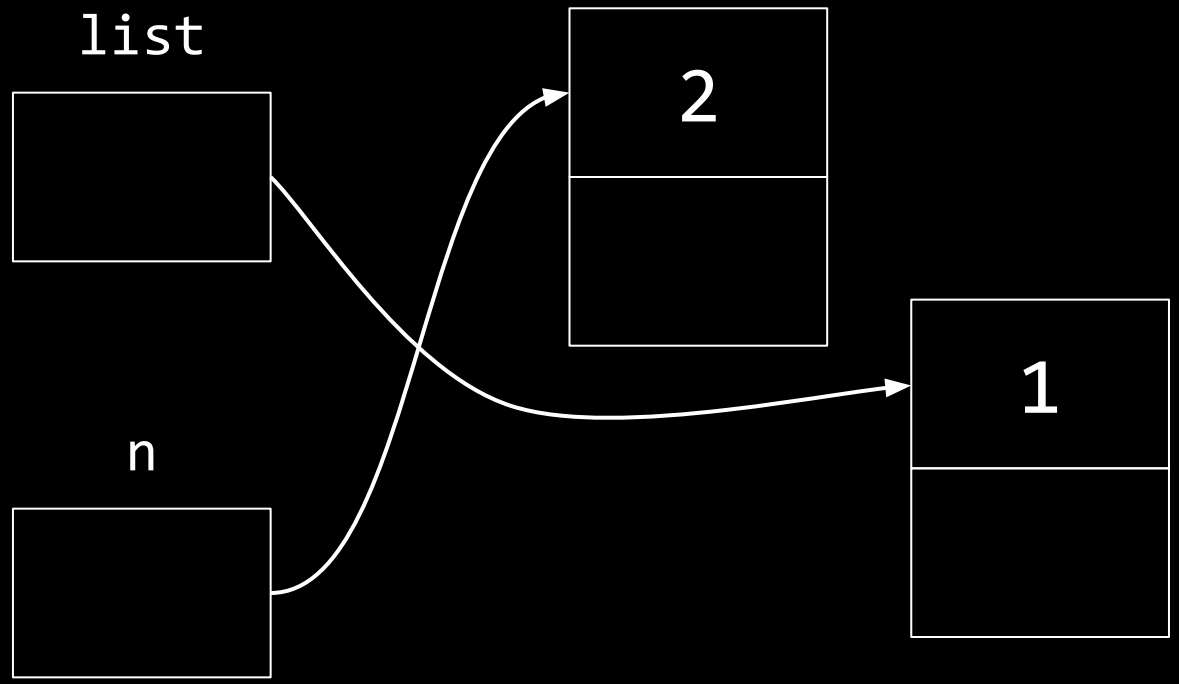
```
n->number = 2;
```

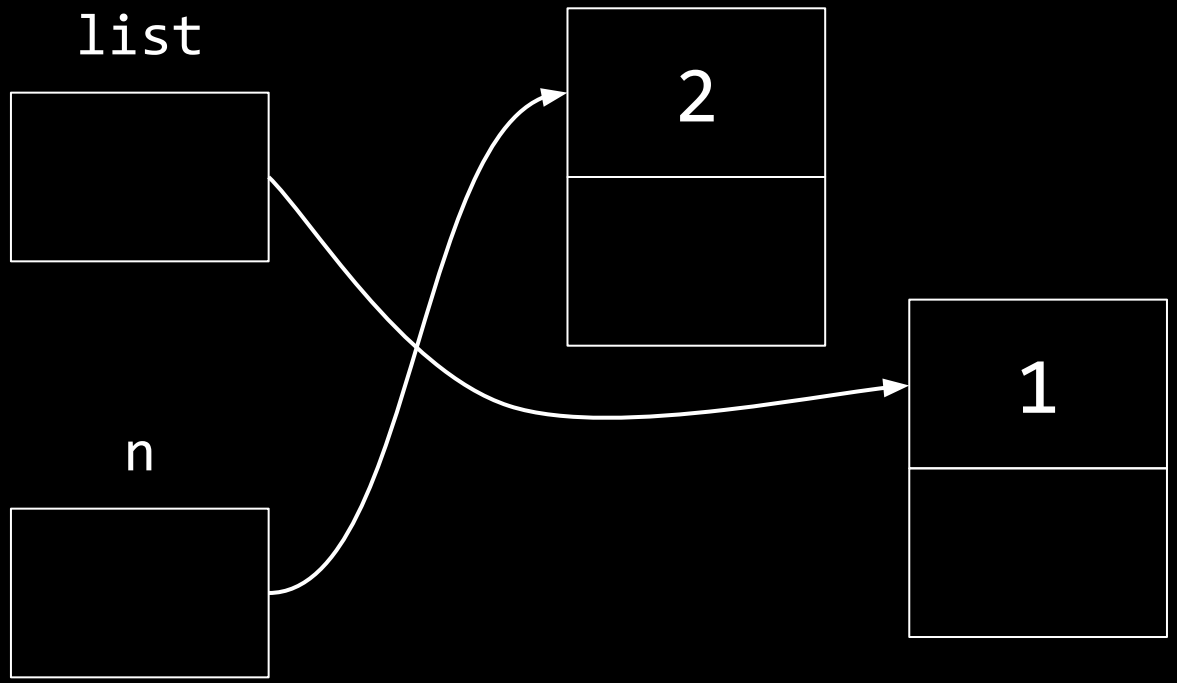


```
n->next = NULL;
```

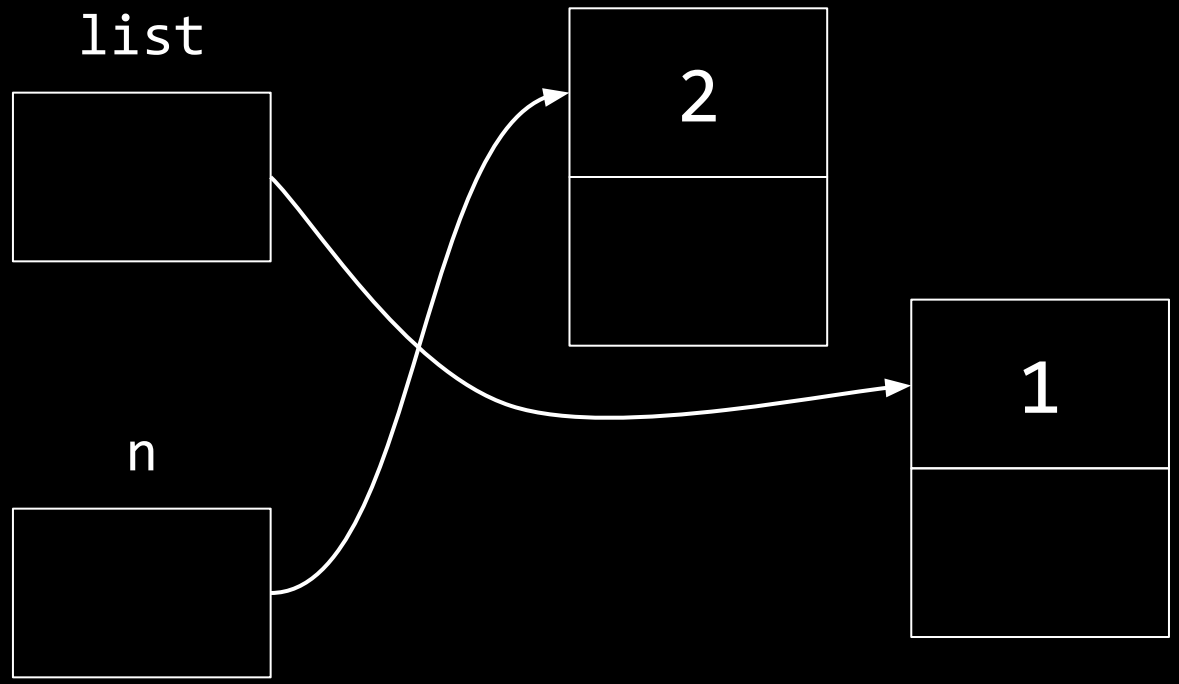


```
n->next = NULL;
```

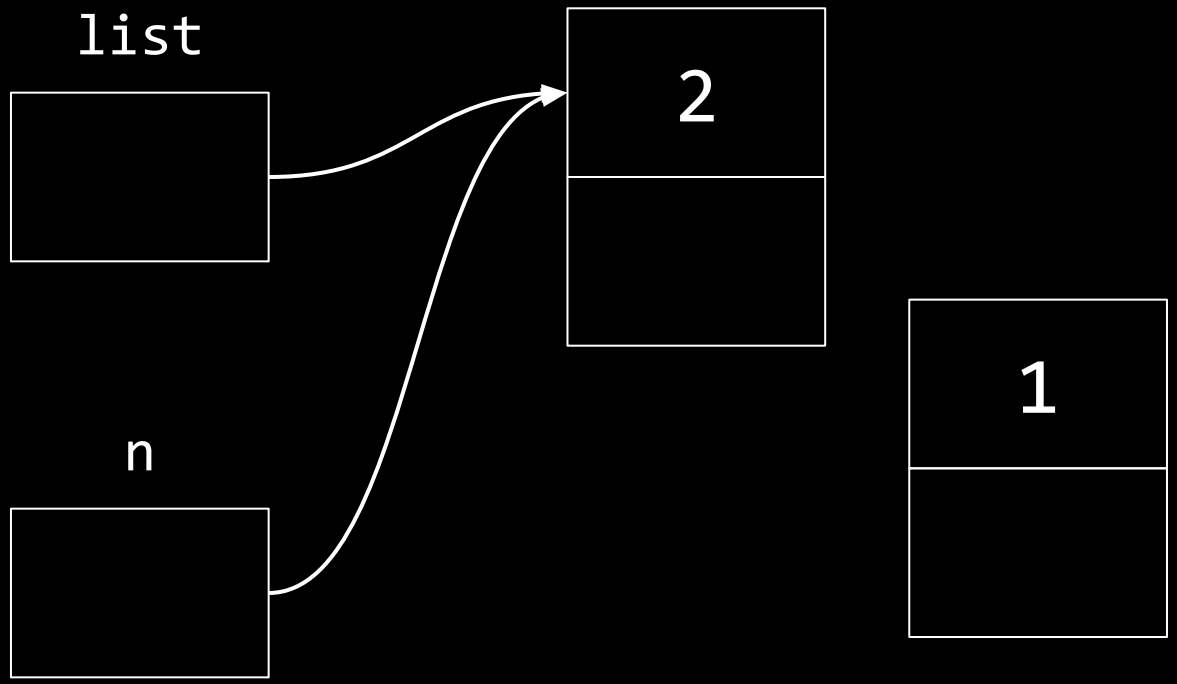



list
2
1
n

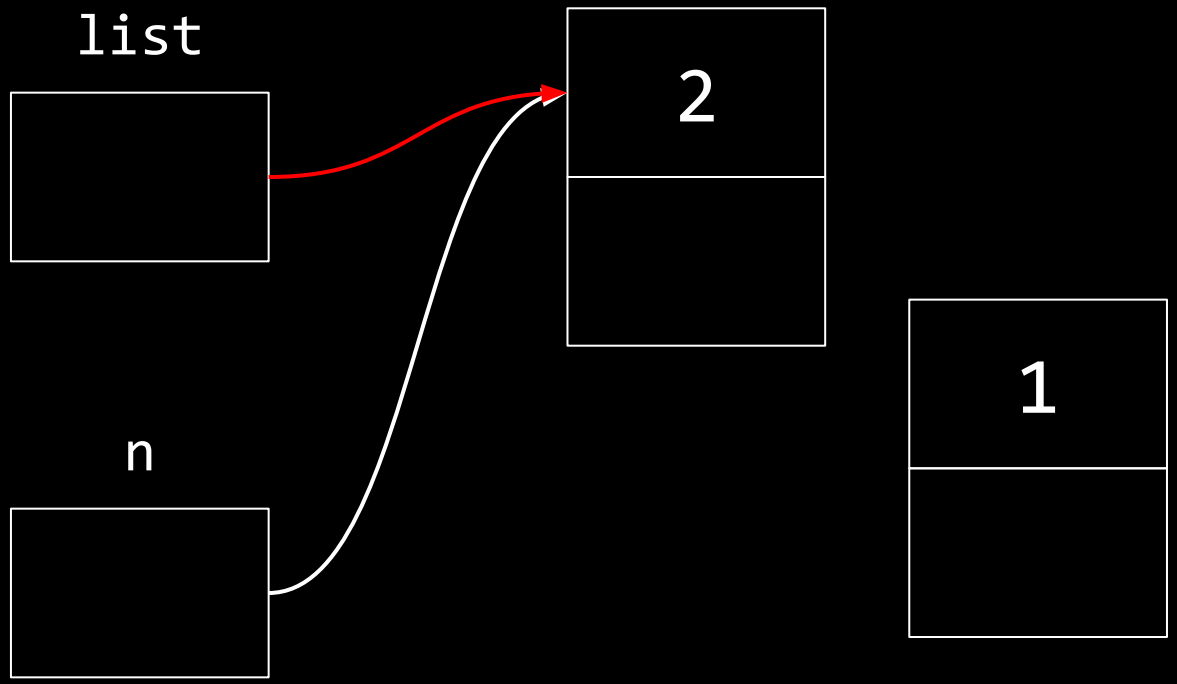
```
list = n;
```

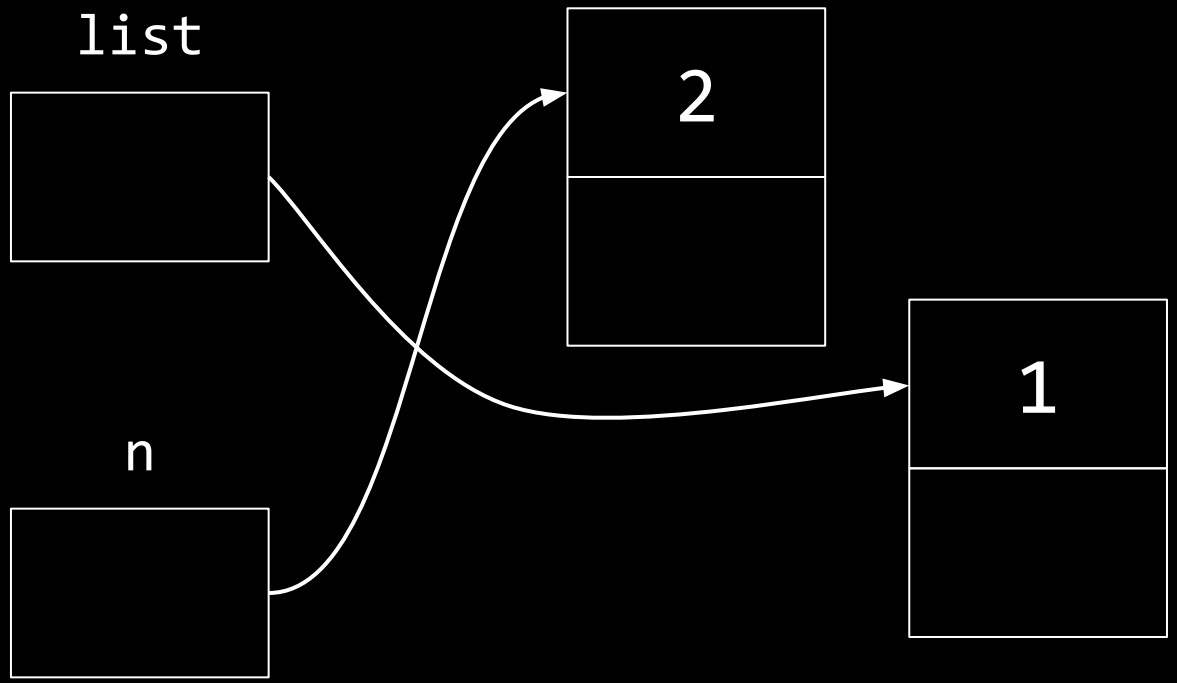


```
list = n;
```

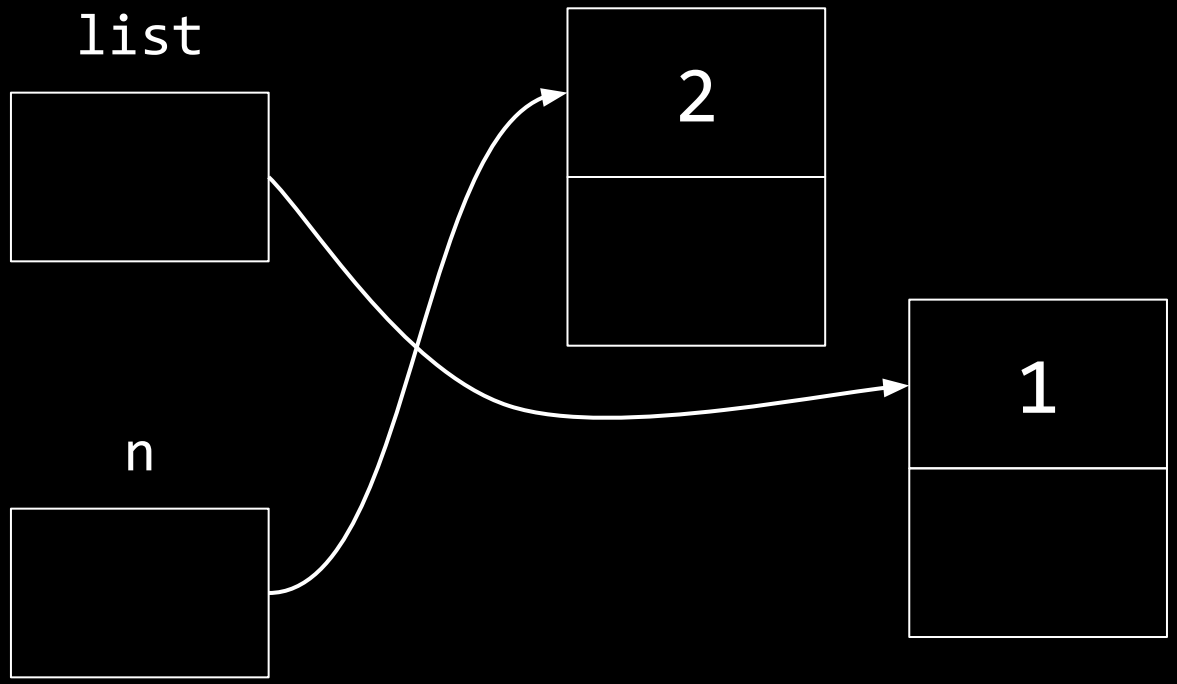


```
list = n;
```

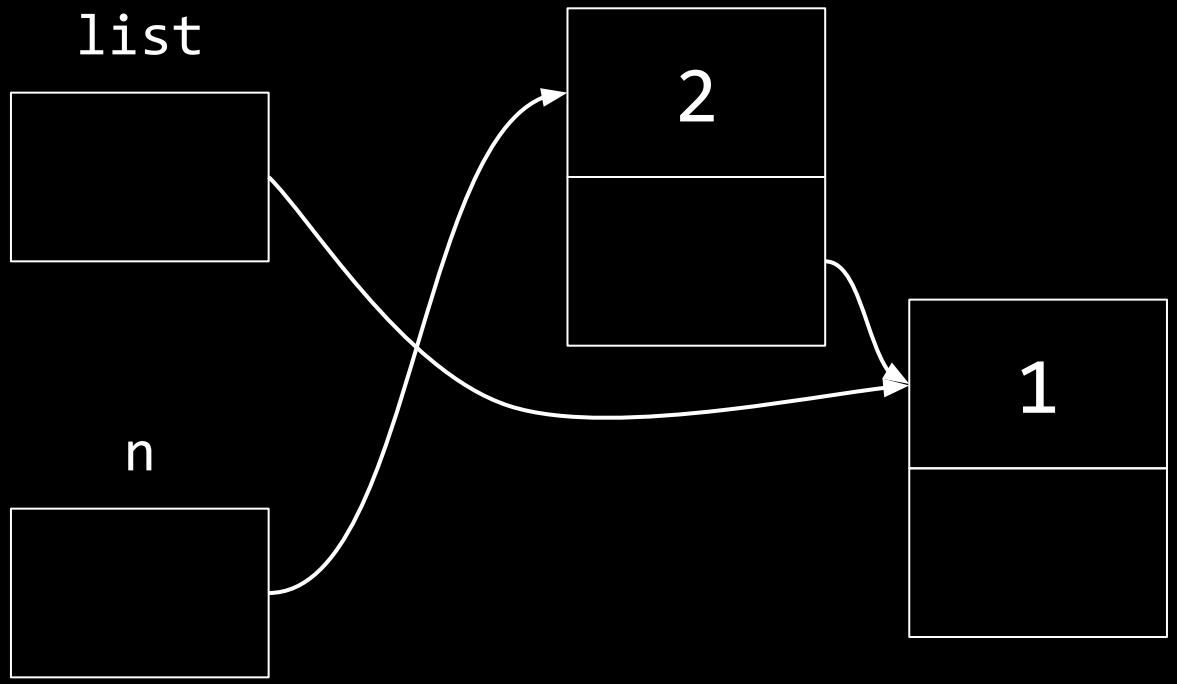



list
2
1
n

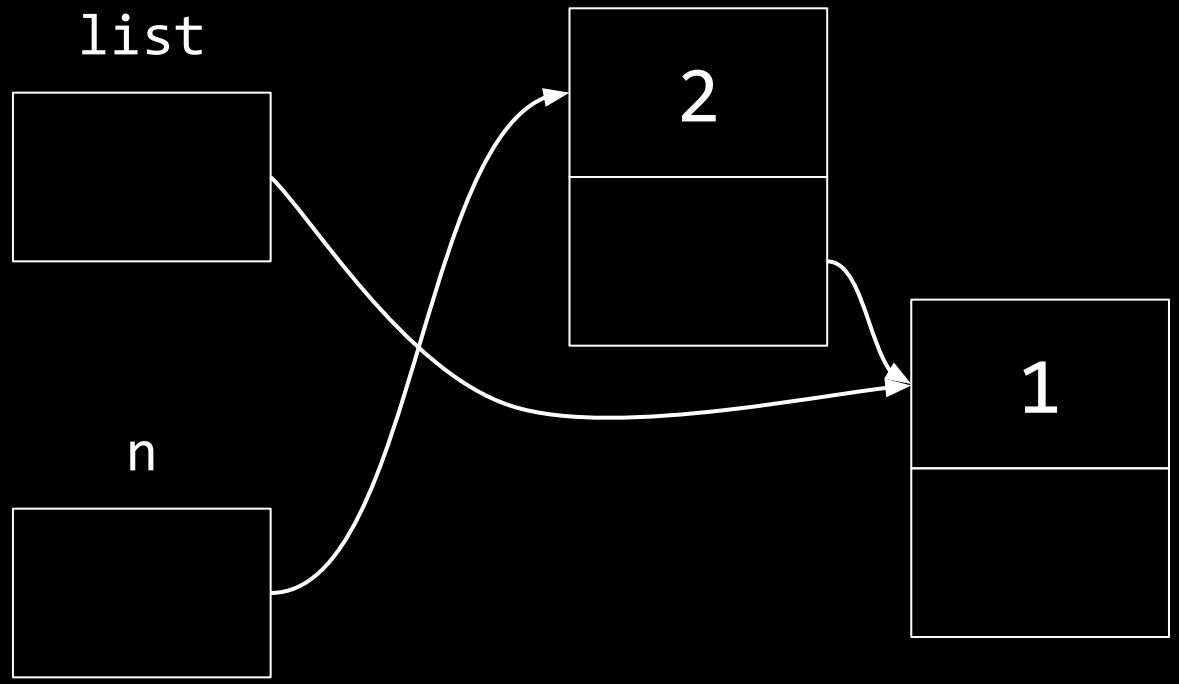
```
n->next = list;
```



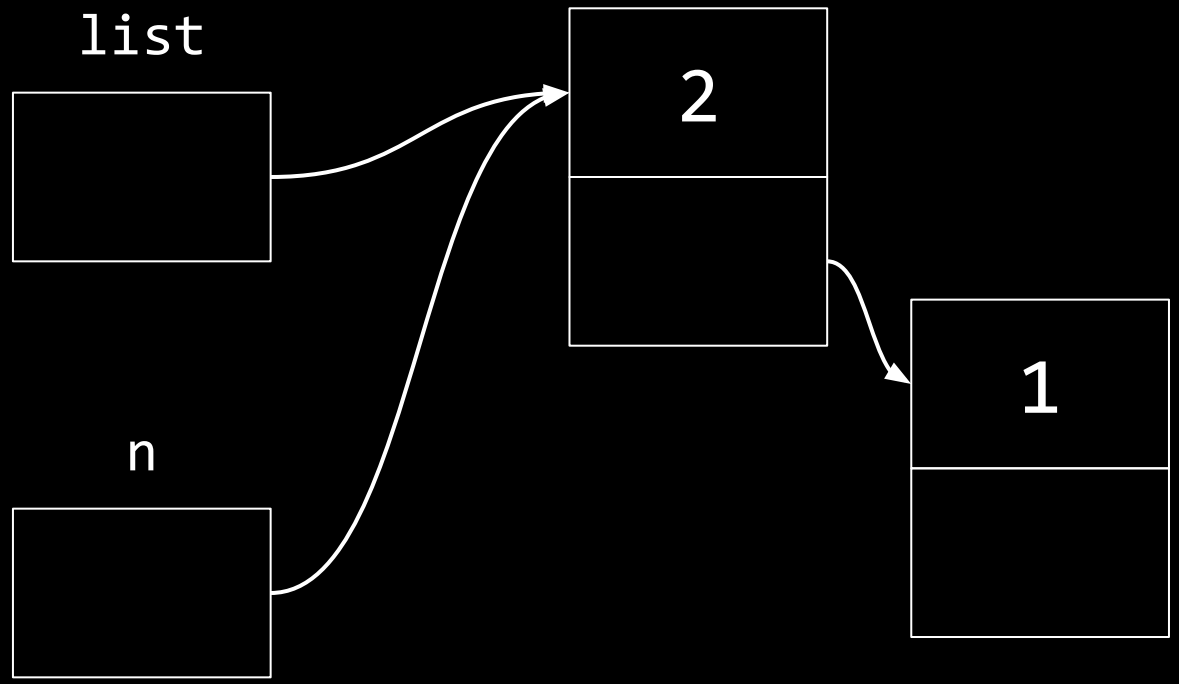
```
n->next = list;
```

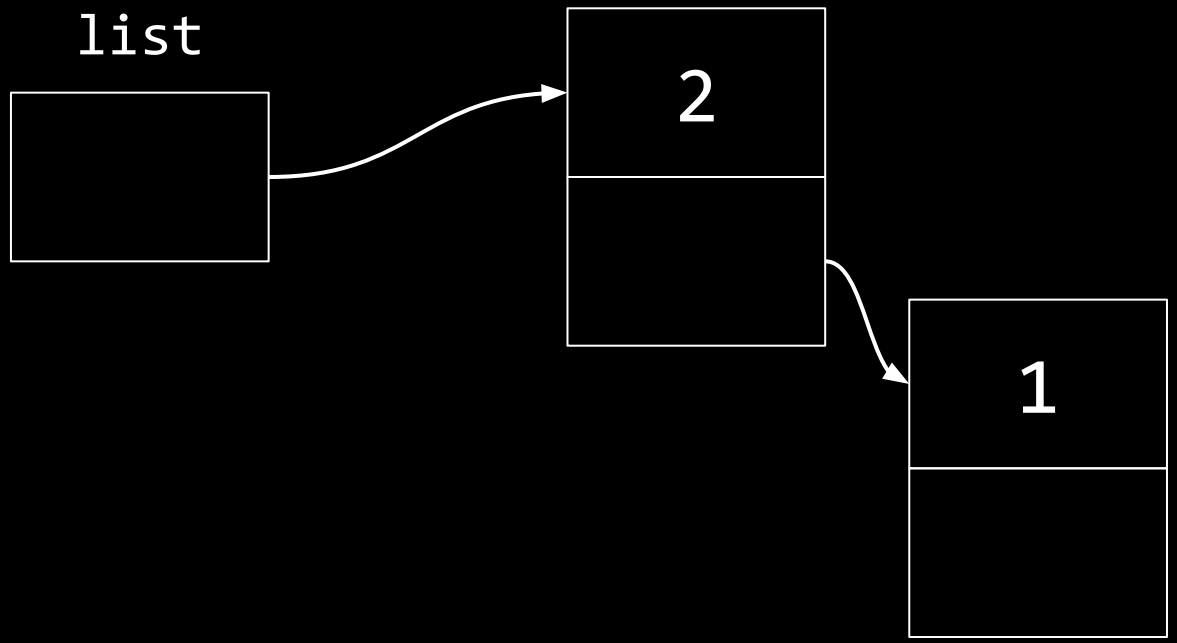


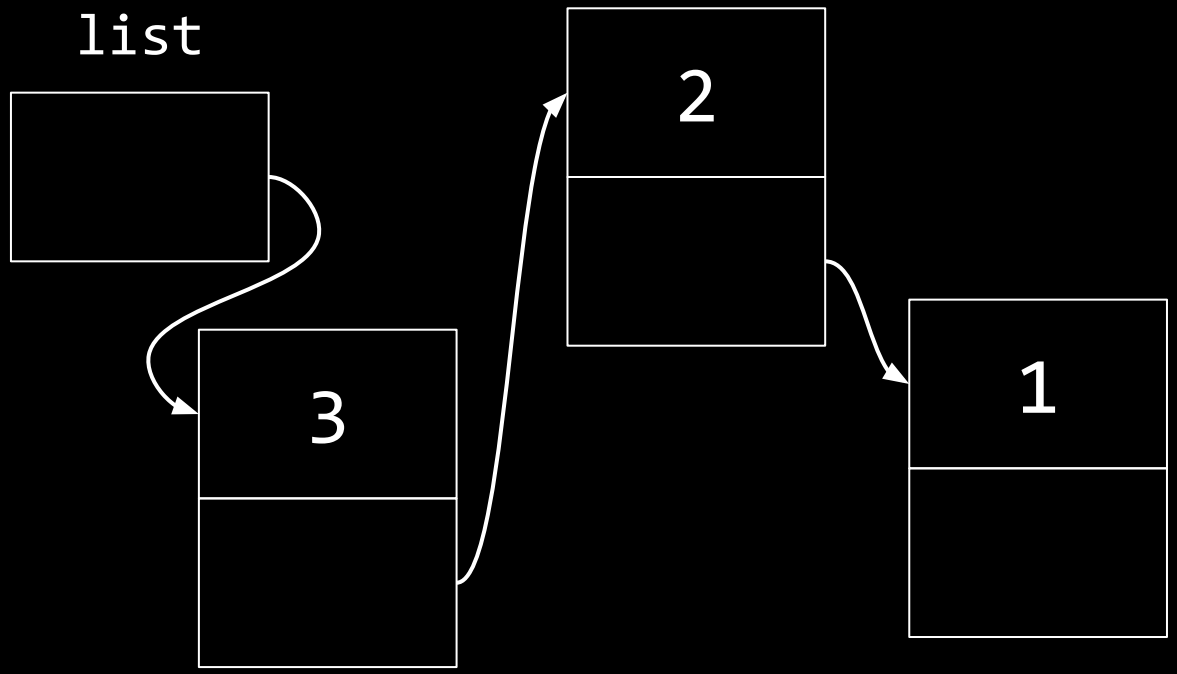
```
list = n;
```

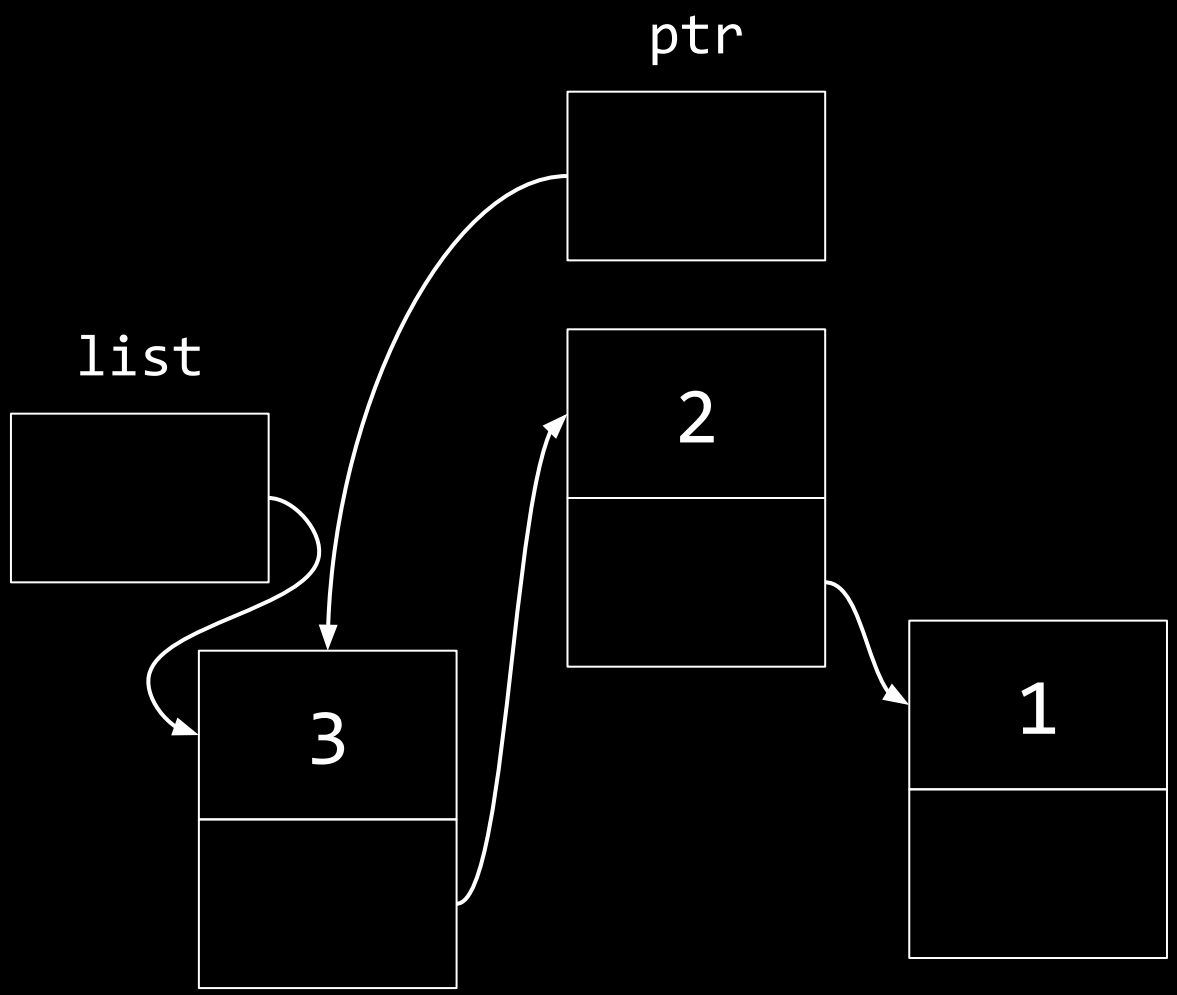


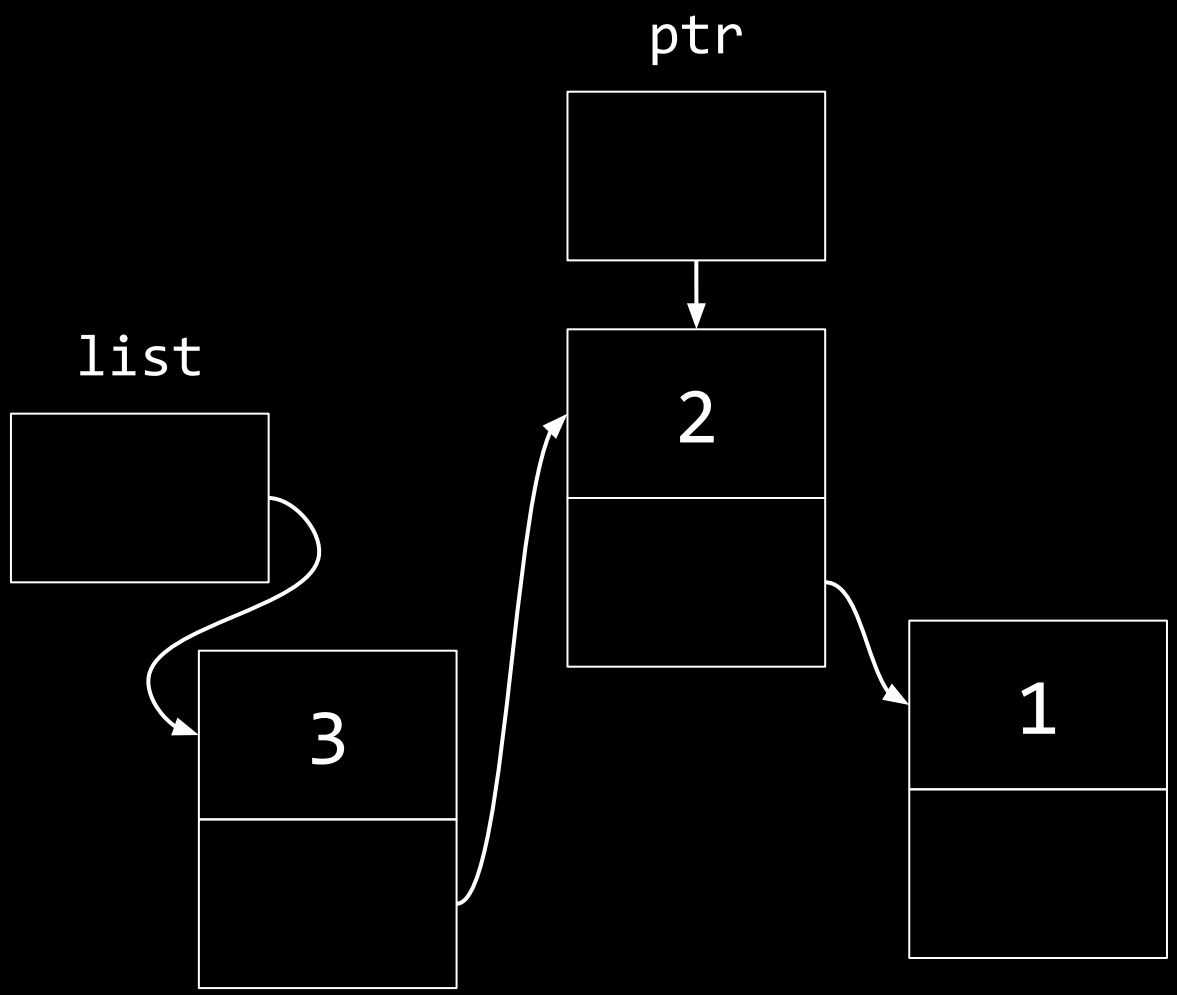
```
list = n;
```

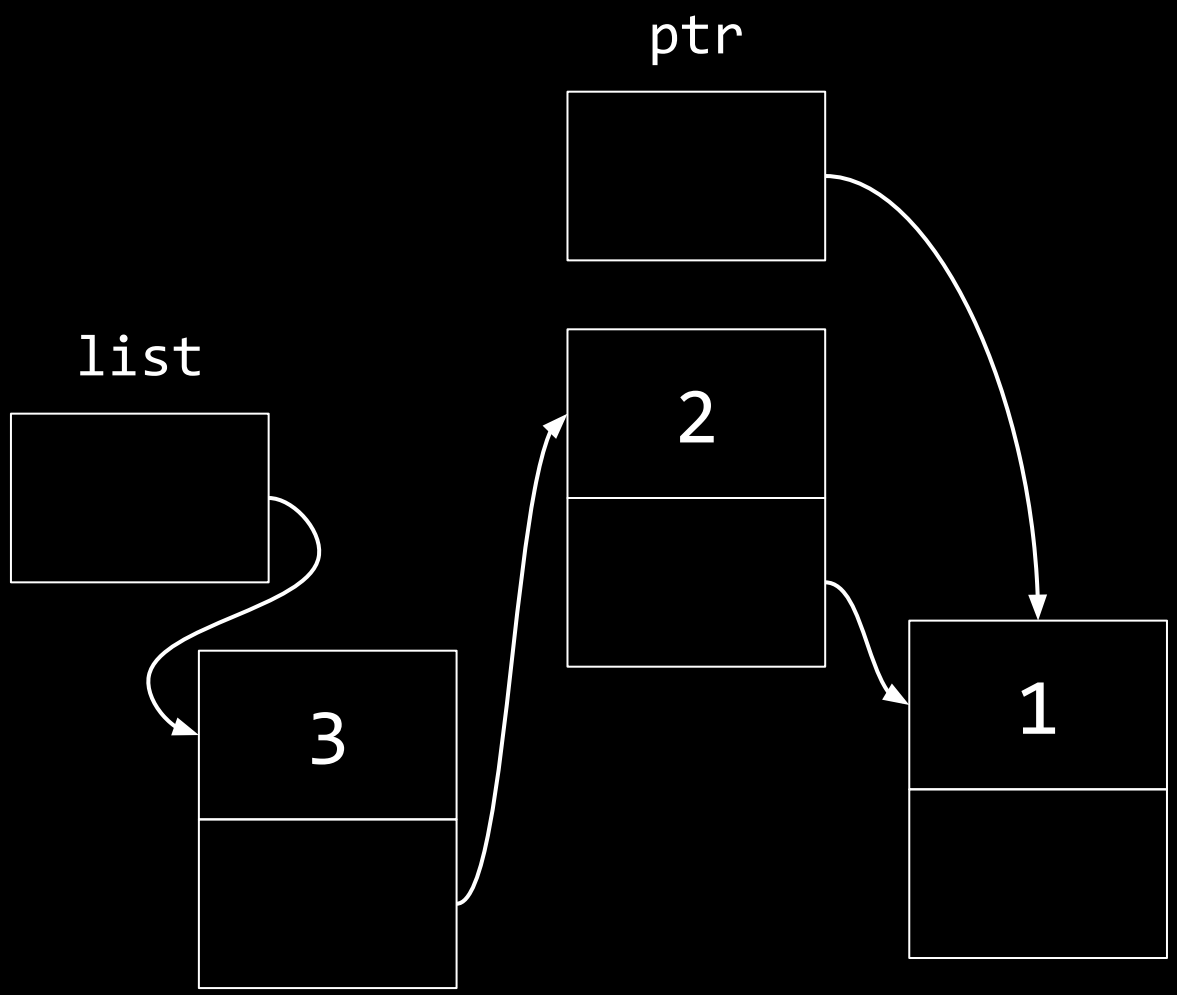


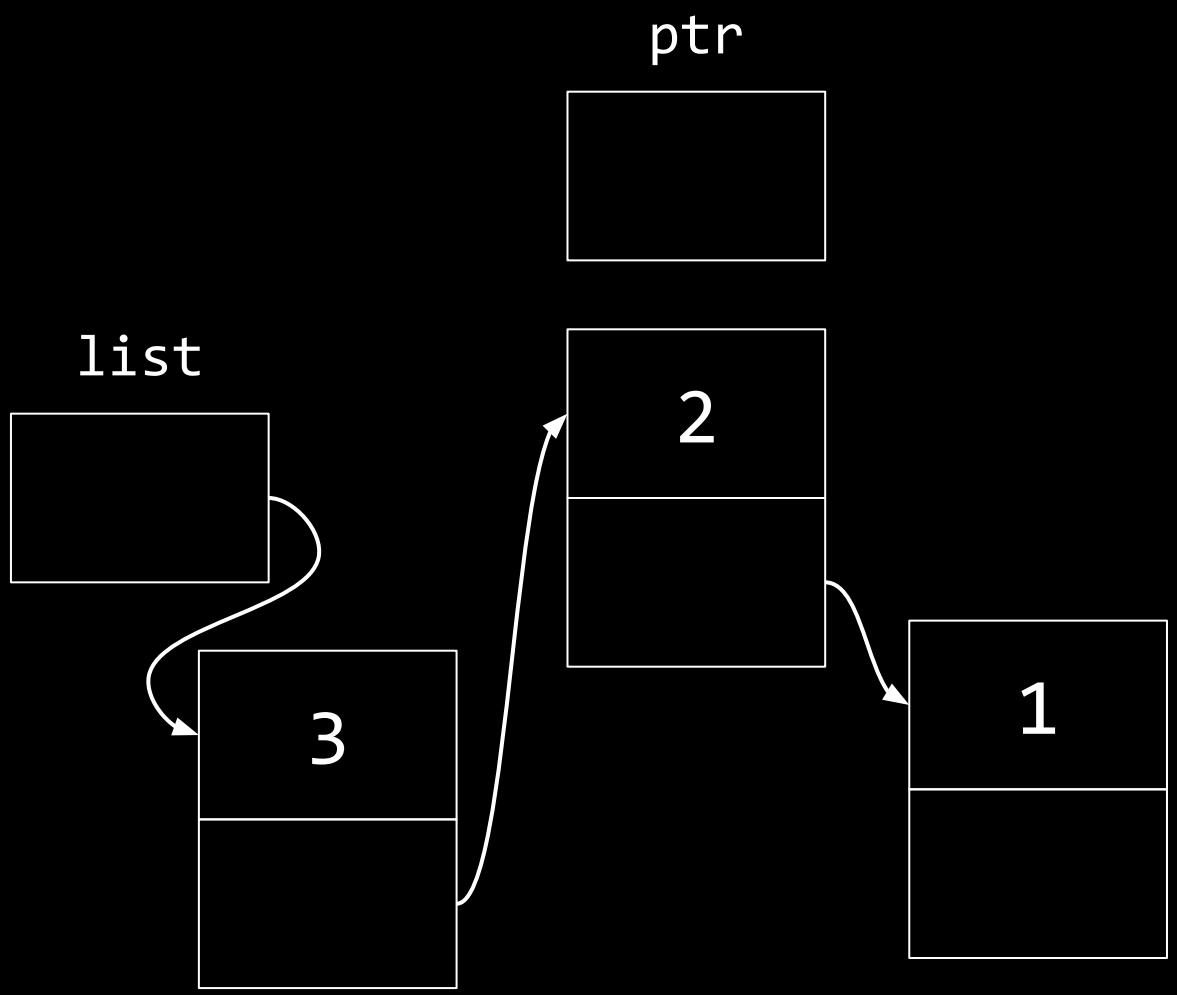

list
2
1

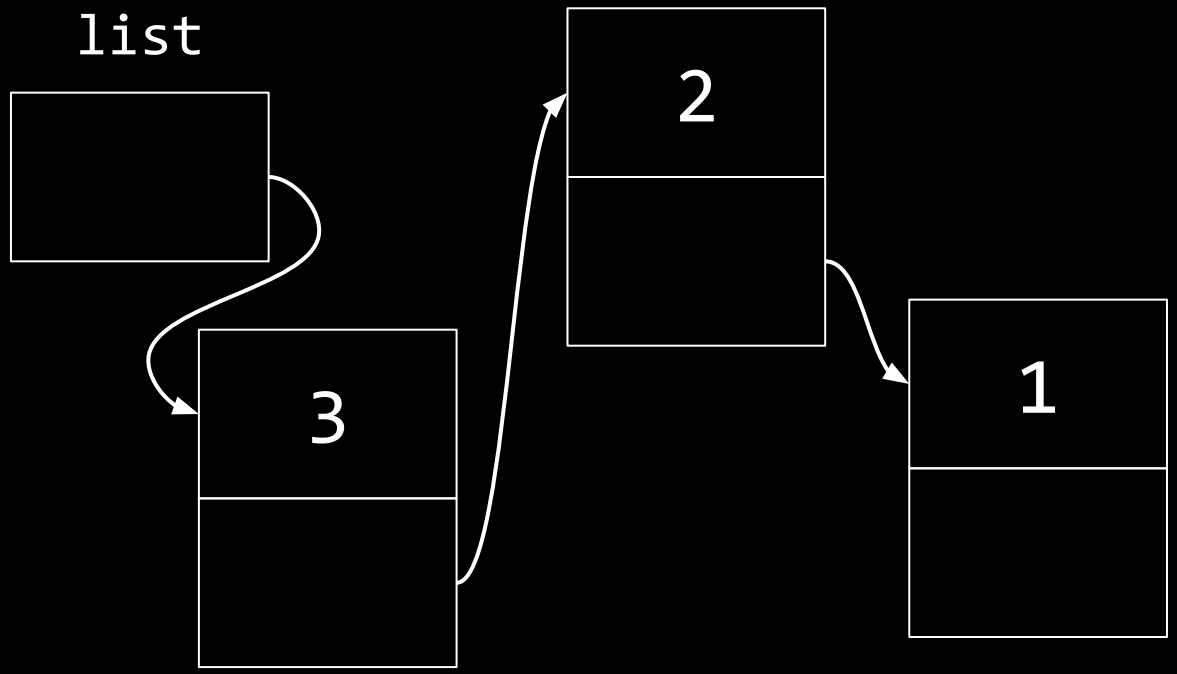

list
2
3
1


ptr
list
2
3
1


ptr
list
2
3
1


ptr
list
2
3
1


ptr
list
2
3
1


list
2
3
1

$O(n^2)$

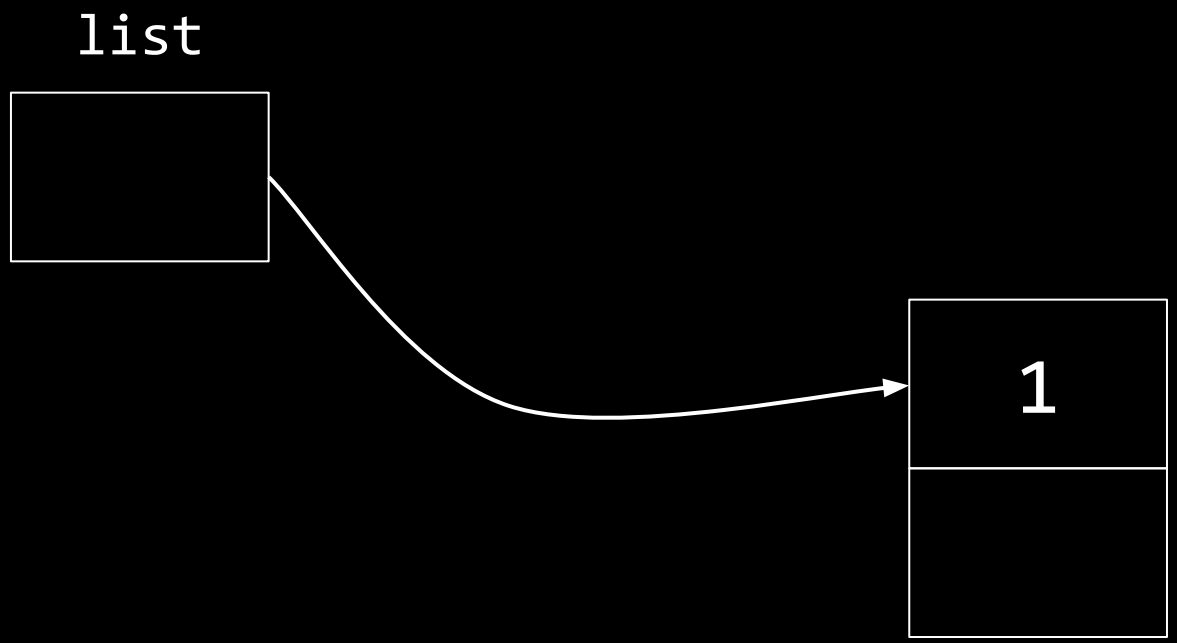
$O(n \log n)$

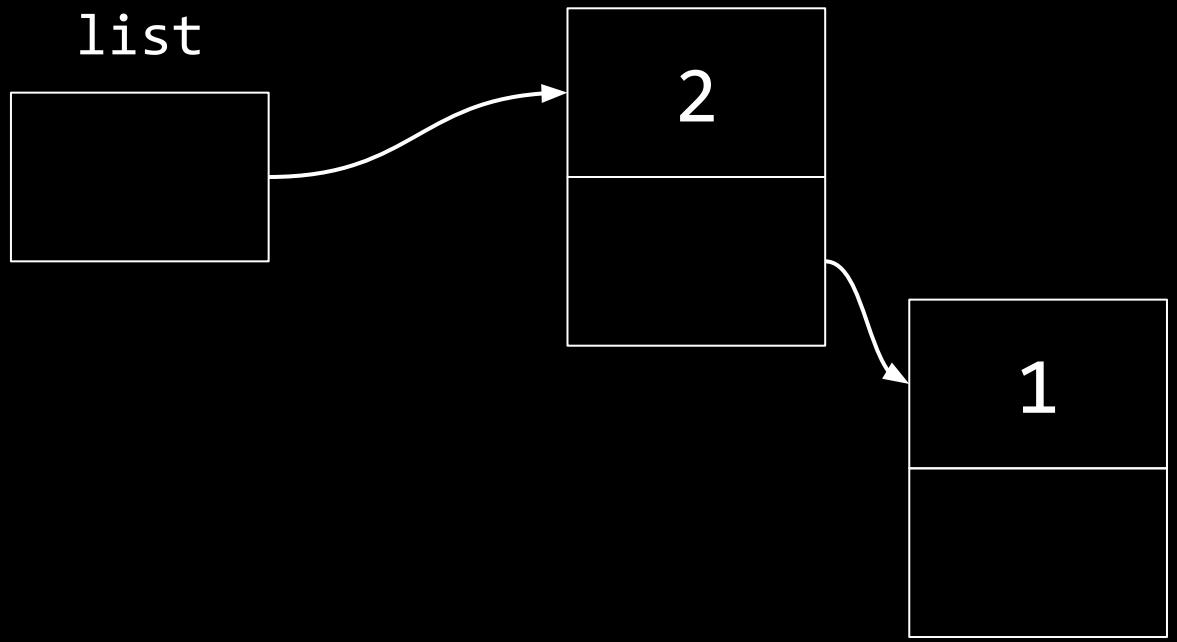
$O(n)$

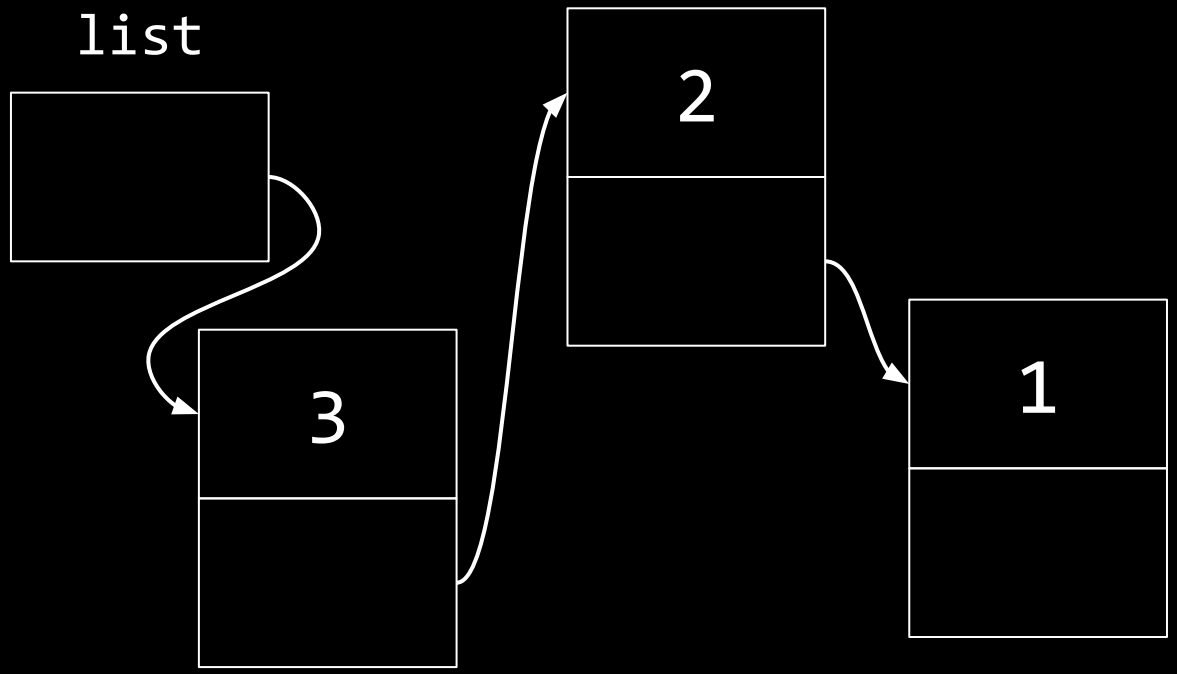
$O(\log n)$

$O(1)$

list


list
1


list
2
1


list
2
3
1

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$

$O(n^2)$

$O(n \log n)$

$O(n)$

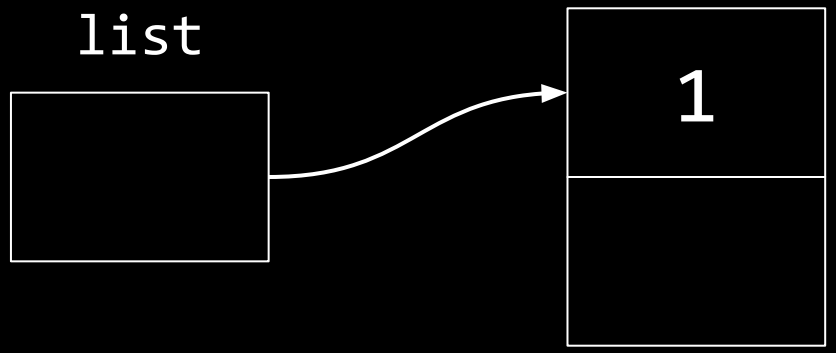
$O(\log n)$

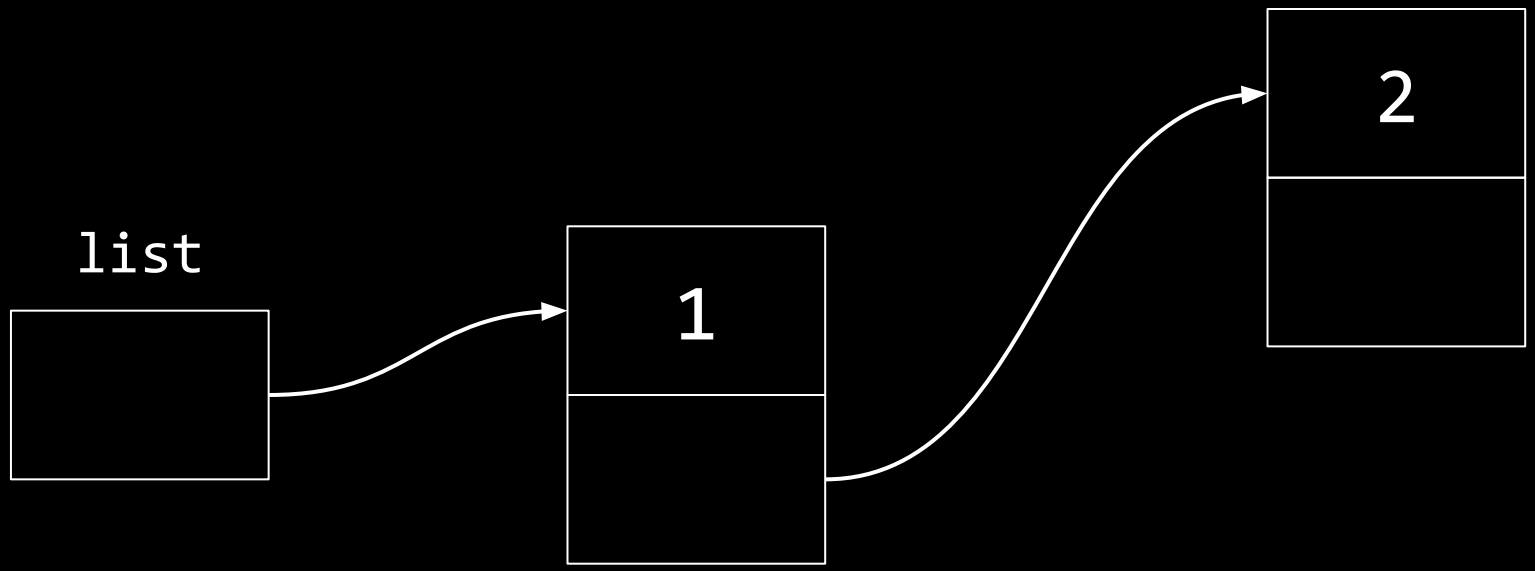
$O(1)$

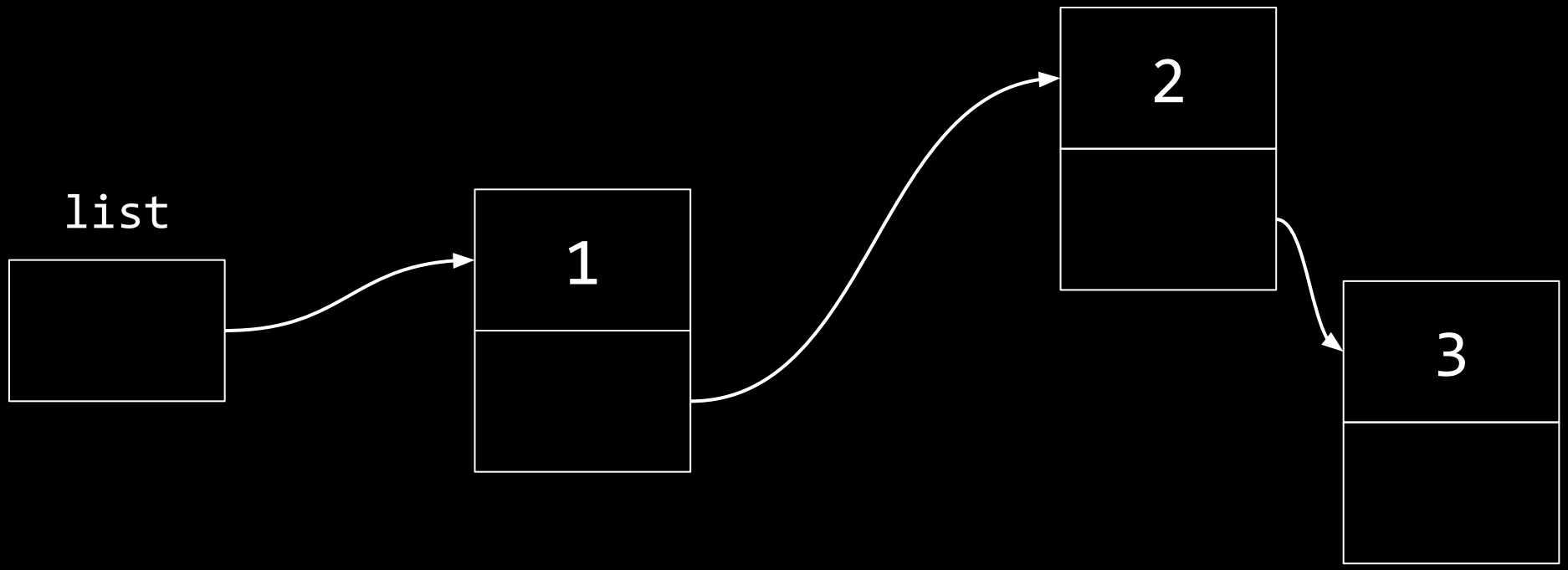
$O(n^2)$	
$O(n \log n)$	
$O(n)$	search
$O(\log n)$	
$O(1)$	insertion

$O(n^2)$	
$O(n \log n)$	
$O(n)$	search, deletion
$O(\log n)$	
$O(1)$	insertion

list


list
1


list
1
2


list
1
2
3

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$

$O(n^2)$

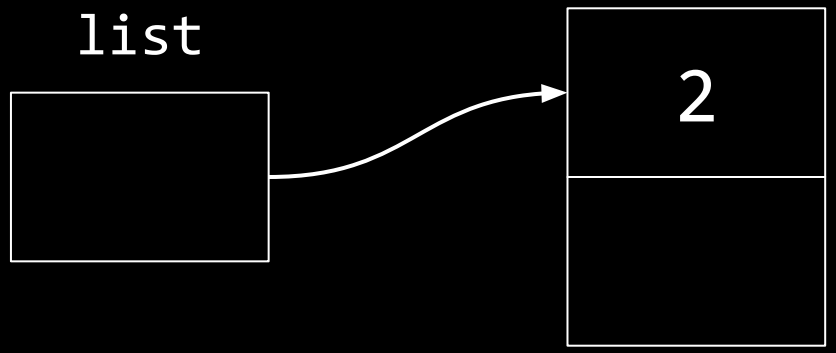
$O(n \log n)$

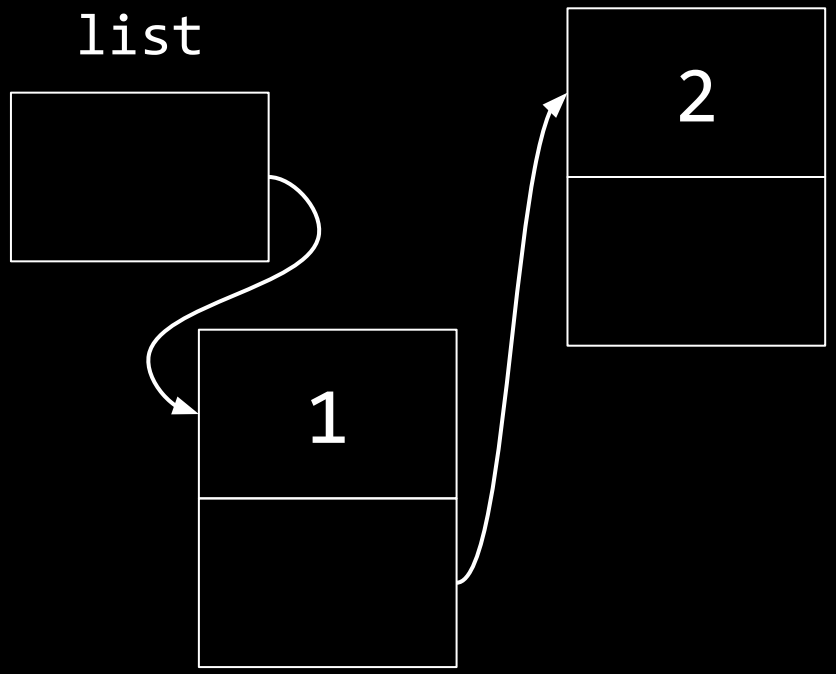
$O(n)$ search, deletion, insertion

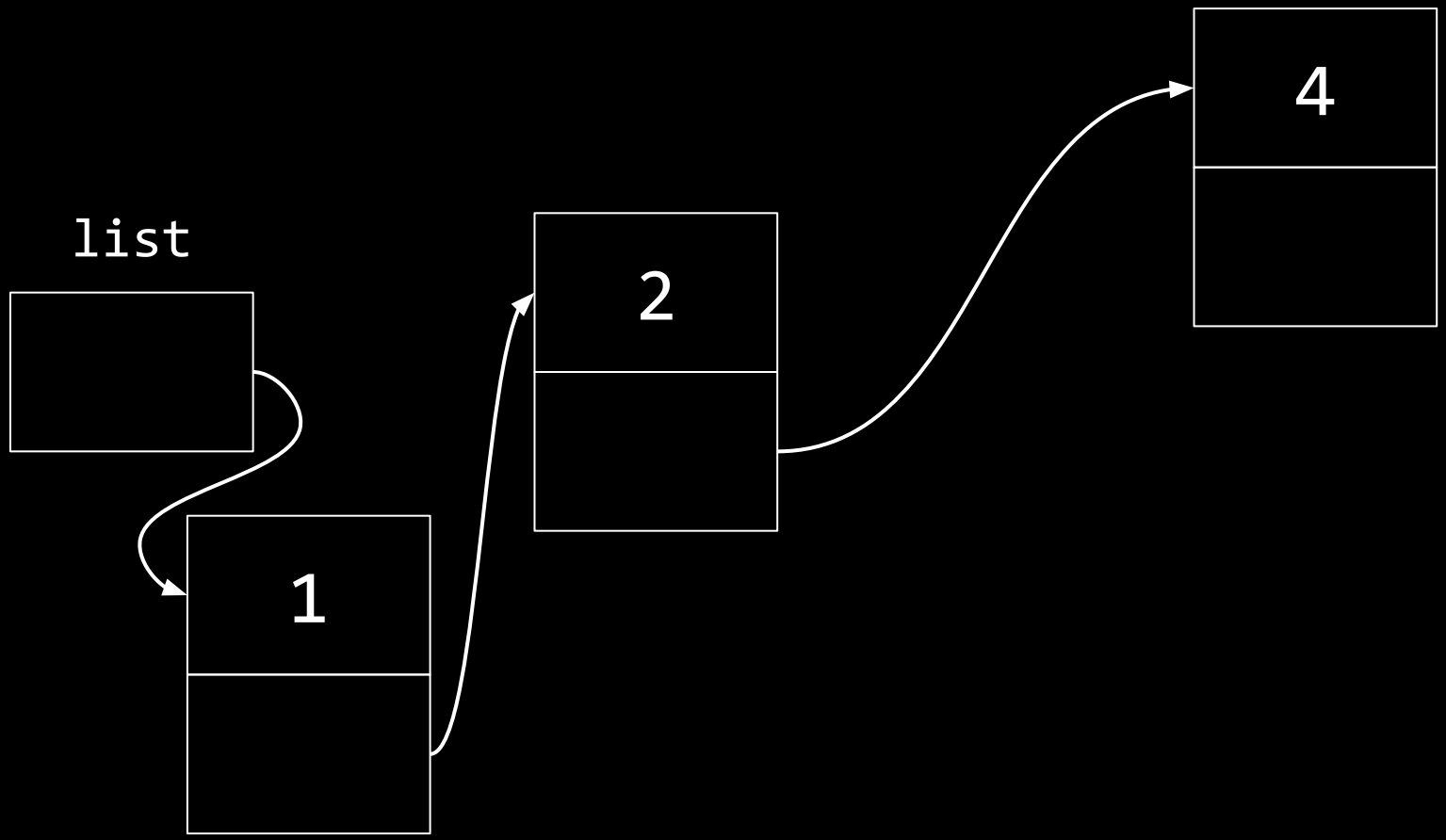
$O(\log n)$

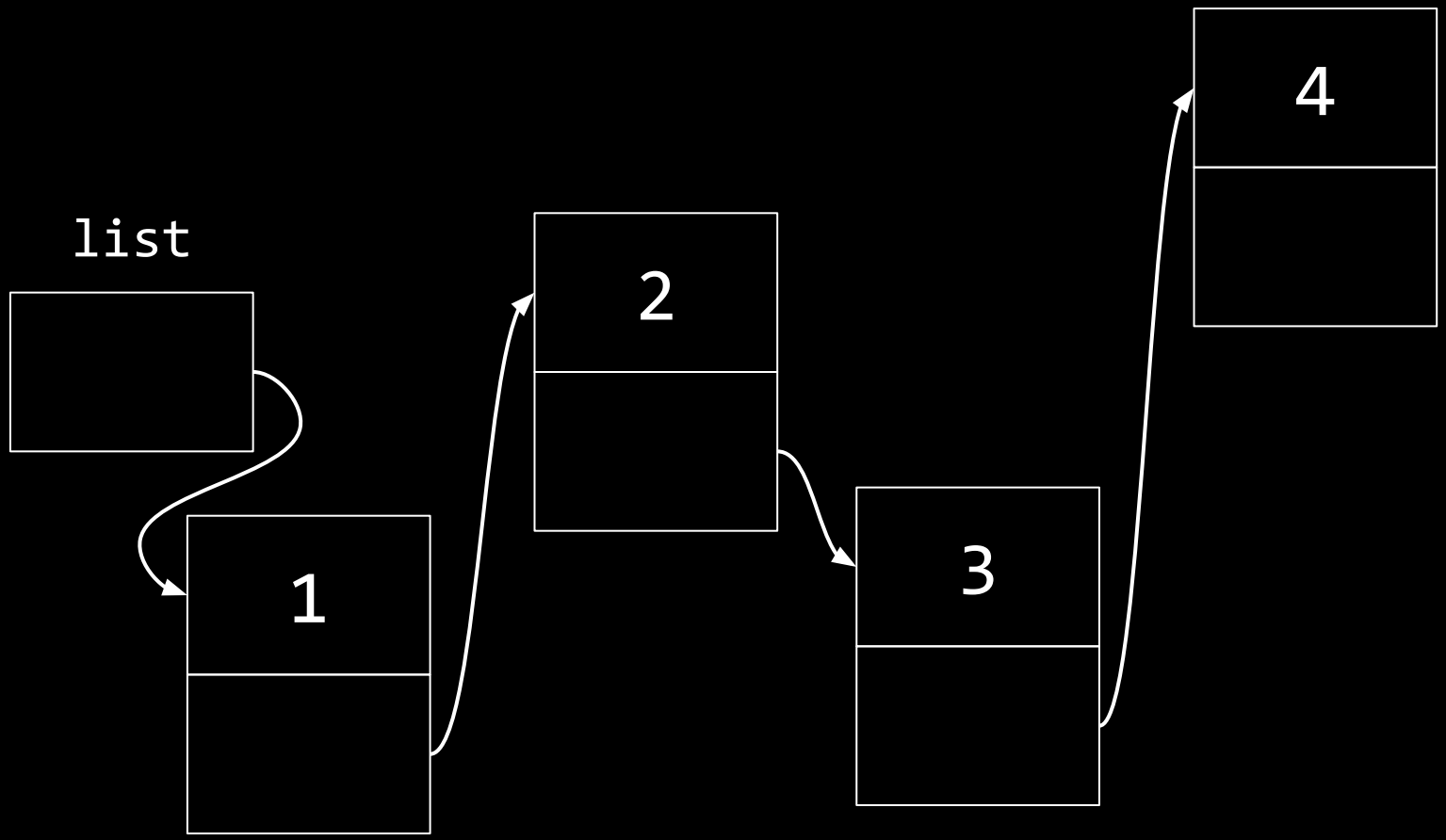
$O(1)$

list


list
2


list
2
1


list
1
2
4


list
1
2
3
4

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$

$O(n^2)$

$O(n \log n)$

$O(n)$ search, deletion, insertion

$O(\log n)$

$O(1)$

arrays

linked lists

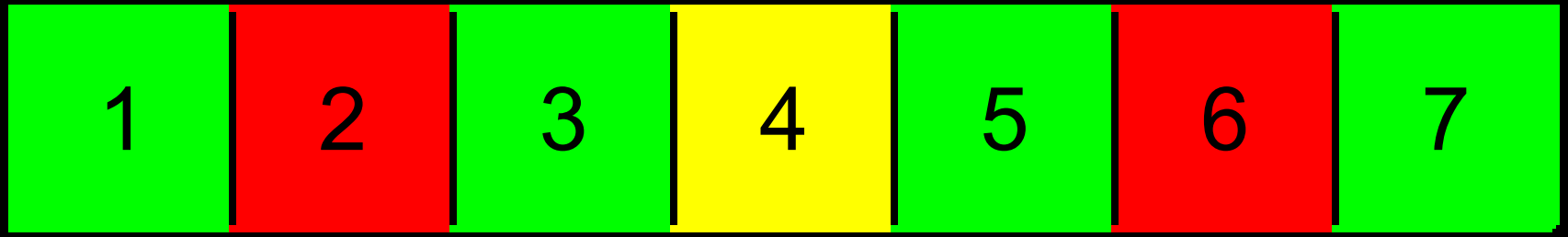
trees

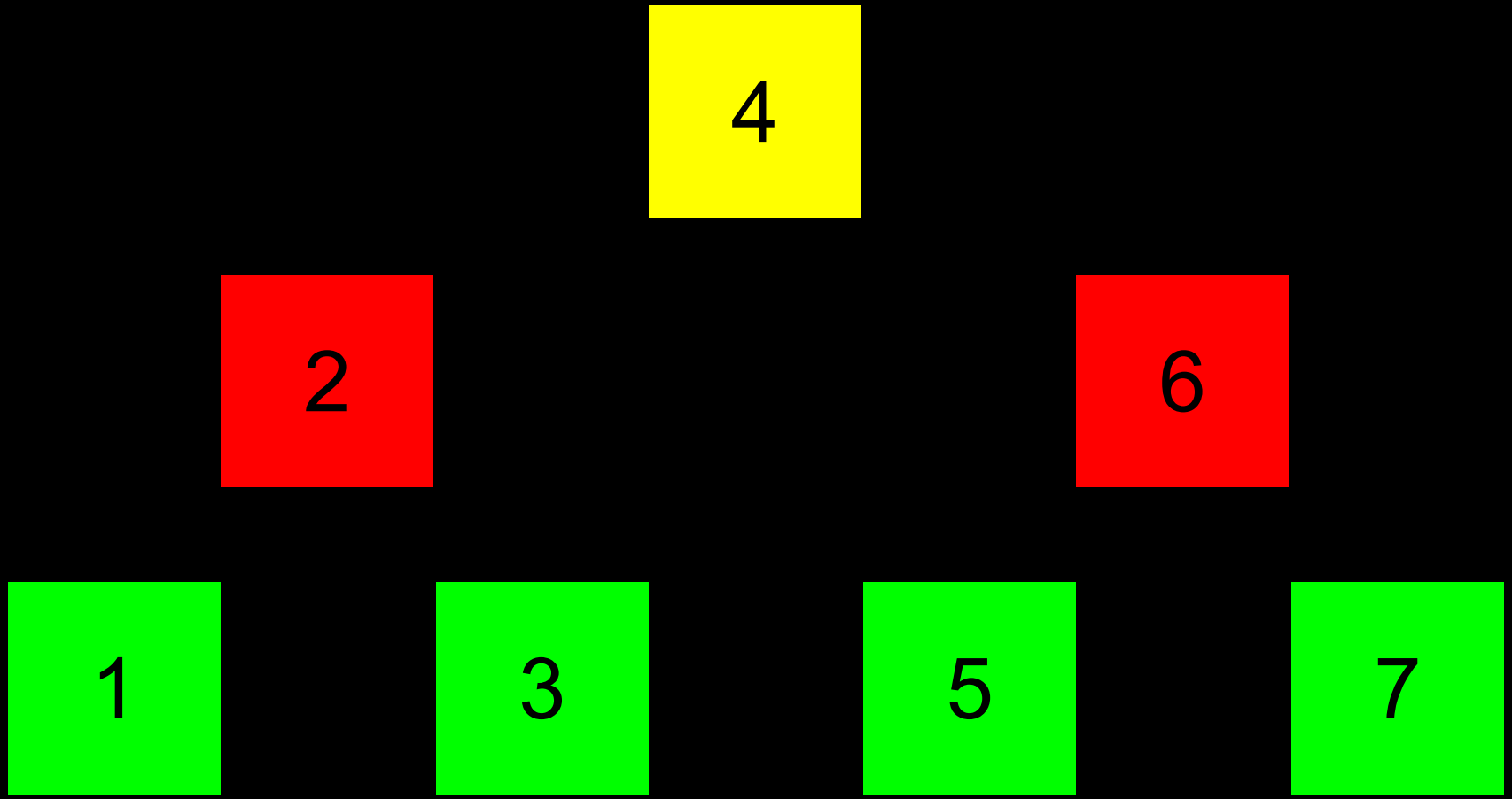
binary search trees

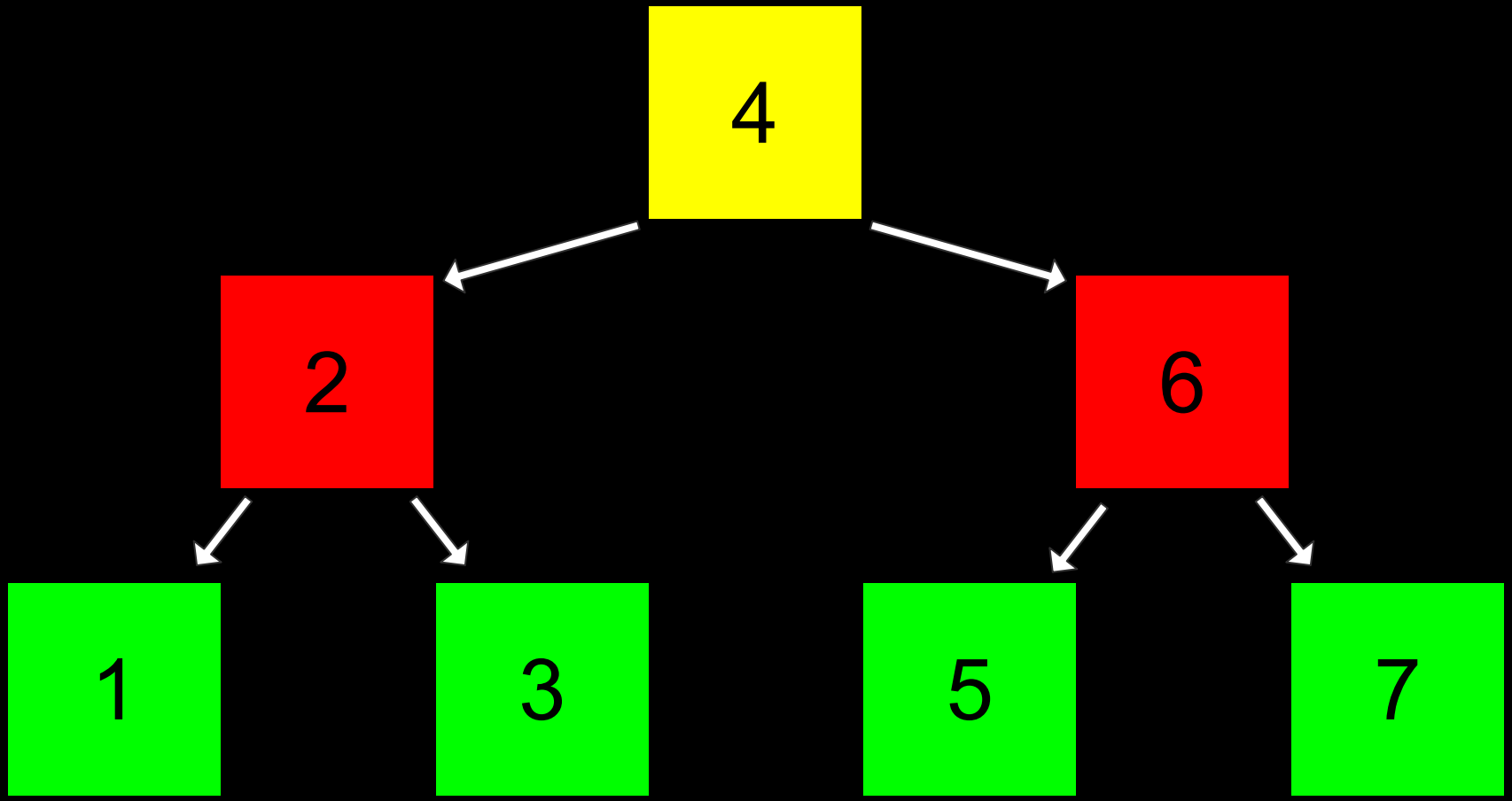
1	2	3	4	5	6	7

1	2	3	4	5	6	7

1	2	3	4	5	6	7


1
2
3
4
5
6
7


4
2
6
1
3
5
7

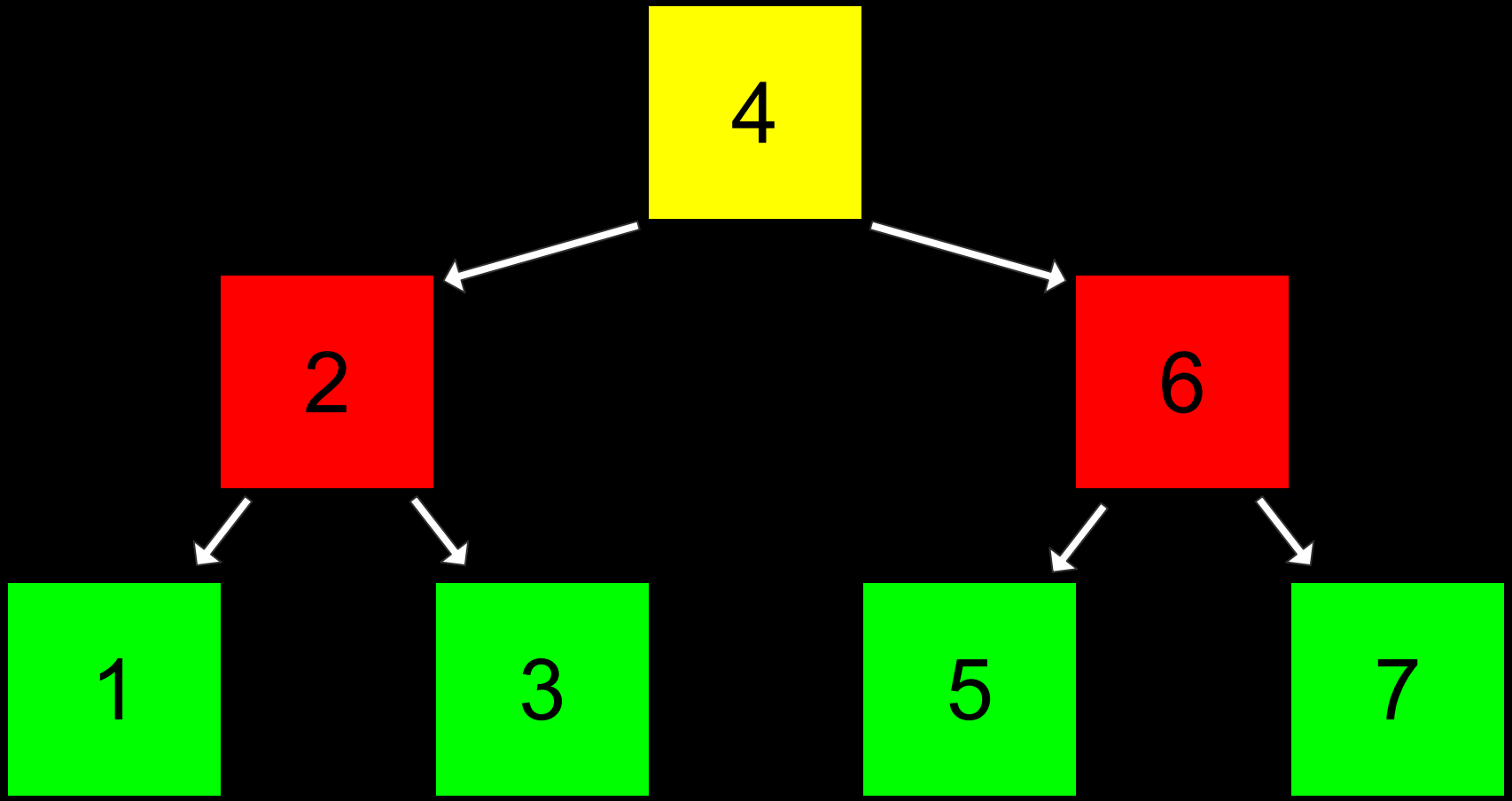

4
2
6
1
3
5
7

```
typedef struct node
{
    int number;
    struct node *next;
} node;
```

```
typedef struct node
{
    int number;

} node;
```

```
typedef struct node
{
    int number;


} node;
```

```
typedef struct node
{
    int number;
    struct node *left;
    struct node *right;
} node;
```


4
2
6
1
3
5
7

```
bool search(node *tree, int number)
{

}
```

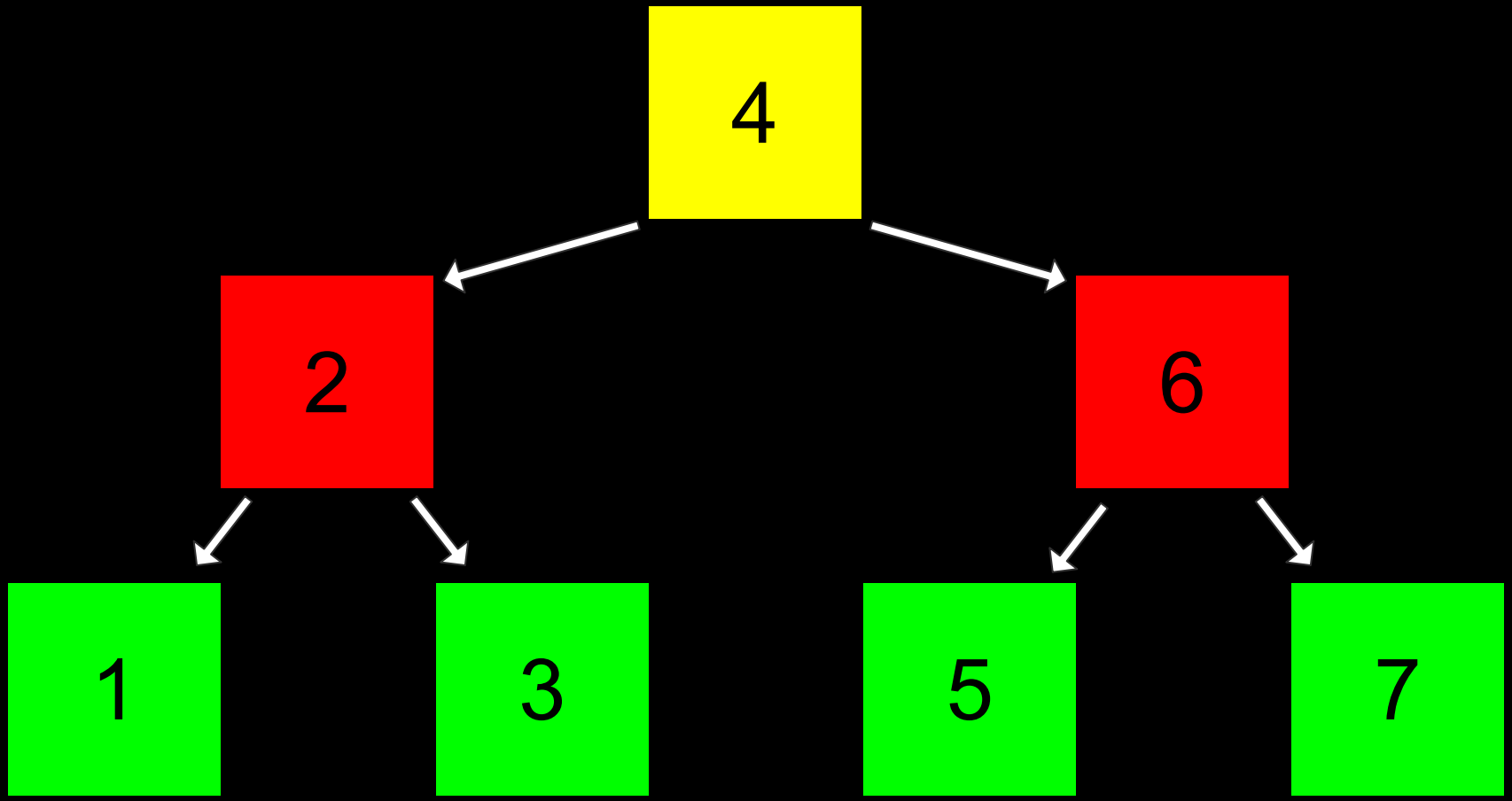
```
bool search(node *tree, int number)
{
    if (tree == NULL)
    {
        return false;
    }



}
```

```
bool search(node *tree, int number)
{
    if (tree == NULL)
    {
        return false;
    }
    else if (number < tree->number)
    {
        return search(tree->left, number);
    }









}
```

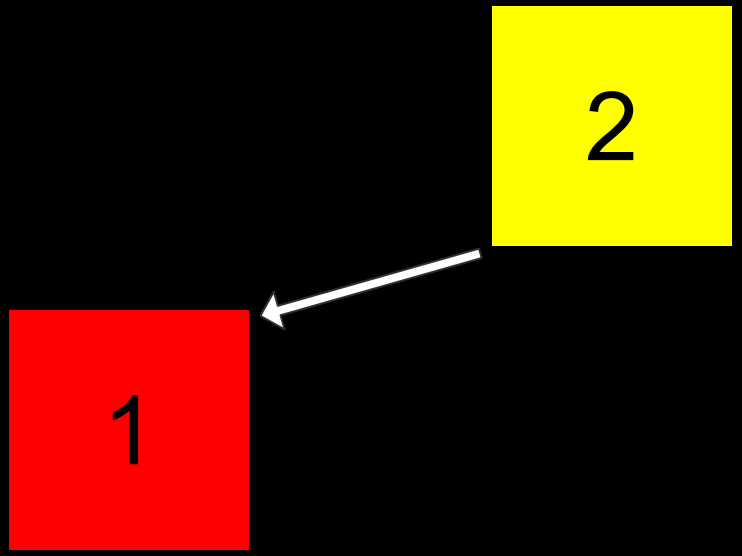
```
bool search(node *tree, int number)
{
    if (tree == NULL)
    {
        return false;
    }
    else if (number < tree->number)
    {
        return search(tree->left, number);
    }
    else if (number > tree->number)
    {
        return search(tree->right, number);
    }




}
```

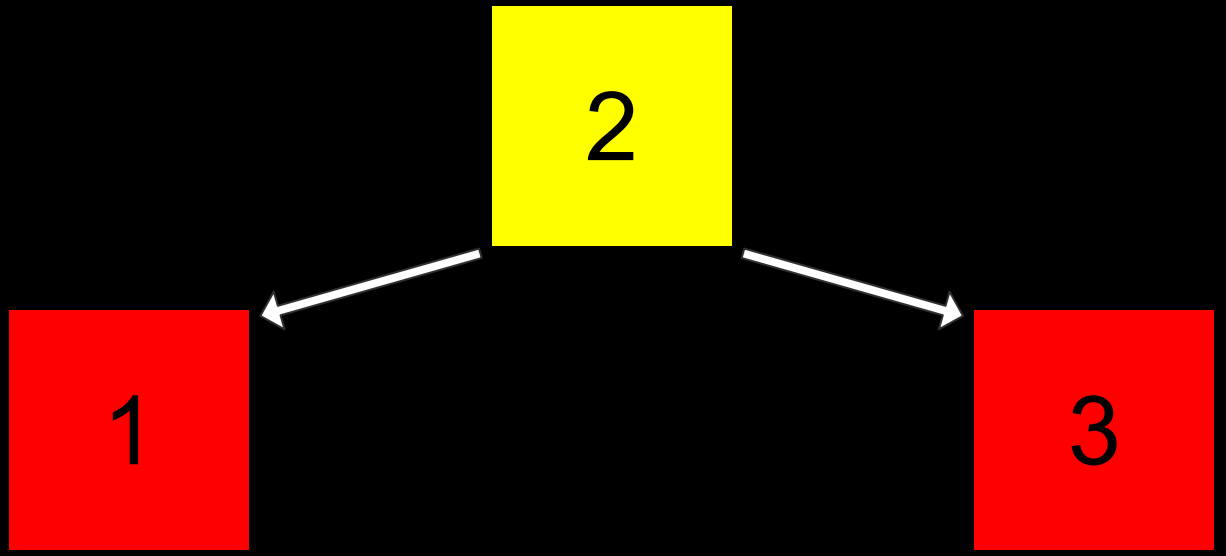
```
bool search(node *tree, int number)
{
    if (tree == NULL)
    {
        return false;
    }
    else if (number < tree->number)
    {
        return search(tree->left, number);
    }
    else if (number > tree->number)
    {
        return search(tree->right, number);
    }
    else if (number == tree->number)
    {
        return true;
    }
}
```

```
bool search(node *tree, int number)
{
    if (tree == NULL)
    {
        return false;
    }
    else if (number < tree->number)
    {
        return search(tree->left, number);
    }
    else if (number > tree->number)
    {
        return search(tree->right, number);
    }
    else
    {
        return true;
    }
}
```

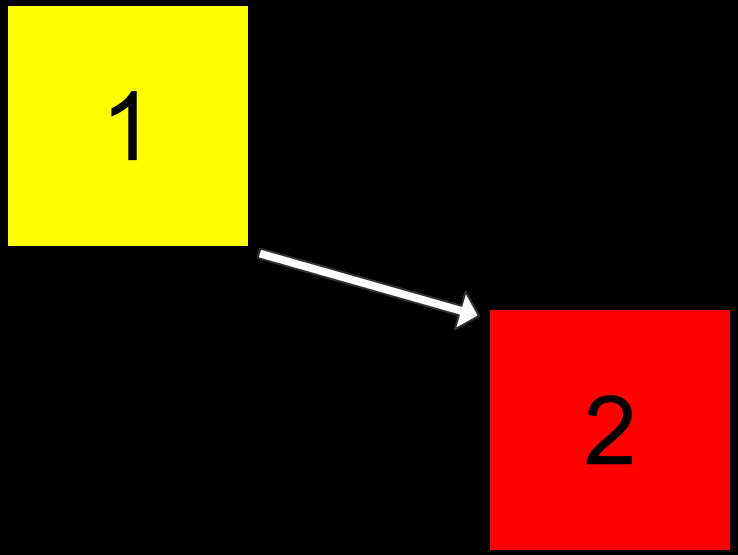

4
2
6
1
3
5
7

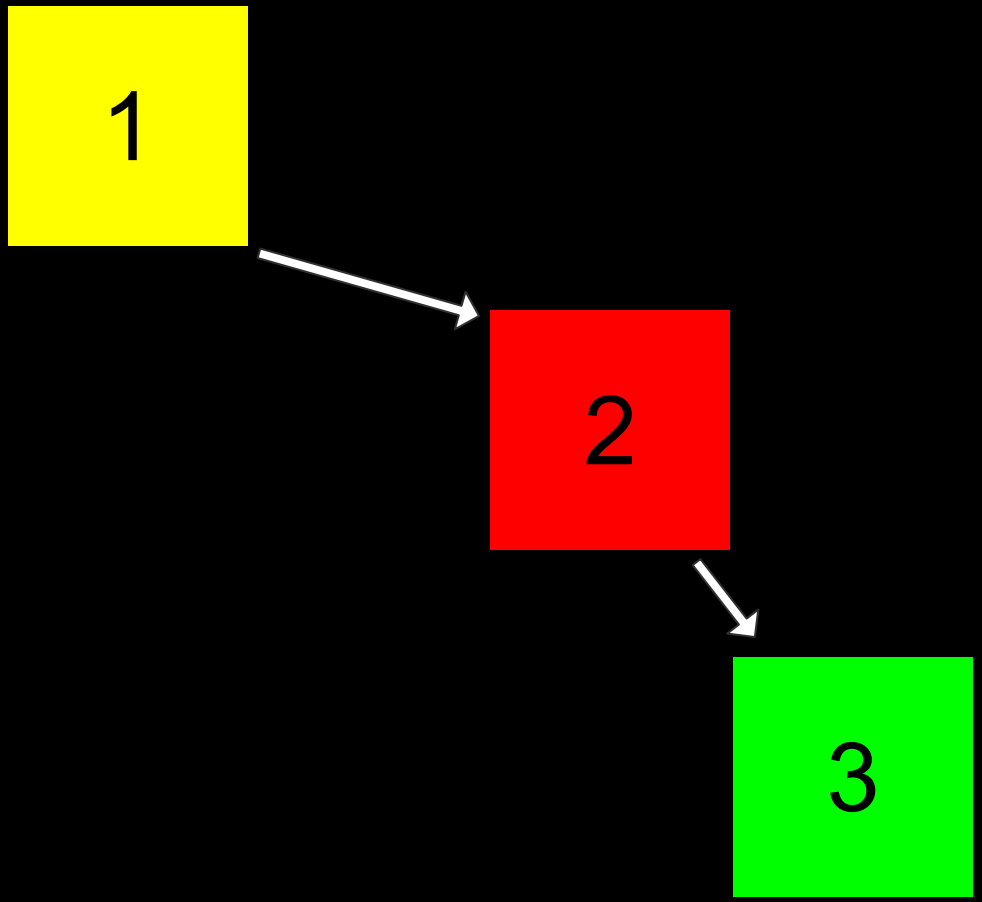
2


2
1


2
1
3

1


1
2


1
2
3

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$

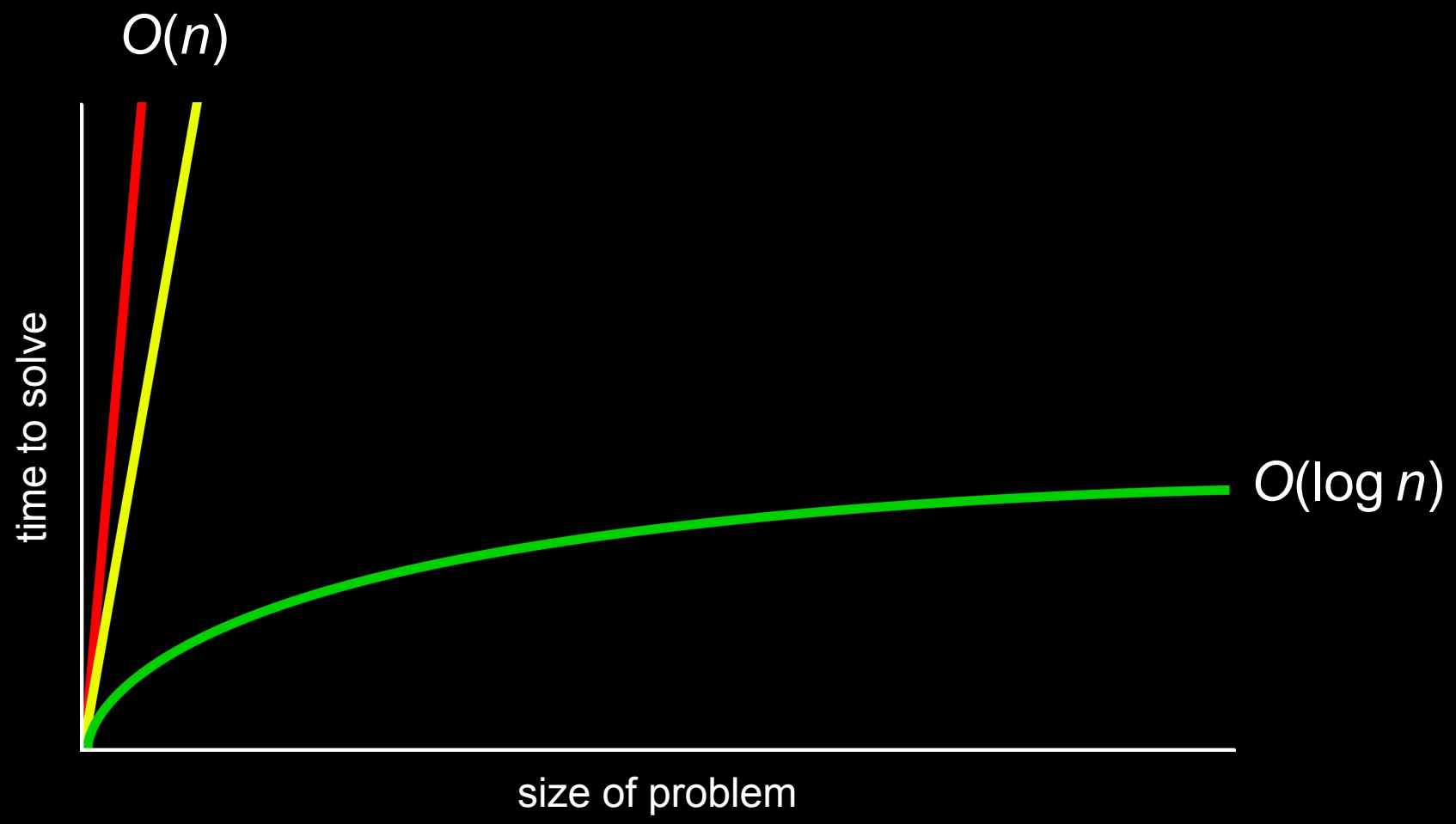
$O(n^2)$

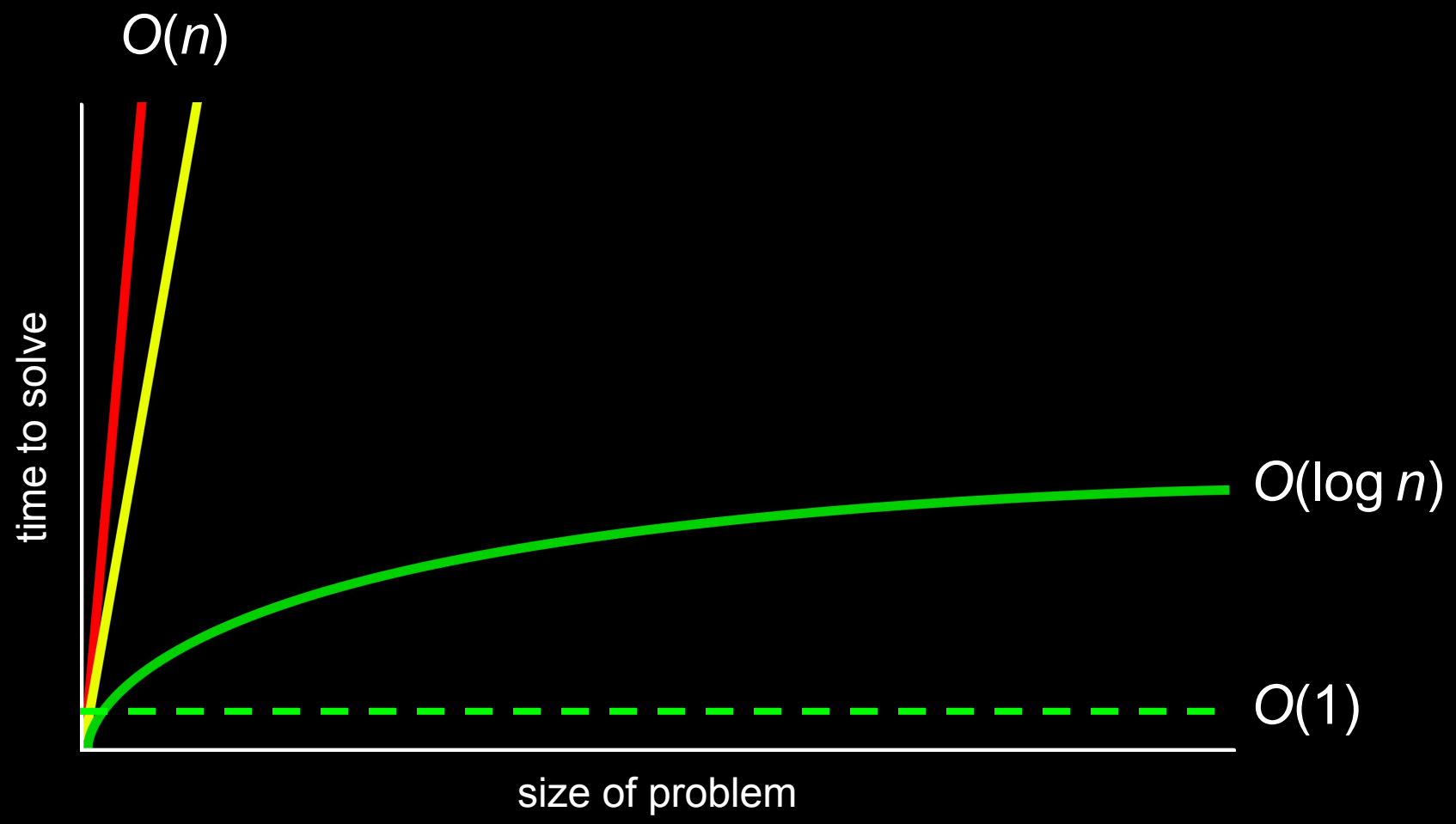
$O(n \log n)$

$O(n)$

$O(\log n)$

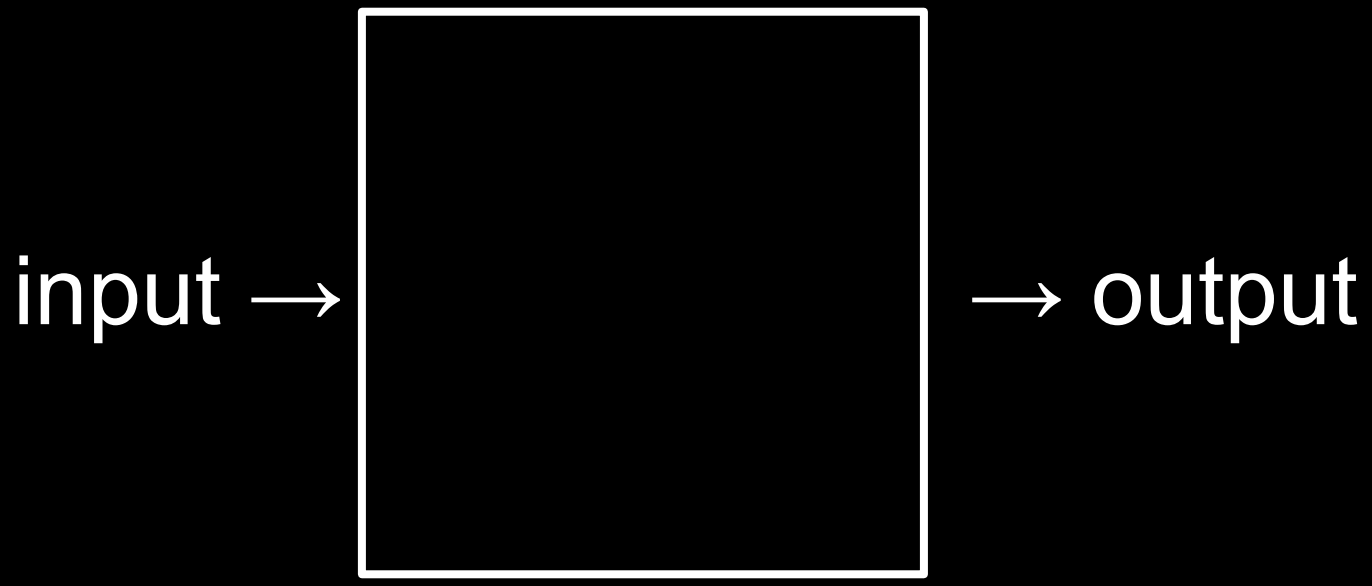
$O(1)$


$O(n)$
time to solve
$O(\log n)$
size of problem


O(n)
time to solve
O(log n)
O(1)
size of problem

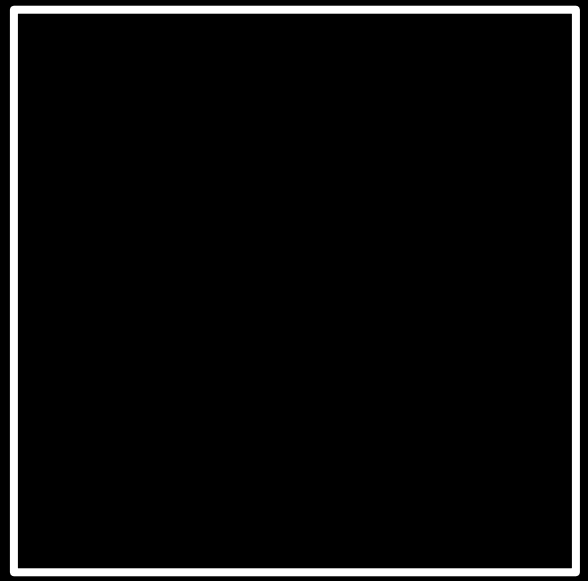
hashing

hash function


input →
→ output

hash function

Mario → [redacted] → 12

Luigi →  → 11

```c
#include <ctype.h>

int hash(char *name)
{
    return toupper(name[0]) - 'A';
}
```

```c
#include <ctype.h>

int hash(const char *name)
{
    return toupper(name[0]) - 'A';
}
```

```c
#include <ctype.h>

unsigned int hash(const char *name)
{
    return toupper(name[0]) - 'A';
}
```

hash tables

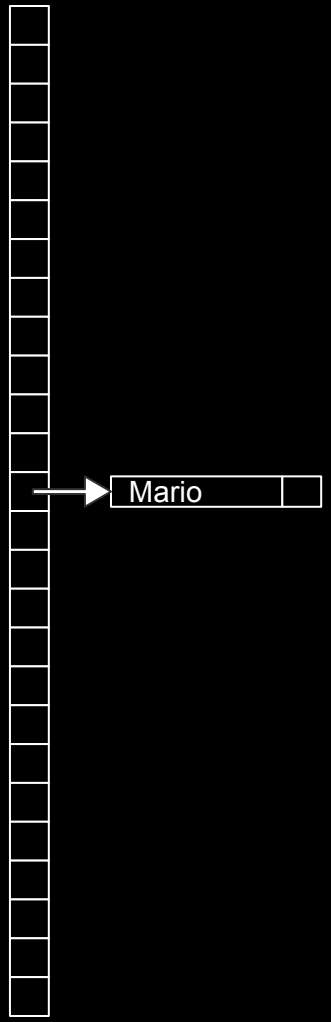
```
node *table[26];
```

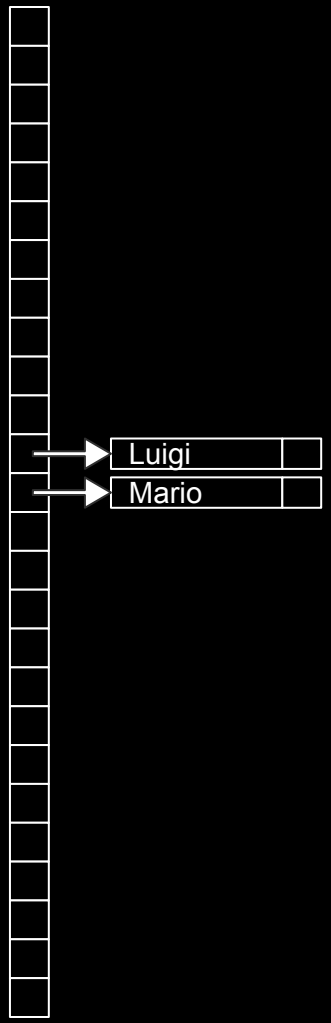
```
typedef struct
{
    char *name;
    char *number;
} person;
```

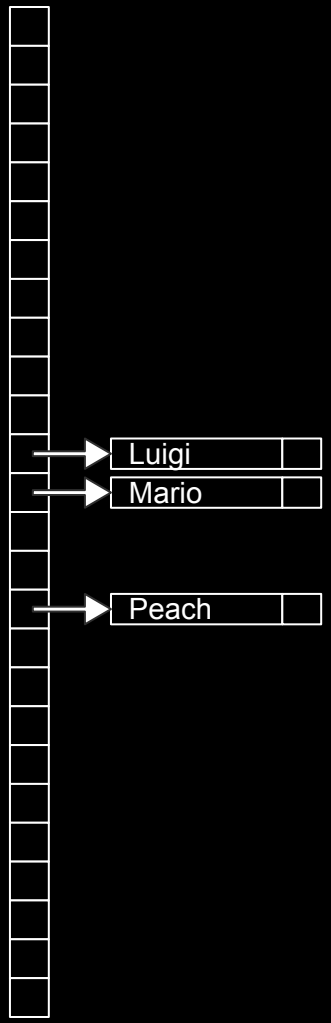
```
typedef struct node
{
    char *name;
    struct node *next;
} node;
```

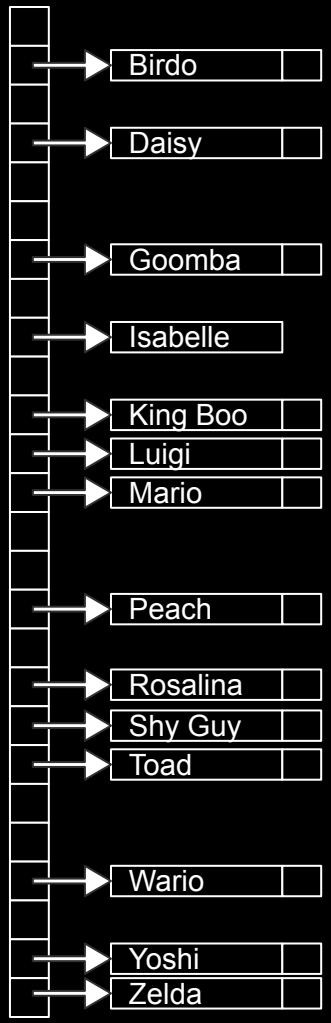
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	

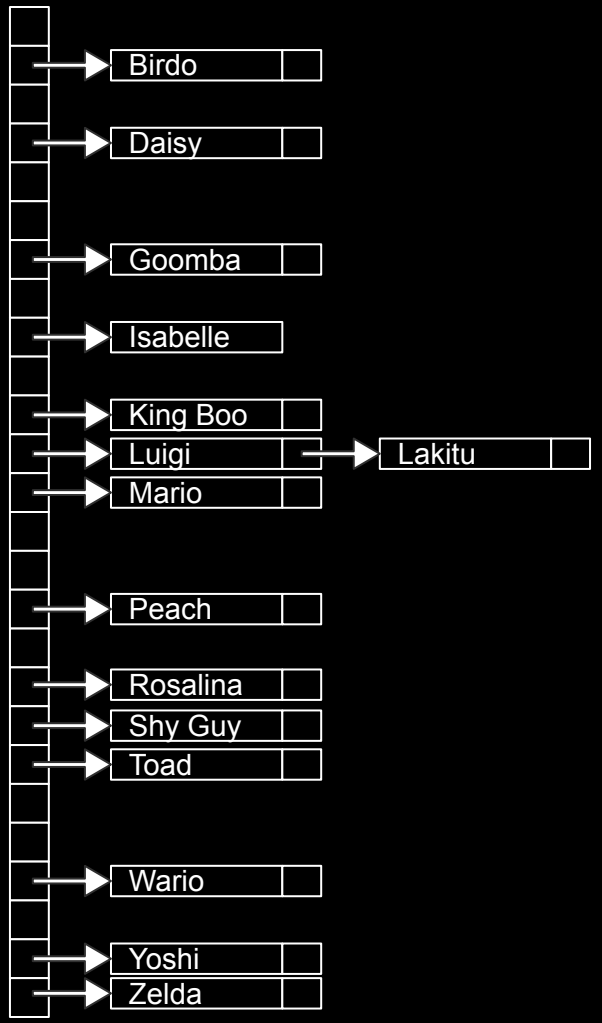
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	
K	
L	
M	
N	
O	
P	
Q	
R	
S	
T	
U	
V	
W	
X	
Y	
Z	


Mario

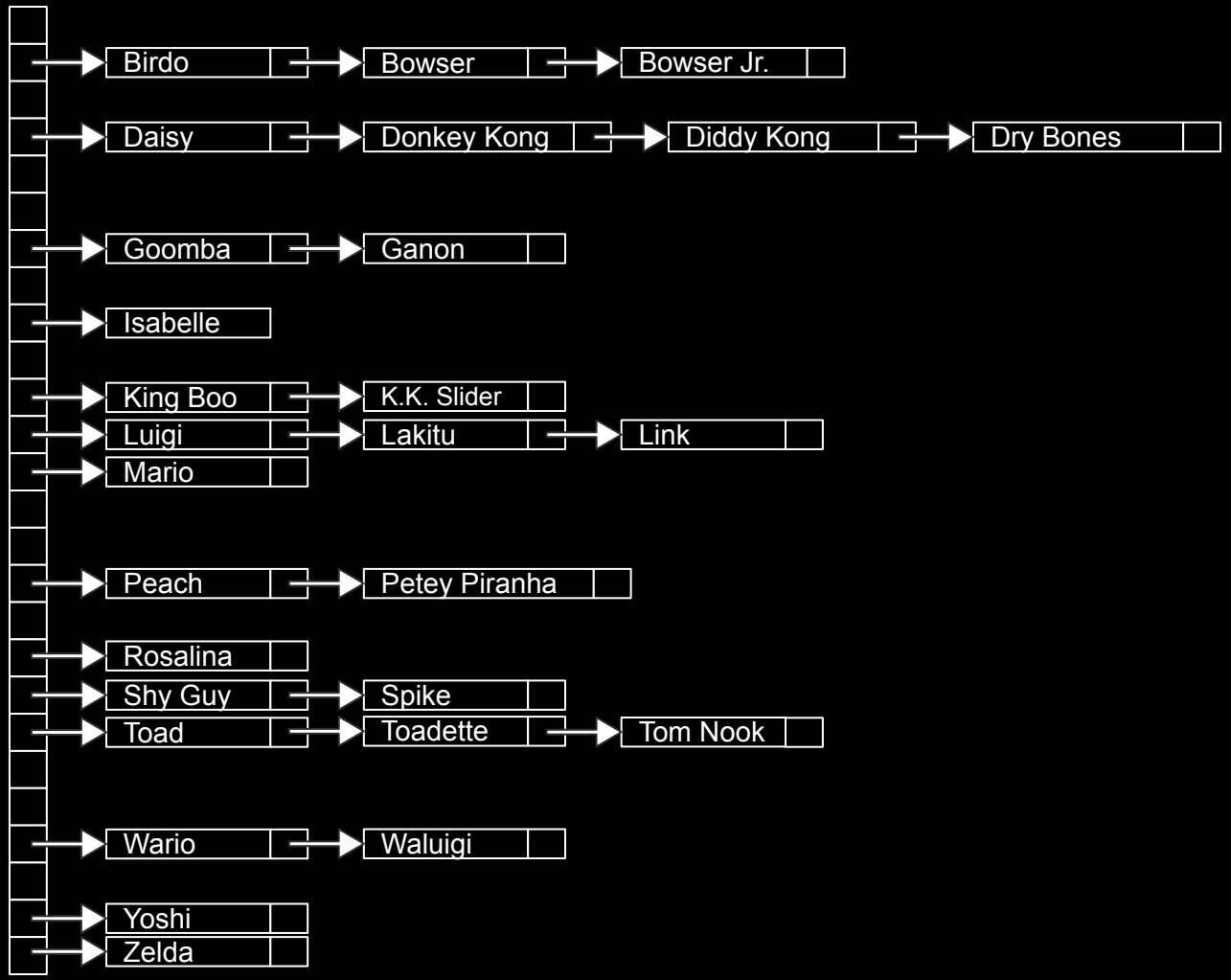

Luigi
Mario


Luigi
Mario
Peach


Birdo
Daisy
Goomba
Isabelle
King Boo
Luigi
Mario
Peach
Rosalina
Shy Guy
Toad
Wario
Yoshi
Zelda


Birdo
Daisy
Goomba
Isabelle
King Boo
Luigi
Lakitu
Mario
Peach
Rosalina
Shy Guy
Toad
Wario
Yoshi
Zelda

Birdo
Daisy
Goomba
Isabelle
King Boo
Luigi
Lakitu
Link
Mario
Peach
Rosalina
Shy Guy
Toad
Wario
Yoshi
Zelda


Birdo
Bowser
Bowser Jr.
Daisy
Donkey Kong
Diddy Kong
Dry Bones
Goomba
Ganon
Isabelle
King Boo
K.K. Slider
Luigi
Lakitu
Link
Mario
Peach
Petey Piranha
Rosalina
Shy Guy
Spike
Toad
Toadette
Tom Nook
Wario
Waluigi
Yoshi
Zelda

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$

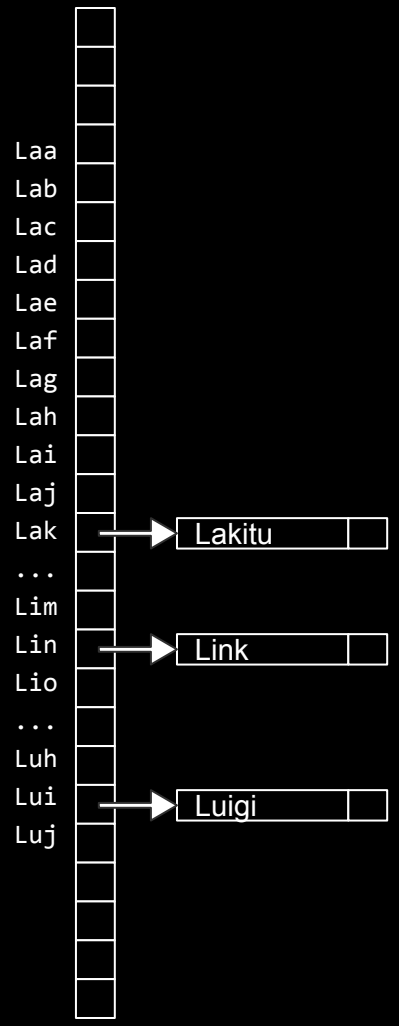
$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$


Laa
Lab
Lac
Lad
Lae
Laf
Lag
Lah
Lai
Laj
Lak
...
Lim
Lin
Lio
...
Luh
Lui
Luj
Lakitu
Link
Luigi

$O(n)$

$O(n/k)$

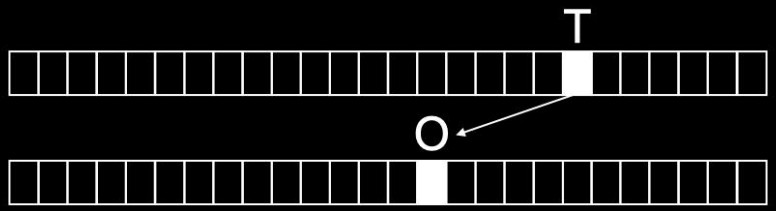
$O(n)$

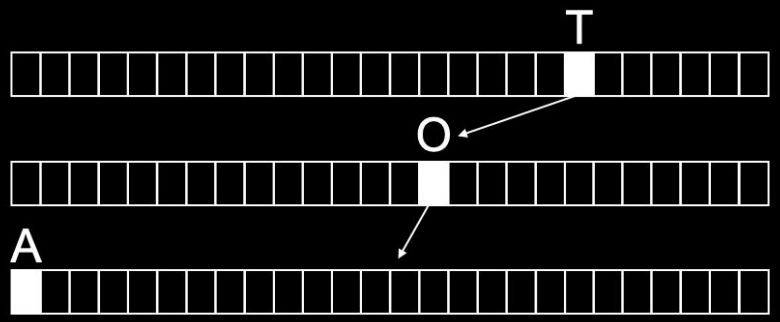
$O(1)$

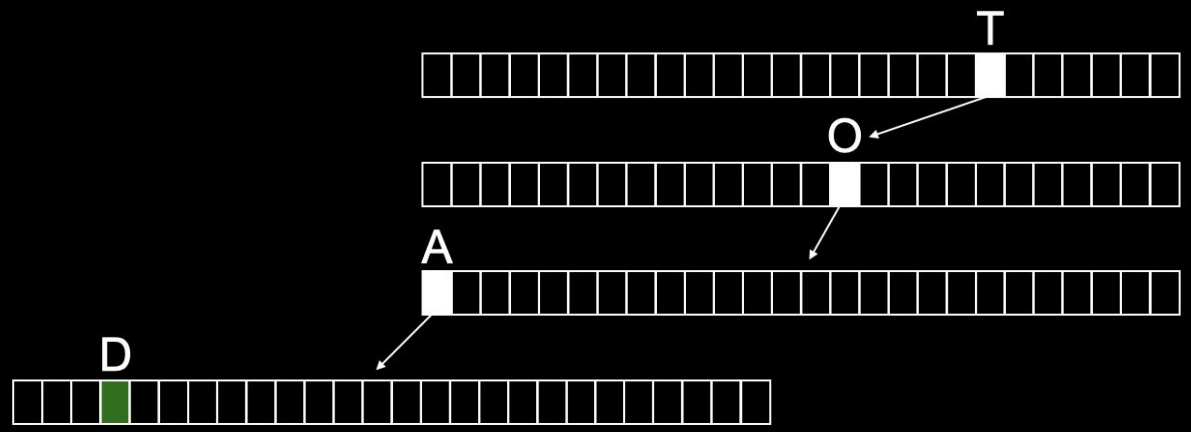
tries

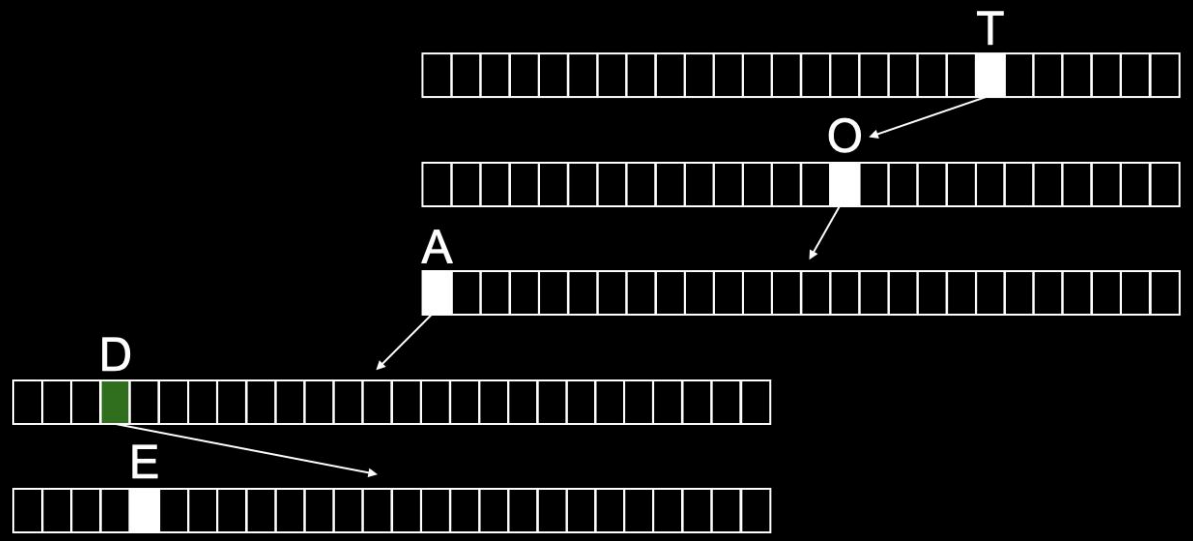
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

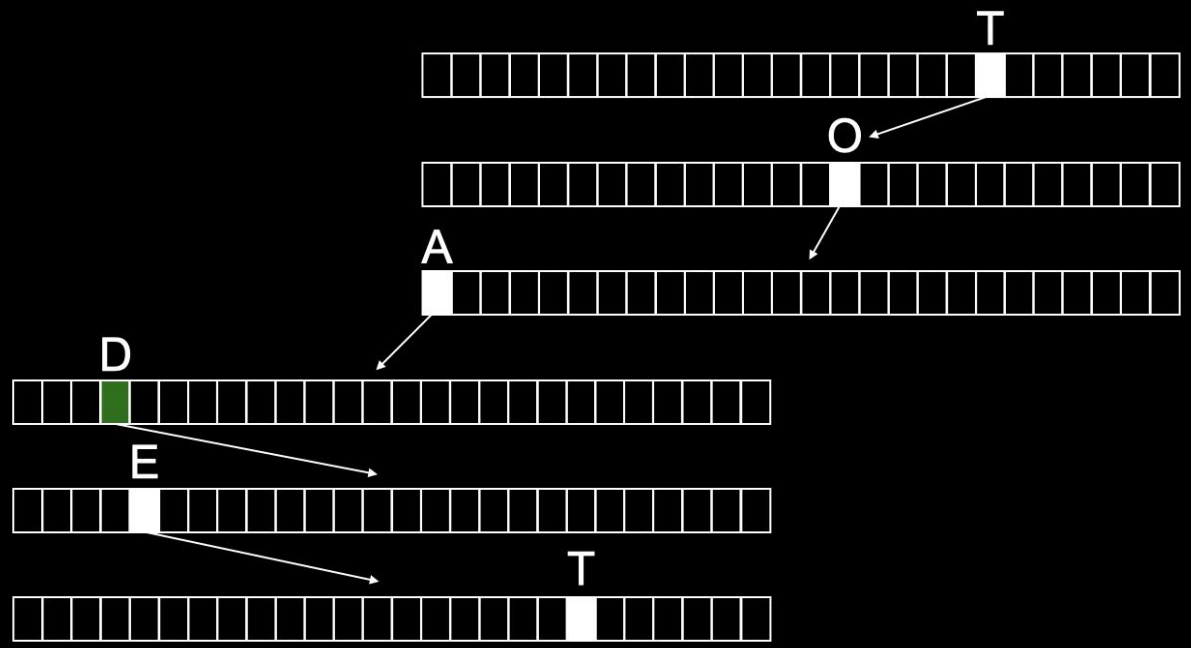
T

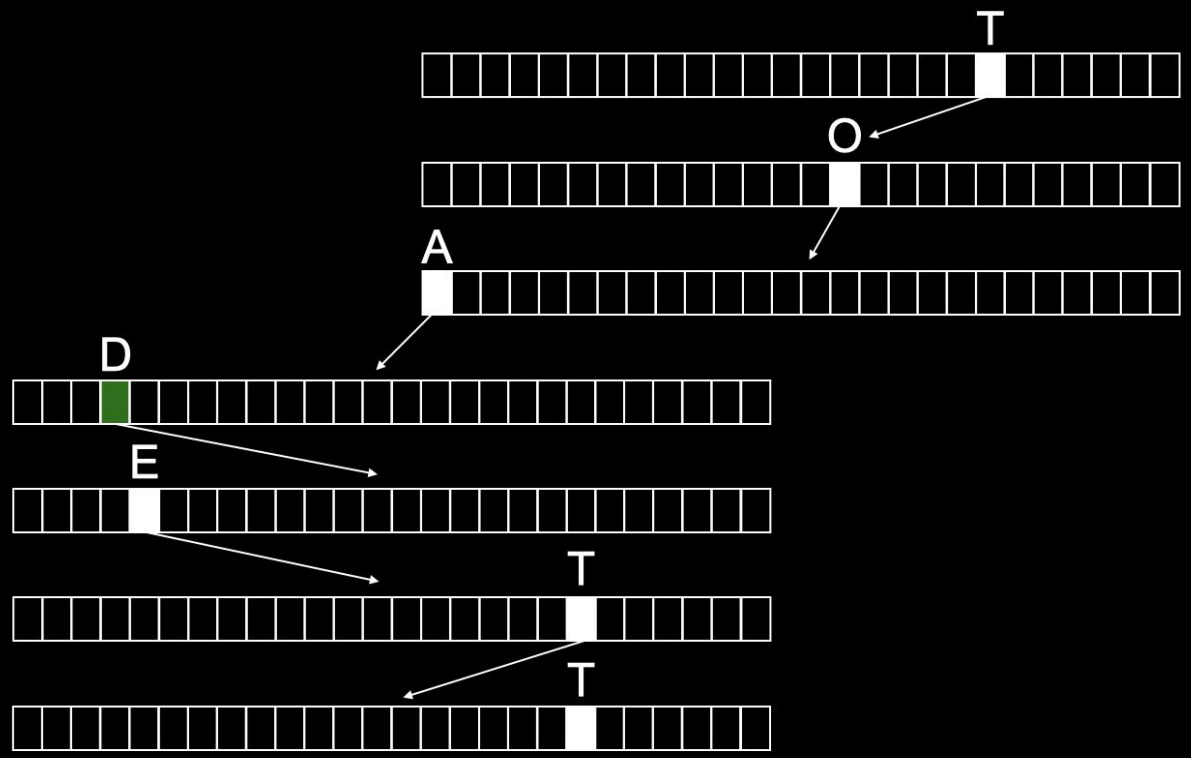

T
O

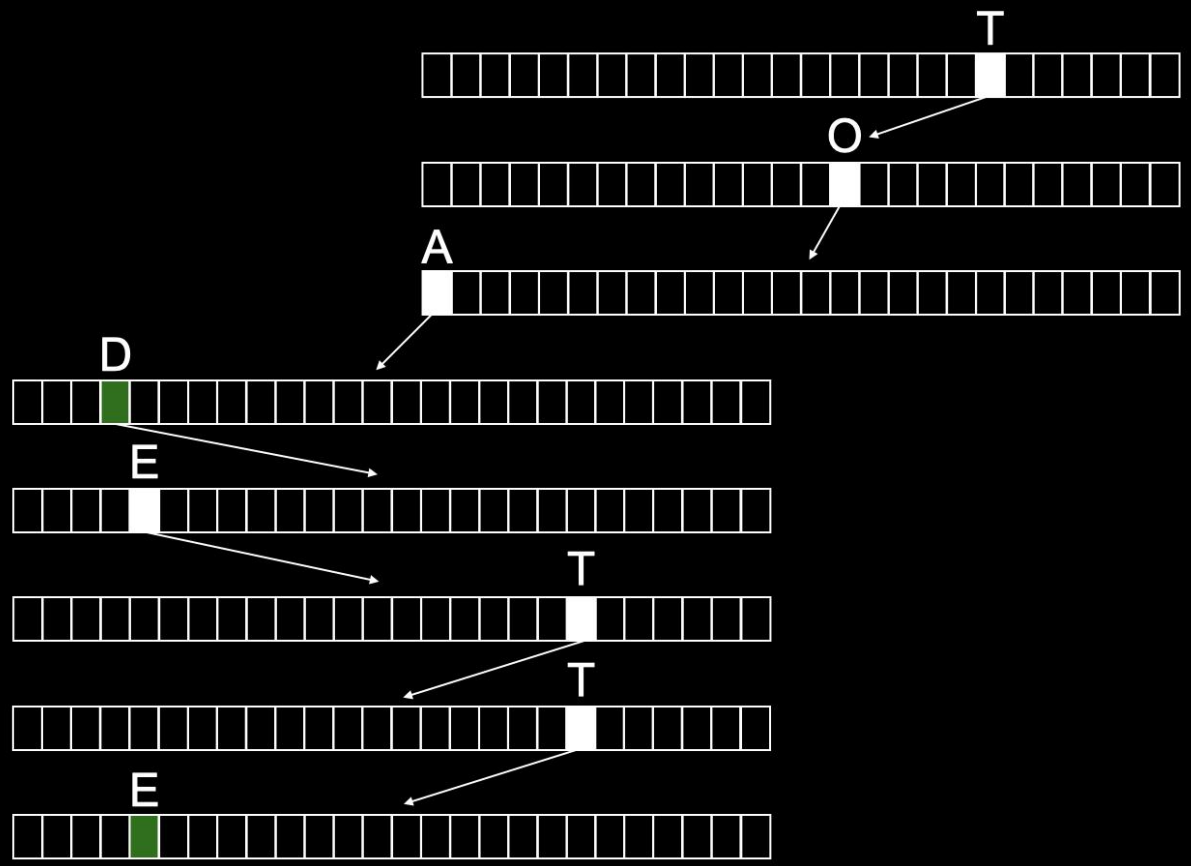

T
O
A

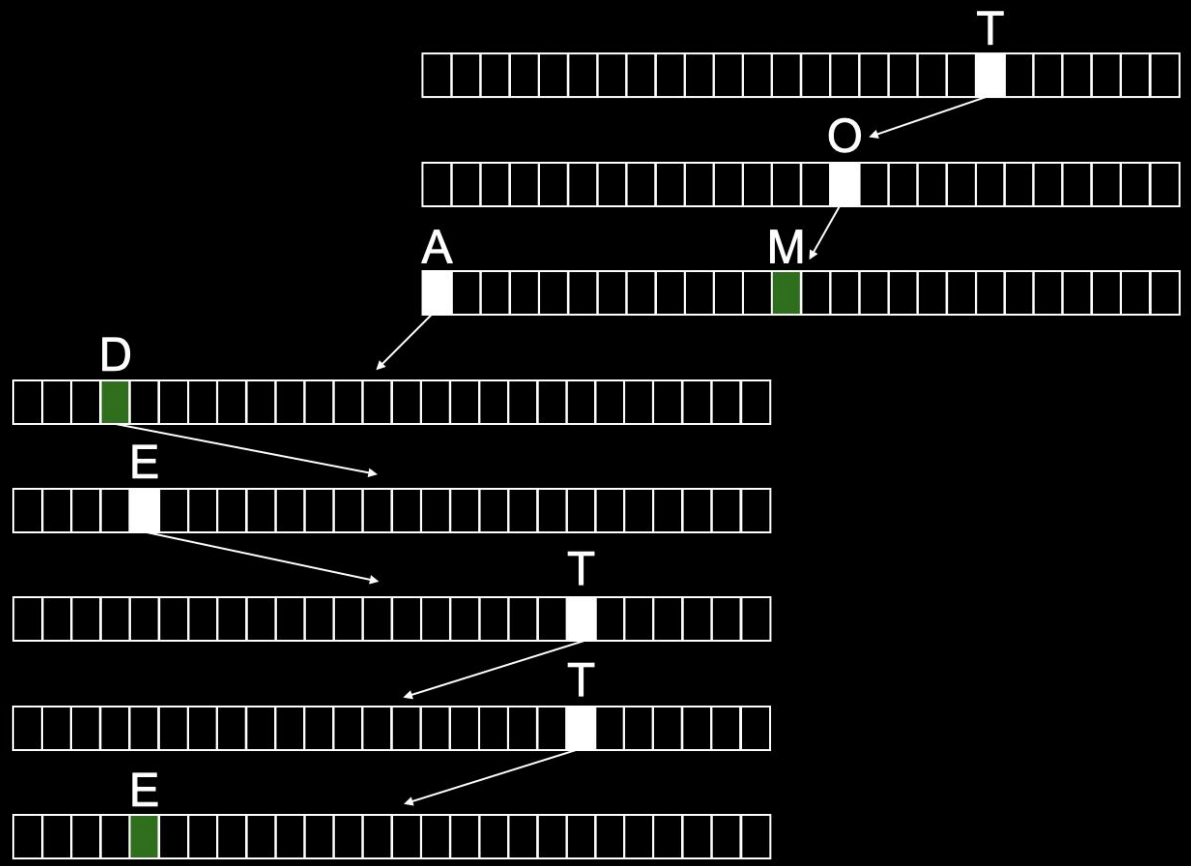

T
O
A
D


T
O
A
D
E


T
O
A
D
E
T


T
O
A
D
E
T
T


T
O
A
D
E
T
T
E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```
typedef struct node
{
    struct node *children[26];
    char *number;
} node;
```

```
node *trie;
```

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$

PICK ME UP
A- E
F- J
K- N
O- Z

This is CS50